Single inclusive gluon production at NLO

October 31, 2026

Contents

1 Review of the vacuum incoming state **1**
1.1 Two outgoing gluons . . . 1
1.1.1 Two ρ part . . . 1
1.1.2 Single ρ part . . . 2
1.2 One outgoing gluon . . . 2
1.2.1 Single ρ part . . . 2
1.2.2 Two ρ part . . . 3
1.2.3 Three ρ part . . . 6
1.3 Final results . . . 8

2 Calculation of the single gluon incoming state **10**
2.1 No outgoing gluon . . . 10
2.2 One outgoing gluon . . . 10
2.2.1 Single ρ part; instantaneous contributions . . . 10
2.2.2 Single ρ part . . . 11
2.2.3 Two ρ part . . . 11
2.3 Two outgoing gluons . . . 12
2.3.1 ρ independent part . . . 12
2.3.2 One ρ part . . . 13
2.4 Normalization . . . 13

3 Parametrization of the Evolution operator (Ω) **14**

4 Production cross section **18**
4.1 Production cross section at the leading order . . . 20
4.2 Production cross section at the Next to leading order . . . 20

A State definitions and normalization **22**

B The CGC Hamiltonian: Eikonal Approximation for QCD Hamiltonian **23**

C Matrix elements **24**

D Fourier Transforms **24**

E BFKL evolution in the coordinate representation **24**

F JIMWLK evolution of the leading order single gluons production cross section. **25**

Corrected copy. Red: as written (to be removed); blue: corrected or added. Your new file (E9 and M8 done). The corrections are explained in `Main_corrections.pdf` (labels E1–E11, X1–X7, G1–G4, M1–M8 refer to it). No limit $\Lambda \to 0$ or $\vee \to \infty$ is taken; the only conditions are $\vee > p^+, k^+ > \Lambda$. The two red terms in the first equations of Appendix F (the $\delta^2(u-v)$ term and the term it leads to) are red in the original and are not corrections.

1 Review of the vacuum incoming state

1.1 Two outgoing gluons

To order g^2, there are three diagrams contributing to the production of two gluons in the final state.

1.1.1 Two ρ part

We first look into the contribution arising from the emission of two gluons from the valence degree of freedom.

diagram `vacuumto2g2rho` (not included)

$$\left|\Psi^1_{gg\rho\rho}\right\rangle = \frac{1}{2}\int_\Lambda^\vee dq^+\, dk^+ \int_\Lambda^{k^+-\Lambda} dp^+ \int d^2\mathbf{q}\, d^2\mathbf{k}\, d^2\mathbf{p}\, \left|g^b_j(k-p)\, g^c_l(p)\right\rangle \frac{\left\langle g^b_j(k-p)\, g^c_l(p)\, |H_g|\, g^a_i(q)\right\rangle \left\langle g^a_i(q)\, |H_g|\, 0\right\rangle}{E_{gg}(p,\, k-p)\, E_g(q)}. \tag{1.1}$$

Using the appropriate expressions (Eqs. (C.2), (C.3)), we obtain

$$\begin{aligned}\left|\Psi^1_{gg\rho\rho}\right\rangle = \frac{1}{2}\int_\Lambda^\vee dk^+\, dq^+ \int_\Lambda^{k^+-\Lambda} dp^+ \int d^2\mathbf{k}\, d^2\mathbf{p}\, d^2\mathbf{q}\, \frac{1}{\frac{\mathbf{q}^2}{2q^+}\left(\frac{\mathbf{p}^2}{2p^+}+\frac{(\mathbf{k}-\mathbf{p})^2}{2(k^+-p^+)}\right)}\Bigg(\delta^{ac}\delta_{il}\delta^{(3)}(p-q)\,\frac{g\left(\mathbf{k}^j-\mathbf{p}^j\right)\rho^b(-\mathbf{k}+\mathbf{p})}{4\pi^{3/2}|k^+-p^+|^{3/2}} \\ + \delta^{ab}\delta_{ji}\delta^{(3)}(k-p-q)\,\frac{g\mathbf{p}^l\rho^c(-\mathbf{p})}{4\pi^{3/2}|p^+|^{3/2}}\Bigg)\left(\frac{g\mathbf{q}^i\rho^a(-\mathbf{q})}{4\pi^{3/2}|q^+|^{3/2}}\right)\left|g^b_j(k-p)\, g^c_l(p)\right\rangle.\end{aligned} \tag{1.2}$$

$$\begin{aligned}\left|\Psi^1_{gg\rho\rho}\right\rangle = \int_\Lambda^\vee dk^+ \int d^2\mathbf{k}\, d^2\mathbf{p} \int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\, \frac{g^2\mathbf{p}^j\left(\mathbf{k}^i-\mathbf{p}^i\right)}{8\pi^3\left(\xi(\mathbf{k}-\mathbf{p})^2+(1-\xi)\mathbf{p}^2\right)}\Bigg(\frac{\rho^a(-\mathbf{k}+\mathbf{p})\rho^b(-\mathbf{p})\sqrt{\xi}}{\mathbf{p}^2\sqrt{1-\xi}} \\ + \frac{\rho^b(-\mathbf{p})\rho^a(-\mathbf{k}+\mathbf{p})\sqrt{1-\xi}}{(\mathbf{k}-\mathbf{p})^2\sqrt{\xi}}\Bigg)\left|g^a_i((1-\xi)k^+,\, \mathbf{k}-\mathbf{p})\, g^b_j(\xi k^+,\, \mathbf{p})\right\rangle.\end{aligned} \tag{1.3}$$

Using the decomposition,

$$\rho^b(-\mathbf{k}+\mathbf{p})\,\rho^a(-\mathbf{p}) = \frac{1}{2}\left[\rho^b(-\mathbf{k}+\mathbf{p}),\, \rho^a(-\mathbf{p})\right] + \frac{1}{2}\left\{\rho^b(-\mathbf{k}+\mathbf{p}),\, \rho^a(-\mathbf{p})\right\} \tag{1.4}$$

$$= \frac{i}{2}f^{bad}\rho^d(-\mathbf{k}) + \frac{1}{2}\left\{\rho^b(-\mathbf{k}+\mathbf{p}),\, \rho^a(-\mathbf{p})\right\}, \tag{1.5}$$

we can separate the contributions

$$\left|\Psi^1_{gg\rho\rho}\right\rangle \equiv \left|\Psi^{1a}_{gg[\rho,\rho]}\right\rangle + \left|\Psi^{1b}_{gg\{\rho,\rho\}}\right\rangle. \tag{1.6}$$

The first term is given by:

$$\left|\Psi^{1a}_{gg[\rho,\rho]}\right\rangle = -\int_\Lambda^\vee dk^+ \int d^2\mathbf{k}\, d^2\mathbf{p} \int_\Lambda^{k^+-\Lambda} dp^+ \, \frac{ig^2 f^{abc}\, p^i (k^j - p^j)\rho^c(-k)\,\sqrt{p^+(k^+-p^+)}}{16\pi^3\ (p^+(k-p)^2 + (k^+-p^+)p^2)} \left[\frac{1}{(k^+-p^+)\,p^2} - \frac{1}{(k-p)^2\,p^+}\right] |g^b_j(k^+-p^+, k-p), g^a_i(p^+,p)\rangle \tag{1.7}$$

$$\left|\Psi^{1a}_{gg[\rho,\rho]}\right\rangle = \int_\Lambda^\vee dk^+ \int d^2\mathbf{k}\, d^2\mathbf{p} \int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\, \frac{-ig^2 f^{abc}\mathbf{p}^i\left(\mathbf{k}^j - \mathbf{p}^j\right)\rho^c(-\mathbf{k})\sqrt{\xi(1-\xi)}}{16\pi^3\,(\xi(\mathbf{k}-\mathbf{p})^2 + (1-\xi)\mathbf{p}^2)} \left(\frac{1}{(1-\xi)\mathbf{p}^2} - \frac{1}{\xi(\mathbf{k}-\mathbf{p})^2}\right) \left|g^b_j((1-\xi)k^+,\,\mathbf{k}-\mathbf{p})\, g^a_i(\xi k^+,\,\mathbf{p})\right\rangle. \tag{1.8}$$

The second term contains the two ρ factors:

$$\left|\Psi^{1b}_{gg\{\rho,\rho\}}\right\rangle = \int_\Lambda^\vee dk^+ \int d^2k\, d^2p \int_\Lambda^{k^+-\Lambda} dp^+ \, \frac{g^2 p^i (k^j - p^j)\left\{\rho^{a(-p),\rho^b(-k+p)}\right\}}{16\pi^3\, p^2(k-p)^2\,\sqrt{p^+(k^+-p^+)}}\, |g^b_j(k^+-p^+,k-p), g^a_i(p^+,p)\rangle \tag{1.9}$$

$$\left|\Psi^{1b}_{gg\{\rho,\rho\}}\right\rangle = \int_\Lambda^\vee dk^+ \int d^2\mathbf{k}\, d^2\mathbf{p} \int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\, \frac{g^2\mathbf{p}^i\left(\mathbf{k}^j - \mathbf{p}^j\right)\left\{\rho^a(-\mathbf{p}),\,\rho^b(-\mathbf{k}+\mathbf{p})\right\}}{16\pi^3\mathbf{p}^2(\mathbf{k}-\mathbf{p})^2\sqrt{\xi(1-\xi)}}\, \left|g^b_j((1-\xi)k^+,\,\mathbf{k}-\mathbf{p})\, g^a_i(\xi k^+,\,\mathbf{p})\right\rangle. \tag{1.10}$$

We transform Eq. (1.10) in the coordinate space using Eq. (D.1)

$$\left|\Psi^{1b}_{gg\{\rho,\rho\}}\right\rangle = -\int_{\mathbf{x},\,\mathbf{y},\,\mathbf{z},\,\mathbf{z}'} \int_\Lambda^\vee dk^+ \int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi \frac{g^2\left\{\rho^a(\mathbf{y}),\,\rho^b(\mathbf{x})\right\}(y-z)^i(x-z')^j}{16\pi^3\sqrt{\xi(1-\xi)}(y-z)^2(x-z')^2}\, \left|g^b_j((1-\xi)k^+,\mathbf{z}')\, g^a_i(\xi k^+,\mathbf{z})\right\rangle. \tag{1.11}$$

1.1.2 Single ρ part

There are two diagrams for this particular section and we first look into a single gluon state from vacuum splitting into two gluons.

diagram `vacuumto2g` (not included)

$$\left|\Psi^2_{gg\rho}\right\rangle = \frac{1}{2}\int_\Lambda^\vee dk^+\, dp^+\, dq^+ \int d^2\mathbf{k}\, d^2\mathbf{p}\, d^2\mathbf{q}\, \left|g^b_j(q)\, g^c_l(p)\right\rangle \frac{\left\langle g^b_j(q)\, g^c_l(p)\left|H_{ggg}\right|\, g^a_i(k)\right\rangle \left\langle g^a_i(k)\left|H_g\right|\, 0\right\rangle}{E_{gg}(p,\,q)\, E_g(k)}. \tag{1.12}$$

By using the relevant matrix elements, Eq. (C.1) and Eq. (C.3), we get

$$\left|\Psi^2_{gg\rho}\right\rangle = \int_\Lambda^\vee dk^+ \int_\Lambda^{k^+-\Lambda} dp^+ \int d^2\mathbf{k}\, d^2\mathbf{p}\, \frac{igf^{acb}}{16\pi^{3/2}\left(\frac{\mathbf{p}^2}{2p^+} + \frac{(\mathbf{k}-\mathbf{p})^2}{2(k^+-p^+)}\right)\left(\frac{\mathbf{k}^2}{2k^+}\right)\sqrt{k^+p^+(k^+-p^+)}} \left(\left[2\mathbf{p}^i - \frac{2p^+}{k^+}\mathbf{k}^i\right]\delta_{jl} + \left[\frac{k^++p^+}{k^+-p^+}(\mathbf{k}^j-\mathbf{p}^j) - \mathbf{k}^j - \mathbf{p}^j\right]\delta_{il} + \left[2\mathbf{k}^l - \frac{2k^+}{p^+}\mathbf{p}^l\right]\delta_{ij}\right)\left(\frac{g\rho^a(-\mathbf{k})\mathbf{k}^i}{4\pi^{3/2}|k^+|^{3/2}}\right)\left|g^b_j(k-p)\, g^c_l(p)\right\rangle. \tag{1.13}$$

$$\left|\Psi^2_{gg\rho}\right\rangle = \int_\Lambda^\vee \frac{dk^+}{k^+} \int_\Lambda^{k^+-\Lambda} dp^+ \int d^2\mathbf{k}\, d^2\mathbf{p}\, \frac{ig^2 f^{abc}\rho^c(-k)k^l}{8\pi^3\, k^2\,\sqrt{p^+(k^+-p^+)}\left(\frac{p^2}{p^+} + \frac{(k-p)^2}{(k^+-p^+)}\right)} \left[\left(\frac{p^l k^+ - p^+k^l}{k^+}\right)\delta_{ij} + \left(\frac{p^+k^j - k^+p^j}{(k^+-p^+)}\right)\delta_{il} + \left(\frac{k^ip^+ - k^+p^i}{p^+}\right)\delta_{jl}\right] |g^b_j(k^+-p^+,k-p), g^a_i(p^+,p)\rangle \tag{1.14}$$

By introducing new variables

$$\mathbf{p} = \xi\mathbf{k} + \widetilde{\mathbf{p}},$$

and Eq. (1.13) becomes

$$\left|\Psi^2_{gg\rho}\right\rangle = -\int_\Lambda^\vee dk^+ \int d^2\mathbf{k}\, d^2\widetilde{\mathbf{p}} \int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\, \frac{ig^2 f^{abc}\rho^a(-\mathbf{k})\sqrt{\xi(1-\xi)}\mathbf{k}^i}{8\pi^3\,(\mathbf{k}^2\xi(1-\xi) + \widetilde{\mathbf{p}}^2)\,\mathbf{k}^2} \left(\widetilde{\mathbf{p}}^i\delta_{jl} - \frac{1}{1-\xi}\widetilde{\mathbf{p}}^j\delta_{il} - \frac{1}{\xi}\widetilde{\mathbf{p}}^l\delta_{ij}\right) \left|g^b_j((1-\xi)k^+,\,(1-\xi)\mathbf{k}-\widetilde{\mathbf{p}})\, g^c_l(\xi k^+,\,\xi\mathbf{k}+\widetilde{\mathbf{p}})\right\rangle. \tag{1.15}$$

The instantaneous contributions are as follows

diagram `vacuumto2ginst` (not included)

$$\left|\Psi^3_{gg\rho}\right\rangle = -\frac{1}{2}\int_\Lambda^\vee dk^+ \int_\Lambda^{k^+-\Lambda} dp^+ \int d^2\mathbf{k}\, d^2\mathbf{p}\, \left|g^b_i(k-p)\, g^c_j(p)\right\rangle \frac{\left\langle g^b_i(k-p)\, g^c_j(p)\left|H_{gg-inst}\right|\, 0\right\rangle}{E_{gg}(k-p,\,p)}. \tag{1.16}$$

By using the relevant matrix elements, Eq. (C.4), and changing the variables we get

$$\left|\Psi^3_{gg\rho}\right\rangle = -\int_\Lambda^\vee dk^+ \int_\Lambda^{k^+-\Lambda} dp^+ \int d^2\mathbf{k}\, d^2\mathbf{p}\, \left|g^b_i(k-p)\, g^c_j(p)\right\rangle \frac{ig^2 f^{abc}(2p^+-k^+)\rho^a(-k)\delta_{ij}}{2(2\pi)^3\sqrt{p^+(k-p)^+}k^{+2}\left(\frac{(k-p)^2}{(k-p)^+} + \frac{p^2}{p^+}\right)} \tag{1.17}$$

Relabeling

$$\left|\Psi^3_{gg\rho}\right\rangle = \int_\Lambda^\vee dk^+ \int_\Lambda^{k^+-\Lambda} dp^+ \int d^2\mathbf{k}\, d^2\mathbf{p}\, \left|g^b_i(k-p)\, g^a_j(p)\right\rangle \frac{ig^2 f^{abc}(2p^+-k^+)\rho^c(-k)\delta_{ij}}{2(2\pi)^3\sqrt{p^+(k-p)^+}k^{+2}\left(\frac{(k-p)^2}{(k-p)^+} + \frac{p^2}{p^+}\right)} \tag{1.18}$$

$$\left|\Psi^3_{gg\rho}\right\rangle = \int_\Lambda^\vee dk^+ \int d^2\mathbf{k}\, d^2\widetilde{\mathbf{p}} \int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\, \frac{ig^2 f^{abc}\rho^a(-\mathbf{k})(1-2\xi)\sqrt{\xi(1-\xi)}}{16\pi^3\,(\mathbf{k}^2\xi(1-\xi) + \widetilde{\mathbf{p}}^2)}\delta_{ij} \left|g^b_i((1-\xi)k^+,\,(1-\xi)\mathbf{k}-\widetilde{\mathbf{p}})\, g^c_j(\xi k^+,\,\xi\mathbf{k}+\widetilde{\mathbf{p}})\right\rangle. \tag{1.19}$$

where $\mathbf{p} = \xi\mathbf{k} + \widetilde{\mathbf{p}}$.

1.2 One outgoing gluon

There is one diagram at order (g), which represents a single outgoing gluon emission. At order (g^3), there are several contributions for a single outgoing gluon state.

diagram `vacuumtog` (not included)

1.2.1 Single ρ part

We first examine the contribution corresponding to the emission of a single gluon from the valence degree of freedom, which results in the leading order contribution.

$$\left|\Psi^{LO}_{g\rho}\right\rangle = -\int_{\Lambda}^{\vee} dk^{+} \int d^{2}\mathbf{k}\, |g^{a}_{i}(\mathbf{k})\rangle \, \frac{\langle g^{a}_{i}(\mathbf{k})\,|\,H_{g}\,|\,0\rangle}{E_{g}(\mathbf{k})}. \tag{1.20}$$

Using the matrix element from Eq. (C.1) and $E_g(k) = \frac{k^2}{2k^+}$

$$\left|\Psi^{LO}_{g\rho}\right\rangle = -\int_{\Lambda}^{\vee} dk^{+} \int d^{2}\mathbf{k}\, \frac{g\,\mathbf{k}^{i}}{2\pi^{3/2}\sqrt{k^{+}}\mathbf{k}^{2}}\, \rho^{a}(-\mathbf{k})\,|g^{a}_{i}(k)\rangle\,. \tag{1.21}$$

Using the transformation equation (Eq. (D.1)) we get the wave function in co-ordinate space.

$$\left|\Psi^{LO}_{g\rho}\right\rangle = \frac{ig}{2\pi^{3/2}}\int_{\Lambda}^{\vee}\frac{dk^{+}}{\sqrt{k^{+}}}\int_{\mathbf{x},\,\mathbf{z}}\frac{(x-z)^{i}}{(x-z)^{2}}\,\rho^{a}(\mathbf{x})\left|g^{a}_{i}(k^{+},\,\mathbf{z})\right\rangle. \tag{1.22}$$

Now, we will examine the contribution at order g^3 with one ρ involvement.

diagram `loop` (not included)

$$\left|\Psi^{1}_{g\rho}\right\rangle = -\frac{1}{2}\int_{\Lambda}^{\vee} dk^{+}\,dp^{+}\,dq^{+}\,dr^{+}\int d^{2}\mathbf{k}\,d^{2}\mathbf{p}\,d^{2}\mathbf{q}\,d^{2}\mathbf{r}\,\left|g^{d}_{l}(r)\right\rangle\frac{\left\langle g^{d}_{l}(r)\,|H_{ggg}|\,g^{c}_{k}(q)\,g^{b}_{j}(p)\right\rangle\left\langle g^{c}_{k}(q)\,g^{b}_{j}(p)\,|H_{ggg}|\,g^{a}_{i}(k)\right\rangle\left\langle g^{a}_{i}(k)\,|H_{g}|\,0\right\rangle}{E_{g}(r)\,E_{gg}(p,\,q)\,E_{g}(k)}, \tag{1.23}$$

Using the matrix elements from Eqs. (C.1) and (C.3), and after the application of the momentum conservation law, which introduces delta functions, both q and r are determined to be $q = k - p$ and $r = k$, we obtain

$$\begin{aligned}\left|\Psi^{1}_{g\rho}\right\rangle = -\int_{\Lambda}^{\vee} dk^{+}\int_{\Lambda}^{k^{+}-\Lambda} dp^{+}\int d^{2}\mathbf{k}\,d^{2}\mathbf{p}\,\frac{g^{3}f^{abc}f^{dbc}\rho^{a}(-k)\,k^{i}}{64\pi^{3/2}\,(2\pi)^{3}k^{+}p^{+}(k^{+}-p^{+})\,\left(\frac{k^{2}}{2k^{+}}\right)^{2}\left(\frac{p^{2}}{2p^{+}}+\frac{(k-p)^{2}}{2(k^{+}-p^{+})}\right)\,k^{+\,3/2}}\\ \left[\left(2p^{l}-\frac{2p^{+}}{k^{+}}k^{l}\right)\delta_{jk}+\left(2k^{j}-\frac{2k^{+}}{p^{+}}p^{j}\right)\delta_{lk}+\left(\frac{(k^{+}+p^{+})}{(k^{+}-p^{+})}(k^{k}-p^{k})-k^{k}-p^{k}\right)\delta_{lj}\right]\times\\ \left[\left(2p^{i}-\frac{2p^{+}}{k^{+}}k^{i}\right)\delta_{jk}+\left(2k^{j}-\frac{2k^{+}}{p^{+}}p^{j}\right)\delta_{ik}+\left(\frac{(k^{+}+p^{+})}{(k^{+}-p^{+})}(k^{k}-p^{k})-k^{k}-p^{k}\right)\delta_{ij}\right]\,|g^{d}_{l}(k^{+},\mathbf{k})\rangle\end{aligned} \tag{1.24}$$

Introducing $p^+ = \xi k^+$ and $p = \xi k + \tilde{p}$, we get

$$\begin{aligned}\left|\Psi^{1}_{g\rho}\right\rangle = -\int_{\Lambda}^{\vee}\frac{dk^{+}}{\sqrt{k^{+}}}\int d^{2}\mathbf{k}\,d^{2}\tilde{\mathbf{p}}\int_{\frac{\Lambda}{k^{+}}}^{1-\frac{\Lambda}{k^{+}}} d\xi\,\frac{g^{3}f^{abc}f^{dbc}\rho^{a}(-k)k^{i}}{8\pi^{3/2}\,(2\pi)^{3}\,\xi(1-\xi)\,k^{4}\,(k^{2}\xi\,(1-\xi)+\tilde{p}^{2})}\\ \left[2\tilde{p}^{l}\delta_{jk}-\frac{2}{\xi}\tilde{p}^{j}\delta_{lk}-\frac{2}{(1-\xi)}\tilde{p}^{k}\delta_{lj}\right]\times\left[2\tilde{p}^{i}\delta_{jk}-\frac{2}{\xi}\tilde{p}^{j}\delta_{ik}-\frac{2}{(1-\xi)}\tilde{p}^{k}\delta_{ij}\right]\,|g^{d}_{l}(k^{+},\mathbf{k})\rangle\end{aligned} \tag{1.25}$$

Simplifying we get

$$\left|\Psi^{1}_{g\rho}\right\rangle = -\int_{\Lambda}^{\vee}\frac{dk^{+}}{\sqrt{k^{+}}}\int d^{2}\mathbf{k}\,d^{2}\tilde{\mathbf{p}}\int_{\frac{\Lambda}{k^{+}}}^{1-\frac{\Lambda}{k^{+}}} d\xi\,\frac{g^{3}N_{c}\rho^{a}(-k)k^{i}\tilde{p}^{2}}{16\pi^{9/2}\,k^{4}\,(k^{2}\xi\,(1-\xi)+\tilde{p}^{2})}\times\left[1+\frac{1}{\xi^{2}}+\frac{1}{(1-\xi)^{2}}\right]\,|g^{a}_{i}(k^{+},\mathbf{k})\rangle \tag{1.26}$$

Integrating wrt $\tilde{p}$, we get

$$\begin{aligned}\left|\Psi^{1}_{g\rho}\right\rangle = -\mu^{2\epsilon}\int_{\Lambda}^{\vee}\frac{dk^{+}}{\sqrt{k^{+}}}\int\frac{d^{d}k}{(2\pi)^{d}}\int_{\frac{\Lambda}{k^{+}}}^{1-\frac{\Lambda}{k^{+}}} d\xi\,\frac{g^{3}\,N_{c}\rho^{a}(-k)k^{i}}{k^{2}\sqrt{\pi}}\times\left[\frac{d}{2(4\pi)^{d/2}}\Gamma\left(-\frac{d}{2}\right)\,\left(\xi(1-\xi)k^{2}\right)^{\frac{d}{2}-1}\right]\times\\ \left[\xi(1-\xi)-2+\frac{1}{\xi(1-\xi)}\right]\,|g^{a}_{i}(k^{+},\mathbf{k})\rangle\end{aligned} \tag{1.27}$$

By performing Γ expansion and setting $\epsilon \to 0$, we get

$$\left|\Psi^{1}_{g\rho}\right\rangle = -\int_{\Lambda}^{\vee}\frac{dk^{+}}{\sqrt{k^{+}}}\int d^{2}k\int_{\frac{\Lambda}{k^{+}}}^{1-\frac{\Lambda}{k^{+}}} d\xi\,\frac{g^{3}\,N_{c}\rho^{a}(-k)k^{i}}{16\pi^{7/2}\,k^{2}}\times\left[-2+\xi(1-\xi)+\frac{1}{\xi(1-\xi)}\right]\times\left[-\frac{2}{\epsilon}+\log\left(\frac{k^{2}\xi(1-\xi)}{\mu^{2}_{\overline{MS}}}\right)\right]\,|g^{a}_{i}(k^{+},\mathbf{k})\rangle \tag{1.28}$$

Integrating over ξ and setting Λ to be 0 wherever possible we get

$$\left|\Psi^{1}_{g\rho}\right\rangle = \int_{\Lambda}^{\vee}\frac{dk^{+}}{\sqrt{k^{+}}}\int d^{2}\mathbf{k}\,\frac{g^{3}N_{c}\rho^{a}(-\mathbf{k})\mathbf{k}^{i}}{32\pi^{7/2}\mathbf{k}^{2}}\left[\left(\frac{11}{3}+4\ln\left(\frac{\Lambda}{k^{+}}\right)\right)\times\left(-\frac{2}{\epsilon}+\ln\left(\frac{\mathbf{k}^{2}}{\mu^{2}_{\overline{MS}}}\right)\right)+2\ln^{2}\left(\frac{\Lambda}{k^{+}}\right)-\frac{67}{9}+\frac{2\pi^{2}}{3}\right]\,|g^{a}_{i}(k^{+},\mathbf{k})\rangle\,. \tag{1.29}$$

Correction. (X1) The exact ξ integral, $\delta = \Lambda/k^+$, replaces the bracket by

$$\begin{aligned}&\left(\frac{11}{3}-8\delta+2\delta^{2}-\frac{4}{3}\delta^{3}+4\ln\frac{\delta}{1-\delta}\right)\left[-\frac{2}{\epsilon}+\log\left(\frac{k^{2}}{\mu^{2}_{\overline{MS}}}\right)\right]-\frac{67}{9}+\frac{2\pi^{2}}{3}+\frac{46}{3}\delta-\frac{4}{3}\delta^{2}+\frac{8}{9}\delta^{3}\\ &+\left(\frac{22}{3}-8\delta+2\delta^{2}-\frac{4}{3}\delta^{3}\right)\ln(1-\delta)-\left(8\delta-2\delta^{2}+\frac{4}{3}\delta^{3}\right)\ln\delta-2\ln^{2}(1-\delta)+2\ln^{2}\delta-4\ln\delta\ln(1-\delta)-8\,\mathrm{Li}_{2}(\delta).\end{aligned}$$

1.2.2 Two ρ part

This subsection contains terms which is of order g^3 with two ρ involvement. We start with the instantaneous interactions and the gluon emitted from the valence degree of freedom can either lose or gain momentum (longitudinal).

diagram `Misha_2rho_single_g` (not included)

$$\left|\Psi^{2}_{g\rho\rho\text{ inst.}}\right\rangle = \int_{\Lambda}^{\vee} dk^{+}\,dp^{+}\int d^{2}\mathbf{k}\,d^{2}\mathbf{p}\,\left|g^{b}_{j}(k)\right\rangle\frac{\left\langle g^{b}_{j}(k)\,|H_{gg-inst}|\,g^{a}_{i}(p)\right\rangle\left\langle g^{a}_{i}(p)\,|H_{g}|\,0\right\rangle}{E_{g}(k)E_{g}(p)}. \tag{1.30}$$

First we consider the process when the longitudinal momentum flows from the valence degree of freedom.

$$\left|\Psi^{2a}_{g\rho\rho\text{ inst.}}\right\rangle = \int_{\Lambda}^{\vee} dk^{+}\int_{\Lambda}^{k^{+}-\Lambda} dp^{+}\int d^{2}\mathbf{k}\,d^{2}\mathbf{p}\,\left|g^{b}_{j}(k)\right\rangle\frac{\left\langle g^{b}_{j}(k)\,|H_{gg-inst}|\,g^{a}_{i}(p)\right\rangle\left\langle g^{a}_{i}(p)\,|H_{g}|\,0\right\rangle}{E_{g}(k)E_{g}(p)}, \tag{1.31}$$

Using the matrix elements from equations (C.1) and (C.5)

$$\left|\Psi^{2a}_{g\rho\rho\text{ inst.}}\right\rangle = \int_\Lambda^\vee dk^+ \int d^2\mathbf{k}\, d^2\mathbf{p} \int_\Lambda^{k^+-\Lambda} dp^+ \frac{ig^3 f^{abc}\rho^c(p-k)\rho^a(-p)p^j(p^+ + k^+)\sqrt{k^+}}{16\pi^{9/2}\, p^+\,(p^+ - k^+)^2\, k^2 p^2}\,\delta_{ij}\,\left|g^b_i(k^+,\,\mathbf{k})\right\rangle. \tag{1.32}$$

Integrating wrt p^+ and setting $\Lambda \to 0$ wherever possible we get

$$\left|\Psi^{2a}_{g\rho\rho\text{ inst.}}\right\rangle = \int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}} \int d^2\mathbf{k}\, d^2\mathbf{p}\, \frac{ig^3 f^{abc}\rho^c(-\mathbf{k}+\mathbf{p})\rho^a(-\mathbf{p})\mathbf{p}^i}{8\pi^{9/2}\mathbf{k}^2\mathbf{p}^2}\left(\frac{k^+}{\Lambda} - 1 - \ln\left(\frac{\Lambda}{k^+}\right)\right)\left|g^b_i(k^+,\mathbf{k})\right\rangle \tag{1.33}$$

Correction. (X2) Exactly, $\frac{k^+}{\Lambda} - 1 - \ln\frac{\Lambda}{k^+} \to E \equiv \frac{k^+}{\Lambda} - \frac{k^+}{k^+-\Lambda} + \ln\frac{k^+-\Lambda}{\Lambda}$, here, in Eq. (1.34), in the one-ρ term below, and in Eq. (1.45).

Using the decomposition principle (see Eq. (1.4)), we split Eq. (1.32) into one ρ and two ρ components.

$$\left|\Psi^{2a}_{g\{\rho,\rho\}\text{ inst.}}\right\rangle = \int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}} \int d^2\mathbf{k}\, d^2\mathbf{p}\, \frac{ig^3 f^{abc}\mathbf{p}^i\,\{\rho^a(-\mathbf{p}),\,\rho^c(-\mathbf{k}+\mathbf{p})\}}{16\pi^{9/2}\mathbf{k}^2\mathbf{p}^2}\left(\frac{k^+}{\Lambda} - 1 - \ln\left(\frac{\Lambda}{k^+}\right)\right)\left|g^b_i(k^+,\mathbf{k})\right\rangle. \tag{1.34}$$

$$\left|\Psi^{2a}_{g\,[\rho,\rho]\text{ inst.}}\right\rangle = -\int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}} \int d^2\mathbf{k}\, d^2\mathbf{p}\, \frac{g^3\, N_c \mathbf{p}^i \rho^a(-k)}{16\pi^{9/2}\mathbf{k}^2\mathbf{p}^2}\left(\frac{k^+}{\Lambda} - 1 - \ln\left(\frac{\Lambda}{k^+}\right)\right)\left|g^a_i(k^+,\mathbf{k})\right\rangle = 0 \tag{1.35}$$

Since,

$$\int d^2p\, \frac{p^i}{p^2} = 0$$

Next, we consider the momentum flowing into the valence degree of freedom.

diagram `Misha_Out_2rho_single_g` (not included)

$$\left|\Psi^{2b}_{g\rho\rho\text{ inst.}}\right\rangle = \int_\Lambda^\vee dk^+ \int_\Lambda^{k^+-\Lambda} dp^+ \int d^2\mathbf{k}\, d^2\mathbf{p}\, \left|g^b_j(p)\right\rangle \frac{\left\langle g^b_j(p)\left|H_{gg-inst}\right|g^a_i(k)\right\rangle \left\langle g^a_i(k)\left|H_g\right|0\right\rangle}{E_g(p)E_g(k)}. \tag{1.36}$$

Using the matrix elements from equations (C.1) and (C.5)

$$\left|\Psi^{2b}_{g\rho\rho\text{ inst.}}\right\rangle = \int_\Lambda^\vee dk^+ \int d^2\mathbf{k}\, d^2\mathbf{p} \int_\Lambda^{k^+-\Lambda} dp^+ \frac{ig^3 f^{abc}\rho^c(k-p)\rho^a(-k)k^i(p^+ + k^+)\sqrt{p^+}}{16\pi^{9/2}\, k^+\,(p^+ - k^+)^2\, k^2 p^2}\,\delta_{ij}\,\left|g^b_j(p^+,\,\mathbf{p})\right\rangle. \tag{1.37}$$

Using the decomposition principle (see Eq. (1.4)), we split Eq. (1.37) into one ρ and two ρ components.

$$\left|\Psi^{2b}_{g\{\rho,\rho\}\text{ inst.}}\right\rangle = \int_\Lambda^\vee dk^+ \int d^2\mathbf{k}\, d^2\mathbf{p} \int_\Lambda^{k^+-\Lambda} dp^+ \frac{ig^3 f^{abc}\{\rho^c(k-p),\rho^a(-k)\}\,k^i(p^+ + k^+)\sqrt{p^+}}{32\pi^{9/2}\, k^+\,(p^+ - k^+)^2\, k^2 p^2}\,\delta_{ij}\,\left|g^b_j(p^+,\,\mathbf{p})\right\rangle. \tag{1.38}$$

Introducing new variables $\xi = \frac{p^+}{k^+}$ and $p \leftrightarrow k$, we get

$$\left|\Psi^{2b}_{g\{\rho,\rho\}\text{ inst.}}\right\rangle = \int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}} \int d^2\mathbf{k}\, d^2\mathbf{p} \int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\, \frac{ig^3 f^{abc}\{\rho^c(-k+p),\rho^a(-p)\}\,p^i}{32\pi^{9/2}\;k^2p^2}\,\frac{(1+\xi)\sqrt{\xi}}{(1-\xi)^2}\,\left|g^b_i(\xi k^+,\,\mathbf{k})\right\rangle. \tag{1.39}$$

The corresponding one ρ term is as follows and it vanishes due to p integration.

$$\left|\Psi^{2b}_{g\,[\rho,\rho]\text{ inst.}}\right\rangle = -\int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}} \int d^2\mathbf{k}\, d^2\mathbf{p} \int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\, \frac{g^3\, N_c\, \rho^b(-k)\, p^i}{32\pi^{9/2}\, k^2p^2} \times \frac{(1+\xi)\sqrt{\xi}}{(1-\xi)^2}\,\left|g^b_i(\xi k^+,\,\mathbf{k})\right\rangle = 0 \tag{1.40}$$

Next, we consider a diagram in which two gluons emitted from the vacuum combine to form a gluon.

diagram `Misha_2rho_3_single_g` (not included)

$$\left|\Psi^3_{g\rho\rho}\right\rangle = -\frac{1}{2}\int_\Lambda^\vee dk^+\,dp^+\,dq^+\,dr^+\int d^2\mathbf{k}\,d^2\mathbf{p}\,d^2\mathbf{q}\,d^2\mathbf{r}\,\left|g^d_m(p)\right\rangle \frac{\left\langle g^d_m(p)\,|H_{ggg}|\,g^c_k(r)\,g^b_j(q)\right\rangle\left\langle g^c_k(r)\,g^b_j(q)\,|H_g|\,g^a_i(k)\right\rangle\left\langle g^a_i(k)\,|H_g|\,0\right\rangle}{E_g(k)\,E_{gg}(q,\,r)\,E_g(p)}. \tag{1.41}$$

Using the matrix elements from (C.1), (C.2) and (C.3), we get

$$\begin{aligned}\left|\Psi^3_{g\rho\rho}\right\rangle = &\int_\Lambda^\vee dk^+\int_\Lambda^{k^+-\Lambda} dp^+\,dq^+\int d^2\mathbf{k}\,d^2\mathbf{p}\,d^2\mathbf{q}\frac{igf^{bcd}}{\left(\frac{\mathbf{q}^2}{2q^+}\right)\left(\frac{\mathbf{p}^2}{2p^+}+\frac{(\mathbf{k}-\mathbf{p})^2}{2(k^+-p^+)}\right)\left(\frac{\mathbf{k}^2}{2k^+}\right)16\pi^{3/2}\sqrt{k^+p^+(k^+-p^+)}}\\ &\left[\left(2\mathbf{p}^l-\frac{2p^+}{k^+}\mathbf{k}^l\right)\delta_{nj}+\left(2\mathbf{k}^j-\frac{2k^+}{p^+}\mathbf{p}^j\right)\delta_{ln}+\left(\frac{k^++p^+}{k^+-p^+}(\mathbf{k}^n-\mathbf{p}^n)-\mathbf{k}^n-\mathbf{p}^n\right)\delta_{lj}\right]\\ &\left(\delta^{ac}\delta^{ni}\delta^{(3)}(q-k+p)\frac{g\mathbf{p}^j\rho^b(-\mathbf{p})}{4\pi^{3/2}|p^+|^{3/2}}+\delta^{ab}\delta^{ni}\delta^{ji}\delta^{(3)}(q-p)\frac{g(\mathbf{k}^n-\mathbf{p}^n)\rho^c(-\mathbf{k}+\mathbf{p})}{4\pi^{3/2}|k^+-p^+|^{3/2}}\right)\left(\frac{g\rho^a(-\mathbf{q})\mathbf{q}^i}{4\pi^{3/2}|q^+|^{3/2}}\right)\left|g^d_l(k^+,\mathbf{k})\right\rangle.\end{aligned} \tag{1.42}$$

Introducing $\xi=\frac{p^+}{k^+}$, we obtain

$$\begin{aligned}\left|\Psi^3_{g\rho\rho}\right\rangle = &\int_\Lambda^\vee dk^+\int d^2\mathbf{k}\,d^2\mathbf{p}\int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\,\frac{ig^3f^{bad}\rho^b(-\mathbf{p})\rho^a(-\mathbf{k}+\mathbf{p})}{16\pi^{9/2}\xi(\mathbf{k}-\mathbf{p})^2\left((1-\xi)\mathbf{p}^2+\xi(\mathbf{k}-\mathbf{p})^2\right)\mathbf{k}^2\sqrt{k^+}}\\ &\left[\left(2\mathbf{p}^l-2\xi\mathbf{k}^l\right)\mathbf{p}\cdot(\mathbf{k}-\mathbf{p})+\left(2\mathbf{k}\cdot\mathbf{p}-\frac{2}{\xi}\mathbf{p}^2\right)(\mathbf{k}-\mathbf{p})^l+\left(\frac{1+\xi}{1-\xi}(\mathbf{k}-\mathbf{p})^2-\mathbf{k}^2+\mathbf{p}^2\right)\mathbf{p}^l\right]\left|g^d_l(k^+,\mathbf{k})\right\rangle.\end{aligned} \tag{1.43}$$

Using the decomposition principle (i.e., (1.4)), we split (1.43) into one ρ and two ρ components.

$$\left|\Psi^3_{g[\rho,\rho]}\right\rangle = -\int_\Lambda^\vee\frac{dk^+}{\sqrt{k^+}}\int d^2\mathbf{k}\,\frac{g^3N_c\rho^a(-\mathbf{k})\mathbf{k}^i}{32\pi^{7/2}\mathbf{k}^2}\left(3\ln\left(\frac{\Lambda}{k^+}\right)\left[-\frac{2}{\epsilon}+\ln\left(\frac{\mathbf{k}^2}{\mu^2_{\overline{MS}}}\right)\right]+\frac{\pi^2}{3}+2\ln^2\left(\frac{\Lambda}{k^+}\right)\right)\left|g^a_i(k^+,\mathbf{k})\right\rangle. \tag{1.44}$$

Correction. (X3) The ξ integral is done with $\Lambda\to 0$. Exactly ($\delta=\Lambda/k^+$) the bracket is

$$3\ln\frac{\delta}{1-\delta}\left[-\frac{2}{\epsilon}+\log\left(\frac{k^2}{\mu^2_{\overline{MS}}}\right)\right]+\frac{\pi^2}{3}+2\ln^2\delta-2\ln^2(1-\delta)-2\ln\delta\ln(1-\delta)-4\,\mathrm{Li}_2(\delta).$$

$$\begin{aligned}\left|\Psi^3_{g\{\rho,\rho\}}\right\rangle = -\int_\Lambda^\vee\frac{dk^+}{\sqrt{k^+}}\int d^2\mathbf{k}\,d^2\mathbf{p}\,\frac{ig^3f^{abc}\left\{\rho^b(-\mathbf{p}),\,\rho^c(-\mathbf{k}+\mathbf{p})\right\}}{32\pi^{9/2}\mathbf{p}^2(\mathbf{k}-\mathbf{p})^2\mathbf{k}^2}\left[\ln\left(\frac{\Lambda}{k^+}\right)\mathbf{k}\cdot\mathbf{p}\,\mathbf{k}^i-\left(\ln\left(\frac{\Lambda}{k^+}\right)\mathbf{k}^2\right.\right.\\ \left.\left.+2\left(\frac{k^+}{\Lambda}-1-\ln\left(\frac{\Lambda}{k^+}\right)\right)(\mathbf{k}-\mathbf{p})^2\right)\mathbf{p}^i\right]\left|g^a_i(k^+,\mathbf{k})\right\rangle.\end{aligned} \tag{1.45}$$

Correction. (X4) Exactly, the anticommutator part of Eq. (1.43) is $\int\frac{dk^+}{\sqrt{k^+}}\int d^2\mathbf{k}\,d^2\mathbf{p}\,\frac{ig^3f^{abc}\{\rho^b(-\mathbf{p}),\rho^c(-\mathbf{k}+\mathbf{p})\}}{32\pi^{9/2}(\mathbf{k}-\mathbf{p})^2\mathbf{k}^2}V^i\left|g^a_i(k^+,\mathbf{k})\right\rangle$ with

$$V^i=(\mathbf{k}-\mathbf{p})^i\Big[\frac{2\,\mathbf{k}\cdot(\mathbf{k}-\mathbf{p})}{\mathbf{p}^2}(\ell-\mathcal{L})-2\Big(\frac{k^+}{\Lambda}-\frac{k^+}{k^+-\Lambda}\Big)\Big]-\frac{2\,\mathbf{p}\cdot(\mathbf{k}-\mathbf{p})}{\Delta}\mathcal{L}\,\mathbf{k}^i+2(\ell+\mathcal{L})\mathbf{p}^i,$$

$\ell=\ln\frac{k^+-\Lambda}{\Lambda}$, $\Delta=(\mathbf{k}-\mathbf{p})^2-\mathbf{p}^2$, $\mathcal{L}=\ln\frac{k^+(\mathbf{k}-\mathbf{p})^2-\Lambda\Delta}{k^+\mathbf{p}^2+\Lambda\Delta}$. In the sum with Eq. (1.34) (with E exact), the part odd under $\mathbf{p}\to\mathbf{k}-\mathbf{p}$ (the only one that survives) is exactly the first term of Eq. (1.91) with $\ln\frac{\Lambda}{k^+}\to\ln\frac{\Lambda}{k^+-\Lambda}$.

We look into an instantaneous contribution which gives two gluons in which one is absorbed to the valence.

diagram `Misha_2rho_insta_single_g` (not included)

$$\left|\Psi^4_{g\rho\rho}\right\rangle = \frac{1}{2}\int_\Lambda^\vee dq^+\, dk^+ \int_\Lambda^{k^+-\Lambda} dp^+ \int d^2\mathbf{q}\, d^2\mathbf{k}\, d^2\mathbf{p}\left|g^d_l(p)\right\rangle \frac{\left\langle g^d_l(p)\left|H_g\right| g^b_j(q)\, g^a_i(k-q)\right\rangle \left\langle g^b_j(q)\, g^a_i(k-q)\left|H_{gg-inst}\right|0\right\rangle}{E_g(p)E_{gg}(q,\,k-q)}. \tag{1.46}$$

Using the matrix elements from Eqs. (C.2) and (C.4) we get,

$$\begin{aligned}\left|\Psi^4_{g\rho\rho}\right\rangle =& \frac{1}{2}\int_\Lambda^\vee dk^+\, dq^+ \int_\Lambda^{k^+-\Lambda} dp^+ \int d^2\mathbf{k}\, d^2\mathbf{p}\, d^2\mathbf{q}\frac{1}{\left(\frac{\mathbf{q}^2}{2q^+}+\frac{(\mathbf{k}-\mathbf{q})^2}{2(k^+-q^+)}\right)\left(\frac{\mathbf{p}^2}{2p^+}\right)}\Bigg(\delta^{dc}\delta_{il}\delta^{(3)}(k-p-q)\frac{g\mathbf{q}^j\rho^b(\mathbf{q})}{4\pi^{3/2}|q^+|^{3/2}}\\ &+\delta^{db}\delta_{lj}\delta^{(3)}(q-p)\frac{g(\mathbf{k}^i-\mathbf{q}^i)\rho^c(\mathbf{k}-\mathbf{q})}{4\pi^{3/2}|k^+-q^+|^{3/2}}\Bigg)\left(\frac{ig^2f^{abc}(k^+-2q^+)\delta_{ij}\rho^a(-\mathbf{k})}{2(2\pi)^3\sqrt{(k^+-q^+)q^+}(k^+)^2}\right)\left|g^d_l(p^+,\mathbf{p})\right\rangle.\end{aligned} \tag{1.47}$$

Integration wrt q, we get

$$\left|\Psi^4_{g\rho\rho}\right\rangle = -\int_\Lambda^\vee dk^+ \int d^2\mathbf{k}\, d^2\mathbf{p}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{\sqrt{p^+}}\frac{ig^3f^{abc}(2p^+-k^+)\,\rho^c(k-p)\rho^a(-k)\,(k^i-p^i)}{64\pi^{9/2}\left(\frac{p^2}{2p^+}+\frac{(k-p)^2}{2(k^+-p^+)}\right)\times\left(\frac{p^2}{2p^+}\right)\,(k^+(k^+-p^+))^2}\left|g^b_i(p^+,\mathbf{p})\right\rangle. \tag{1.48}$$

Introducing $\xi=\frac{p^+}{k^+}$, we get

$$\left|\Psi^4_{g\rho\rho}\right\rangle = -\int_\Lambda^\vee\frac{dk^+}{\sqrt{k^+}}\int d^2\mathbf{k}\, d^2\mathbf{p}\int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\frac{ig^3f^{abc}\rho^c(\mathbf{k}-\mathbf{p})\rho^a(-\mathbf{k})(\mathbf{k}^i-\mathbf{p}^i)(2\xi-1)\xi^{3/2}}{16\pi^{9/2}\left(\xi(\mathbf{k}-\mathbf{p})^2+(1-\xi)\mathbf{p}^2\right)\,\mathbf{p}^2(1-\xi)}\left|g^b_i(\xi k^+,\mathbf{p})\right\rangle. \tag{1.49}$$

Next, we look into another contribution where a gluon emitted from the vacuum splits into two gluons, with one of them being absorbed.

diagram `Misha_2rho_loop_single_g` (not included)

$$\left|\Psi^5_{g\rho\rho}\right\rangle = -\frac{1}{2}\int_\Lambda^\vee dk^+\, dp^+\, dq^+\, dr^+\int d^2\mathbf{k}\, d^2\mathbf{p}\, d^2\mathbf{q}\, d^2\mathbf{r}\left|g^d_m(k)\right\rangle\frac{\left\langle g^d_m(k)\left|H_g\right| g^c_k(r)\, g^b_j(q)\right\rangle\left\langle g^c_k(r)\, g^b_j(q)\left|H_{ggg}\right| g^a_i(p)\right\rangle\left\langle g^a_i(p)\left|H_g\right|0\right\rangle}{E_g(p)\,E_{gg}(q,\,r)\,E_g(k)}. \tag{1.50}$$

Using the matrix elements from Eqs. (C.1),(C.2) and (C.3), we obtain the following

$$\begin{aligned}\left|\Psi^5_{g\rho\rho}\right\rangle =& -\int_\Lambda^\vee dk^+\, dp^+\int_\Lambda^{k^+-\Lambda} dq^+\int d^2\mathbf{k}\, d^2\mathbf{p}\, d^2\mathbf{q}\frac{1}{\left(\frac{\mathbf{p}^2}{2p^+}\right)\left(\frac{(\mathbf{k}-\mathbf{q})^2}{2(k^+-q^+)}+\frac{\mathbf{q}^2}{2q^+}\right)\left(\frac{\mathbf{k}^2}{2k^+}\right)}\Bigg[\delta^{dc}\delta^{km}\delta^{(3)}(k-p-q)\frac{g\mathbf{q}^j\rho^b(\mathbf{q})}{4\pi^{3/2}|q^+|^{3/2}}\\ &+\delta^{db}\delta^{jm}\delta^{(3)}(p-q)\frac{g(\mathbf{k}^k-\mathbf{q}^k)\rho^c(\mathbf{k}-\mathbf{q})}{4\pi^{3/2}|k^+-q^+|^{3/2}}\Bigg]\times\Bigg[\left(2\mathbf{q}^i-\frac{2q^+}{k^+}\mathbf{k}^i\right)\delta_{jk}+\left(2\mathbf{k}^j-\frac{2k^+}{q^+}\mathbf{q}^j\right)\delta_{ik}+\Bigg(\frac{k^++q^+}{k^+-q^+}(\mathbf{k}^k-\mathbf{q}^k)\\ &-\mathbf{k}^k-\mathbf{q}^k\Bigg)\delta_{ij}\Bigg]\frac{igf^{abc}}{16\pi^{3/2}\sqrt{k^+q^+(k^+-q^+)}}\left(\frac{g\mathbf{k}^i\rho^a(-\mathbf{k})}{4\pi^{3/2}|k^+|^{3/2}}\right)\left|g^d_m(p^+,\mathbf{p})\right\rangle.\end{aligned} \tag{1.51}$$

Integrating wrt q and introducing $p^+=\xi k^+$ we get

$$\begin{aligned}\left|\Psi^5_{g\rho\rho}\right\rangle =& \int_\Lambda^\vee\frac{dk^+}{\sqrt{k^+}}\int d^2\mathbf{k}\, d^2\mathbf{p}\int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\,\frac{ig^3f^{abc}\rho^c(\mathbf{k}-\mathbf{p})\rho^a(-\mathbf{k})\xi^{3/2}}{16\pi^{9/2}\mathbf{k}^2\mathbf{p}^2\left(\xi(\mathbf{k}-\mathbf{p})^2+(1-\xi)\mathbf{p}^2\right)(1-\xi)}\\ &\times\left[\left((2\xi-1)\mathbf{k}^2-\mathbf{p}^2-\frac{1+\xi}{1-\xi}(\mathbf{k}-\mathbf{p})^2\right)\mathbf{k}^i+\left(\frac{\xi-1}{\xi}\mathbf{k}\cdot\mathbf{p}+\frac{1-\xi^2}{\xi}\mathbf{k}^2\right)2\mathbf{p}^i\right]\left|g^b_i(\xi k^+,\,\mathbf{p})\right\rangle.\end{aligned} \tag{1.52}$$

By combining Eqs. . (1.49) and (1.52), we split them into one ρ and two ρ components using the decomposition principle (Eq. (1.4)), we get

$$\left|\Psi^4_{g\rho\rho}\right\rangle+\left|\Psi^5_{g\rho\rho}\right\rangle=\left|\Psi^{4+5}_{g\rho\rho}\right\rangle+\left|\Psi^{4+5}_{g\rho}\right\rangle \tag{1.53}$$

We write it by doing a transformation as $p\leftrightarrow k$ and we get

$$\begin{aligned}\left|\Psi^{4+5}_{g\rho\rho}\right\rangle =& \int_\Lambda^\vee\frac{dk^+}{\sqrt{k^+}}\int d^2\mathbf{k}\, d^2\mathbf{p}\int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\,\frac{ig^3f^{abc}\left\{\rho^c(-\mathbf{k}+\mathbf{p}),\,\rho^a(-\mathbf{p})\right\}\xi^{3/2}}{32\pi^{9/2}\mathbf{k}^2\mathbf{p}^2\left(\xi(\mathbf{k}-\mathbf{p})^2+(1-\xi)\mathbf{k}^2\right)(1-\xi)}\\ &\left(\left[\frac{2}{\xi}\left(\mathbf{p}^2-\mathbf{k}\cdot\mathbf{p}\right)-\mathbf{p}^2+2\mathbf{k}\cdot\mathbf{p}\right]\mathbf{k}^j-\left[\mathbf{k}^2+\frac{1+\xi}{1-\xi}(\mathbf{k}-\mathbf{p})^2\right]\mathbf{p}^j\right)\left|g^b_j(\xi k^+,\,\mathbf{k})\right\rangle.\end{aligned} \tag{1.54}$$

and

$$\begin{aligned}\left|\Psi^{4+5}_{g\rho}\right\rangle =& \int_\Lambda^\vee\frac{dk^+}{\sqrt{k^+}}\int d^2\mathbf{k}\int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\,\frac{g^3N_c\rho^a(-\mathbf{k})\mathbf{k}^i}{32\pi^{7/2}\sqrt{\xi}(1-\xi)\mathbf{k}^2}\Bigg((\xi-2)\ln(\xi)+(\xi+1)\ln(1-\xi)-(\xi-2)\\ &\left[-\frac{2}{\epsilon}+\ln\left(\frac{\mathbf{k}^2}{\mu^2_{\overline{MS}}}\right)\right]\Bigg)\left|g^a_i(\xi k^+,\,\mathbf{k})\right\rangle.\end{aligned} \tag{1.55}$$

A convenient way to represent (1.54) using (1.39), we get

$$\begin{aligned}\left|\Psi^{4+5}_{g\rho\rho}\right\rangle =& -\left|\Psi^{2b}_{g\{\rho,\rho\}\text{ inst.}}\right\rangle\\ &+\int_\Lambda^\vee\frac{dk^+}{\sqrt{k^+}}\int d^2\mathbf{k}\, d^2\mathbf{p}\int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\,\frac{ig^3f^{abc}\left\{\rho^c(-\mathbf{k}+\mathbf{p}),\,\rho^a(-\mathbf{p})\right\}\xi^{3/2}}{32\pi^{9/2}\mathbf{k}^2\mathbf{p}^2\left(\xi(\mathbf{k}-\mathbf{p})^2+(1-\xi)\mathbf{k}^2\right)(1-\xi)}\\ &\left(\left[\frac{2}{\xi}\left(\mathbf{p}^2-\mathbf{k}\cdot\mathbf{p}\right)-\mathbf{p}^2+2\mathbf{k}\cdot\mathbf{p}\right]\mathbf{k}^i+\frac{1}{\xi}\mathbf{k}^2\mathbf{p}^i\right)\left|g^b_i(\xi k^+,\,\mathbf{k})\right\rangle.\end{aligned} \tag{1.56}$$

The benefit of this final representation becomes clear in the next subsection, where we combine $\left|\Psi^{4+5}_{g\rho\rho}\right\rangle$ and $\left|\Psi^{2b}_{g\{\rho,\rho\}\text{ inst.}}\right\rangle$. In this form, the mutual cancellation between the contributions becomes immediately transparent.

1.2.3 Three ρ part

We will consider three contributions in which a gluon emitted from the vacuum is absorbed, followed by the emission of another gluon. There are three different possible time orderings for this process. All these contributions are at order g^3.

$$\left|\Psi^6_{g\rho\rho\rho}\right\rangle = -\frac{1}{2}\int_\Lambda^\vee dk^+\, dp^+\, dq^+\, dr^+\int d^2\mathbf{k}\, d^2\mathbf{p}\, d^2\mathbf{q}\, d^2\mathbf{r}\left|g^d_l(k)\right\rangle\frac{\left\langle g^d_l(k)\left|H_g\right| g^b_j(r)\, g^c_k(q)\right\rangle\left\langle g^b_j(r)\, g^c_k(q)\left|H_g\right| g^a_i(p)\right\rangle\left\langle g^a_i(p)\left|H_g\right|0\right\rangle}{E_g(k)\,E_{gg}(r,\,q)\,E_g(p)}. \tag{1.57}$$

diagram `Misha_2rho_loopVal1_single_g` (not included)

diagram `Misha_2rho_loopVal2_single_g` (not included)

Using the matrix elements from Eqs. . (C.1) and (C.2) we get,

$$\begin{aligned}\left|\Psi^6_{g\rho\rho\rho}\right\rangle = -\frac{1}{2}\int_\Lambda^\vee dk^+\,dp^+\,dq^+\,dr^+\int d^2\mathbf{k}\,d^2\mathbf{p}\,d^2\mathbf{q}\,d^2\mathbf{r}\frac{1}{\left(\frac{\mathbf{p}^2}{2p^+}\right)\left(\frac{\mathbf{r}^2}{2r^+}+\frac{\mathbf{q}^2}{2q^+}\right)\left(\frac{\mathbf{k}^2}{2k^+}\right)}\left(\delta^{dc}\delta_{kl}\delta^{(3)}(q-k)\,\frac{g\mathbf{r}^j\rho^b(\mathbf{r})}{4\pi^{3/2}|r^+|^{3/2}}\right.\\ \left.+\,\delta^{db}\delta_{jl}\delta^{(3)}(k-r)\,\frac{g\mathbf{q}^k\rho^c(\mathbf{q})}{4\pi^{3/2}|q^+|^{3/2}}\right)\left(\delta^{ab}\delta_{ij}\delta^{(3)}(r-p)\,\frac{g\mathbf{q}^k\rho^c(-\mathbf{q})}{4\pi^{3/2}|q^+|^{3/2}}+\delta^{ac}\delta_{ik}\delta^{(3)}(p-q)\,\frac{g\mathbf{r}^j\rho^b(-\mathbf{r})}{4\pi^{3/2}|r^+|^{3/2}}\right)\\ \left(\frac{g\rho^a(-\mathbf{p})\mathbf{p}^i}{4\pi^{3/2}|p^+|^{3/2}}\right)\left|g_i^a(k^+,\mathbf{k})\right\rangle.\end{aligned}\tag{1.58}$$

Simplifying we get

$$\begin{aligned}\left|\Psi^6_{g\rho\rho\rho}\right\rangle = -\int_\Lambda^\vee dk^+\,dp^+\int d^2k\,d^2p\,\frac{g^3\,p^2k^i\rho^b(p)}{64\pi^{9/2}\,\left(\frac{k^2}{2k^+}\right)\left(\frac{k^2}{2k^+}+\frac{p^2}{2p^+}\right)|p^+|^3|k^+|^{3/2}}\\ \left[\frac{\rho^b(-p)\rho^a(-k)}{\frac{k^2}{2k^+}}+\frac{\rho^a(-k)\rho^b(-p)}{\frac{p^2}{2p^+}}\right]\left|g_i^a(k^+,\mathbf{k})\right\rangle.\end{aligned}\tag{1.59}$$

Using the decomposition principle (i.e., (1.4)) we get

$$\left|\Psi^6_{g\rho\rho\rho}\right\rangle = -\int_\Lambda^\vee dk^+\,dp^+\int d^2k\,d^2p\,\frac{2\,g^3\,k^i\rho^b(p)}{64\pi^{9/2}\,\left(\frac{k^2}{2k^+}\right)|p^+|^2|k^+|^{3/2}}\left[\frac{\rho^b(-p)\rho^a(-k)}{\frac{k^2}{2k^+}}+\frac{if^{abc}\rho^c(-k-p)}{\left(\frac{k^2}{2k^+}+\frac{p^2}{2p^+}\right)}\right]\left|g_i^a(k^+,\mathbf{k})\right\rangle.\tag{1.60}$$

Using the decomposition principle, we will write it as $\rho, \rho\rho, \rho\rho\rho$ terms i.e.,

$$\left|\Psi^6_{g\rho\rho\rho}\right\rangle = \left|\Psi^{6\,a}_{g\,[\rho,\rho]}\right\rangle + \left|\Psi^{6\,b}_{g\,\{\rho,\rho\}}\right\rangle + \left|\Psi^{6\,c}_{g\rho\rho\rho}\right\rangle$$

$$\left|\Psi^{6\,a}_{g\,[\rho,\rho]}\right\rangle = \int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}}\,dp^+\int d^2k\,d^2p\frac{g^3N_ck^i\rho^a(-k)}{16\pi^{9/2}p^+\left(\frac{p^+}{k^+}k^2+p^2\right)\,k^2}\,\left|g_i^a(k^+,\mathbf{k})\right\rangle.\tag{1.61}$$

Integrating wrt p and setting $p^+=\xi k^+$, we get

$$\left|\Psi^{6\,a}_{g\,[\rho,\rho]}\right\rangle = \mu^{2\epsilon}\int_\Lambda^\vee\frac{dk^+}{\sqrt{k^+}}\int_{\frac{\Lambda}{k^+}}^{\frac{\vee}{k^+}}d\xi\int\frac{d^dk}{(2\pi)^d}\,\frac{g^3\,N_c\rho^a(-k)k^i}{\sqrt{\pi}\xi k^2}\times\frac{1}{(4\pi)^{d/2}}\,\Gamma\left[1-\frac{d}{2}\right]\,\left(\frac{1}{\xi k^2}\right)^{1-d/2}\,\left|g_i^a(k^+,\mathbf{k})\right\rangle.\tag{1.62}$$

Expanding the Γ function and setting $\epsilon\to 0$, we get

$$\left|\Psi^{6\,a}_{g\,[\rho,\rho]}\right\rangle = -\int_\Lambda^\vee\frac{dk^+}{\sqrt{k^+}}\int_{\frac{\Lambda}{k^+}}^{\frac{\vee}{k^+}}d\xi\int d^2k\,\frac{g^3N_c\rho^a(-k)k^i}{16\pi^{7/2}\,\xi k^2}\left[-\frac{2}{\epsilon}+\log\left(\frac{\xi k^2}{\mu^2_{\overline{MS}}}\right)\right]\left|g_i^a(k^+,\mathbf{k})\right\rangle.\tag{1.63}$$

Integrating wrt ξ, we get

$$\begin{aligned}\left|\Psi^{6\,a}_{g\,[\rho,\rho]}\right\rangle = -\int_\Lambda^\vee\frac{dk^+}{\sqrt{k^+}}\int d^2k\,\frac{g^3N_c\rho^a(-k)k^i}{16\pi^{7/2}\,k^2}\,\log\left[\frac{\vee}{\Lambda}\right]\left[-\frac{2}{\epsilon}+\log\left[\frac{k^2}{\mu^2_{\overline{MS}}}\right]\right]\,\left|g_i^a(k^+,\mathbf{k})\right\rangle\\ -\int_\Lambda^\vee\frac{dk^+}{\sqrt{k^+}}\int d^2k\,\frac{g^3N_c\rho^a(-k)k^i}{32\pi^{7/2}\,k^2}\,\left[\log^2\left[\frac{\vee}{k^+}\right]-\log^2\left[\frac{\Lambda}{k^+}\right]\right]\,\left|g_i^a(k^+,\mathbf{k})\right\rangle\end{aligned}\tag{1.64}$$

Lets look at 2 ρ contribution

$$\left|\Psi^{6\,b}_{g\,\{\rho,\rho\}}\right\rangle = -\int_\Lambda^\vee\frac{dk^+}{\sqrt{k^+}}\,dp^+\int d^2k\,d^2p\frac{ig^3f^{abc}k^i\{\rho^b(p),\rho^c(-k-p)\}}{16\pi^{9/2}p^+\left(\frac{p^+}{k^+}k^2+p^2\right)\,k^2}\,|g_i^a(k^+,\mathbf{k})\rangle\tag{1.65}$$

Introducing $\xi=\frac{p^+}{k^+}$, we get

$$\left|\Psi^{6\,b}_{g\,\{\rho,\rho\}}\right\rangle = -\int_\Lambda^\vee\frac{dk^+}{\sqrt{k^+}}\int_{\frac{\Lambda}{k^+}}^{\frac{\vee}{k^+}}d\xi\int d^2k\,d^2p\frac{ig^3f^{abc}k^i\{\rho^b(p),\rho^c(-k-p)\}}{16\pi^{9/2}\,\xi\,(\xi k^2+p^2)\,k^2}\,\left|g_i^a(k^+,\mathbf{k})\right\rangle.\tag{1.66}$$

Integrating wrt ξ and doing $p\to -p$, we get

$$\left|\Psi^{6\,b}_{g\,\{\rho,\rho\}}\right\rangle = -\int_\Lambda^\vee\frac{dk^+}{\sqrt{k^+}}\int d^2k\,d^2p\frac{ig^3f^{abc}k^i\{\rho^b(-p),\rho^c(-k+p)\}}{16\pi^{9/2}\,k^2p^2}\,\left[\log\left(\frac{\vee}{\Lambda}\right)+\log\left(\frac{\left(\frac{\Lambda}{k^+}\right)k^2+p^2}{\left(\frac{\vee}{k^+}\right)k^2+p^2}\right)\right]\,\left|g_i^a(k^+,\mathbf{k})\right\rangle.\tag{1.67}$$

Simplifying

$$\left|\Psi^{6\,b}_{g\,\{\rho,\rho\}}\right\rangle = -\int_\Lambda^\vee\frac{dk^+}{\sqrt{k^+}}\int d^2k\,d^2p\frac{ig^3f^{abc}k^i\{\rho^b(-p),\rho^c(-k+p)\}}{16\pi^{9/2}\,k^2p^2}\,\left[\log\left(\frac{\vee}{\Lambda}\right)+\log\left(\frac{\Lambda\,k^2+k^+p^2}{\vee k^2+k^+p^2}\right)\right]\,\left|g_i^a(k^+,\mathbf{k})\right\rangle.\tag{1.68}$$

Now at $\rho\rho\rho$ term

$$\left|\Psi^{6\,c}_{g\rho\rho\rho}\right\rangle = -\int_\Lambda^\vee dk^+\,dp^+\int d^2k\,d^2p\,\frac{g^3\,k^i\rho^b(p)\rho^b(-p)\rho^a(-k)\,\sqrt{k^+}}{8\pi^{9/2}\,|p^+|^2\,k^4}\,\left|g_i^a(k^+,\mathbf{k})\right\rangle.\tag{1.69}$$

Integrating wrt p^+, we get

$$\left|\Psi^{6\,c}_{g\rho\rho\rho}\right\rangle = \int_\Lambda^\vee dk^+\int d^2k\,d^2p\,\frac{g^3\,k^i\rho^b(p)\rho^b(-p)\rho^a(-k)\,\sqrt{k^+}}{8\pi^{9/2}\,k^4}\,\left[\frac{1}{\vee}-\frac{1}{\Lambda}\right]\,\left|g_i^a(k^+,\mathbf{k})\right\rangle.\tag{1.70}$$

Using the argument when $\vee$ is very large or Λ is very small, we get (X5) Not needed: keep $\frac{1}{\vee}-\frac{1}{\Lambda}$. It cancels exactly against the term $\frac{2k^+}{k^2}\left(\frac{1}{\Lambda}-\frac{1}{\vee}\right)$ of Eq. (1.74).

$$\left|\Psi^{6\,c}_{g\rho\rho\rho}\right\rangle = -\int_\Lambda^\vee dk^+\int d^2k\,d^2p\,\frac{g^3\,k^i\rho^b(p)\rho^b(-p)\rho^a(-k)\,\sqrt{k^+}}{8\pi^{9/2}\,k^4\,\Lambda}\,\left|g_i^a(k^+,\mathbf{k})\right\rangle.\tag{1.71}$$

Finally, we look into a term where a gluon emitted from the vacuum is absorbed back, and then another gluon is emitted from the vacuum. This term is a contribution of the norm from the first order perturbation theory.

diagram `Misha_2rho_loopVal3_single_g` (not included)

$$\left|\Psi^7_{g\rho\rho\rho}\right\rangle = \int_\Lambda^\vee dk^+\, dp^+ \int d^2\mathbf{k}\, d^2\mathbf{p}\ \left|g^a_i(k)\right\rangle \frac{\left|\left\langle g^b_j(p)\left|H_g\right|0\right\rangle\right|^2 \left\langle g^a_i(k)\left|H_g\right|0\right\rangle (2E_g(p)+E_g(k))}{2E_g^2(k)E_g^2(p)}. \tag{1.72}$$

After inserting the relevant matrix element (Eq. (C.1)) and changing the variables $\xi = \frac{p^+}{k^+}$, we get

$$\left|\Psi^7_{g\rho\rho\rho}\right\rangle = \int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}} \int d^2\mathbf{k}\, d^2\mathbf{p} \int_{\frac{\Lambda}{k^+}}^{\frac{\vee}{k^+}} d\xi\, \frac{g^3\mathbf{k}^i\rho^b(\mathbf{p})\rho^b(-\mathbf{p})\rho^a(-\mathbf{k})}{16\pi^{9/2}\xi^2\mathbf{k}^2} \left(\frac{\xi}{\mathbf{p}^2} + \frac{2}{\mathbf{k}^2}\right) \left|g^a_i(k^+,\mathbf{k})\right\rangle. \tag{1.73}$$

Integration wrt ξ, we get

$$\left|\Psi^7_{g\rho\rho\rho}\right\rangle = \int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}} \int d^2\mathbf{k}\, d^2\mathbf{p}\, \frac{g^3\mathbf{k}^i\rho^b(\mathbf{p})\rho^b(-\mathbf{p})\rho^a(-\mathbf{k})}{16\pi^{9/2}\mathbf{k}^2} \left[\frac{\log\left(\frac{\vee}{\Lambda}\right)}{p^2} + \frac{2k^+}{k^2}\left(\frac{1}{\Lambda}-\frac{1}{\vee}\right)\right] \left|g^a_i(k^+,\mathbf{k})\right\rangle. \tag{1.74}$$

Using the argument when $\vee$ is very large or Λ is very small, we get (X5) Not needed; Eq. (1.95) follows exactly from Eqs. (1.74) and the exact line before Eq. (1.71).

$$\left|\Psi^7_{g\rho\rho\rho}\right\rangle \approx \int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}} \int d^2\mathbf{k}\, d^2\mathbf{p}\, \frac{g^3\mathbf{k}^i\rho^b(\mathbf{p})\rho^b(-\mathbf{p})\rho^a(-\mathbf{k})}{16\pi^{9/2}\mathbf{k}^2} \left[\frac{\log\left(\frac{\vee}{\Lambda}\right)}{p^2} + \frac{2k^+}{k^2\,\Lambda}\right] \left|g^a_i(k^+,\mathbf{k})\right\rangle. \tag{1.75}$$

1.3 Final results

Now, we collect all the terms corresponding to one ρ, two ρ and three ρ components. Thus the NLO wave function is decomposed as follows

$$\left|\Psi^{\rm NLO}\right\rangle = \mathbb{N}\left|0\right\rangle + \left|\Psi^{\rm LO}_{g\rho}\right\rangle + \left|\Psi_{gg\rho}\right\rangle + \left|\Psi_{gg\rho\rho}\right\rangle + \left|\Psi_{g\rho}\right\rangle + \left|\Psi_{g\rho\rho}\right\rangle + \left|\Psi_{g\rho\rho\rho}\right\rangle. \tag{1.76}$$

where

$$\left|\Psi^{LO}_{g\rho}\right\rangle = -\int_\Lambda^\vee dk^+ \int d^2\mathbf{k}\, \frac{g\,\mathbf{k}^i}{2\pi^{3/2}\sqrt{k^+}\mathbf{k}^2}\, \rho^a(-\mathbf{k})\left|g^a_i(k)\right\rangle. \tag{1.77}$$

Using the transformation equation (Eq. (D.1)) we get the wave function in co-ordinate space.

$$\left|\Psi^{LO}_{g\rho}\right\rangle = \frac{ig}{2\pi^{3/2}}\int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}} \int_{\mathbf{x},\mathbf{z}} \frac{(x-z)^i}{(x-z)^2}\,\rho^a(\mathbf{x})\left|g^a_i(k^+,\mathbf{z})\right\rangle. \tag{1.78}$$

$|\Psi_{gg\rho}\rangle$ is obtained from (1.8), (1.15), (1.19)

$$|\Psi_{gg\rho}\rangle = \left|\Psi^{1a}_{gg[\rho,\rho]}\right\rangle + \left|\Psi^2_{gg\rho}\right\rangle + \left|\Psi^3_{gg\rho}\right\rangle, \tag{1.79}$$

$$\begin{aligned}|\Psi_{gg\rho}\rangle = \int_\Lambda^\vee dk^+ \int d^2\mathbf{k}\, d^2\widetilde{\mathbf{p}} \int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\, \frac{ig^2 f^{abc}\rho^a(-\mathbf{k})\sqrt{\xi(1-\xi)}}{8\pi^3\left(\mathbf{k}^2\xi(1-\xi)+\widetilde{\mathbf{p}}^2\right)\mathbf{k}^2} \\ \left[\frac{\mathbf{k}^2\left(\xi\mathbf{k}^l+\widetilde{\mathbf{p}}^l\right)\left((1-\xi)\mathbf{k}^j-\widetilde{\mathbf{p}}^j\right)\left(\xi((1-\xi)\mathbf{k}-\widetilde{\mathbf{p}})^2-(1-\xi)(\xi\mathbf{k}+\widetilde{\mathbf{p}})^2\right)}{2\xi(1-\xi)(\xi\mathbf{k}+\widetilde{\mathbf{p}})^2((1-\xi)\mathbf{k}-\widetilde{\mathbf{p}})^2} + \frac{\delta^{jl}}{2}\left((1-2\xi)\mathbf{k}^2-2\mathbf{k}\cdot\widetilde{\mathbf{p}}\right) + \frac{1}{\xi}\mathbf{k}^j\widetilde{\mathbf{p}}^l \right. \\ \left. + \frac{1}{1-\xi}\mathbf{k}^l\widetilde{\mathbf{p}}^j\right] \left|g^b_j((1-\xi)k^+,\,(1-\xi)\mathbf{k}-\widetilde{\mathbf{p}})\,g^c_l(\xi k^+,\,\xi\mathbf{k}+\widetilde{\mathbf{p}})\right\rangle.\end{aligned} \tag{1.80}$$

Rearranging

$$\begin{aligned}|\Psi_{gg\rho}\rangle = -\int_\Lambda^\vee dk^+ \int d^2\mathbf{k}\, d^2\mathbf{p} \int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\, \frac{ig^2 f^{abc}\rho^c(-\mathbf{k})\sqrt{\xi(1-\xi)}}{8\pi^3\left(\mathbf{k}^2\xi(1-\xi)+\mathbf{p}^2\right)\mathbf{k}^2} \\ \left[\frac{\mathbf{k}^2\left(\xi\mathbf{k}^i+\mathbf{p}^i\right)\left((1-\xi)\mathbf{k}^j-\mathbf{p}^j\right)\left(\xi((1-\xi)\mathbf{k}-\mathbf{p})^2-(1-\xi)(\xi\mathbf{k}+\mathbf{p})^2\right)}{2\xi(1-\xi)(\xi\mathbf{k}+\mathbf{p})^2((1-\xi)\mathbf{k}-\mathbf{p})^2} + \frac{\delta^{ji}}{2}\left((1-2\xi)\mathbf{k}^2-2\mathbf{k}\cdot\mathbf{p}\right) + \frac{1}{\xi}\mathbf{k}^j\mathbf{p}^i \right. \\ \left. + \frac{1}{1-\xi}\mathbf{k}^i\mathbf{p}^j\right] \left|g^b_j((1-\xi)k^+,\,(1-\xi)\mathbf{k}-\mathbf{p})\,g^a_i(\xi k^+,\,\xi\mathbf{k}+\mathbf{p})\right\rangle.\end{aligned} \tag{1.81}$$

We can transform Eq. (1.80) into coordinate space using Eqs. . (D.2) and (D.6).

$$\begin{aligned}|\Psi_{gg\rho}\rangle = \int_\Lambda^\vee dk^+ \int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi \int_{\mathbf{x},\mathbf{z},\mathbf{w}} \frac{ig^2 f^{abc}\rho^c(\mathbf{x})\sqrt{\xi(1-\xi)}}{8\pi^3\left((1-\xi)(x-w)^2+\xi(x-z)^2\right)} \left(\frac{\delta^{ij}}{2(z-w)^2}\left((x-z)^2-(x-w)^2\right)\right. \\ \left. + \frac{(x-w)^i(z-w)^j}{\xi(z-w)^2} + \frac{(x-z)^j(z-w)^i}{(1-\xi)(z-w)^2} + \frac{(x-z)^j(x-w)^i}{2\xi(x-z)^2} - \frac{(x-z)^j(x-w)^i}{2(1-\xi)(x-w)^2}\right) \left|g^b_j(\xi k^+,\,\mathbf{z})\,g^a_i((1-\xi)k^+,\,\mathbf{w})\right\rangle\end{aligned} \tag{1.82}$$

$|\Psi_{gg\rho\rho}\rangle$ is obtained from (1.10)

$$|\Psi_{gg\rho\rho}\rangle \equiv \left|\Psi_{gg\{\rho,\rho\}}\right\rangle \tag{1.83}$$

$$|\Psi_{gg\rho\rho}\rangle = \int_\Lambda^\vee dk^+ \int d^2\mathbf{k}\, d^2\mathbf{p} \int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\, \frac{g^2\mathbf{p}^i\left(\mathbf{k}^j-\mathbf{p}^j\right)\left\{\rho^a(-\mathbf{p}),\,\rho^b(-\mathbf{k}+\mathbf{p})\right\}}{16\pi^3\mathbf{p}^2(\mathbf{k}-\mathbf{p})^2\sqrt{\xi(1-\xi)}} \left|g^b_j((1-\xi)k^+,\,\mathbf{k}-\mathbf{p})\,g^a_i(\xi k^+,\,\mathbf{p})\right\rangle. \tag{1.84}$$

We transform Eq. (1.84) into coordinate space using Eq. (D.1).

$$|\Psi_{gg\rho\rho}\rangle = -\int_\Lambda^\vee dk^+ \int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi \int_{\mathbf{x},\mathbf{y},\mathbf{z},\mathbf{w}} \frac{g^2\left\{\rho^a(\mathbf{y}),\,\rho^b(\mathbf{x})\right\}(y-z)^i(x-w)^j}{16\pi^3\sqrt{\xi(1-\xi)}(y-z)^2(x-w)^2} \left|g^b_j((1-\xi)k^+,\mathbf{w})\,g^a_i(\xi k^+,\mathbf{z})\right\rangle. \tag{1.85}$$

We split $|\Psi_{g\rho}\rangle$ into two parts:

$$|\Psi_{g\rho}\rangle = \left|\Psi^A_{g\rho}\right\rangle + \left|\Psi^B_{g\rho}\right\rangle$$

$\left|\Psi^A_{g\rho}\right\rangle$ is obtained from (1.29), (1.44), (1.64) and $\left|\Psi^B_{g\rho}\right\rangle$ is obtained from (1.55)

$$\begin{aligned}\left|\Psi^A_{g\rho}\right\rangle = \int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}} \int d^2\mathbf{k}\, \frac{g^3N_c\rho^a(-k)k^i}{32\pi^{7/2}\,k^2} \left[-\frac{2}{\epsilon}+\log\left(\frac{k^2}{\mu^2_{\overline{MS}}}\right)\right] \times \left[\frac{11}{3}+\log\left(\frac{\Lambda}{k^+}\right)-2\log\left(\frac{\vee}{\Lambda}\right)\right] \left|g^a_i(k^+,k)\right\rangle \\ + \int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}} \int d^2\mathbf{k}\, \frac{g^3N_c\rho^a(-k)k^i}{32\pi^{7/2}\,k^2} \left[\frac{\pi^2}{3}-\frac{67}{9}+\log^2\left(\frac{\Lambda}{k^+}\right)-\log^2\left(\frac{\vee}{k^+}\right)\right] \left|g^a_i(k^+,k)\right\rangle\end{aligned} \tag{1.86}$$

$$\left|\Psi^B_{g\rho}\right\rangle = \int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}} \int d^2\mathbf{k} \int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\, \frac{g^3N_c\rho^a(-\mathbf{k})\mathbf{k}^i}{32\pi^{7/2}\sqrt{\xi}(1-\xi)\mathbf{k}^2} \left((\xi-2)\ln(\xi)+(\xi+1)\ln(1-\xi)-(\xi-2)\right.$$

$$\left[-\frac{2}{\epsilon}+\ln\left(\frac{\mathbf{k}^2}{\mu^2_{\overline{MS}}}\right)\right]\Bigg)\left|g_i^a(\xi k^+,\,\mathbf{k})\right\rangle. \tag{1.87}$$

Adding we get

$$\begin{aligned}
|\Psi_{g\rho}\rangle = &\int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}}\int d^2\mathbf{k}\,\frac{g^3N_c\rho^a(-k)k^i}{32\pi^{7/2}\,k^2}\left[-\frac{2}{\epsilon}+\log\left(\frac{k^2}{\mu^2_{\overline{MS}}}\right)\right]\times\left[\frac{11}{3}+\log\left(\frac{\Lambda}{k^+}\right)-2\,\log\left(\frac{\vee}{\Lambda}\right)\right]|g_i^a(k^+,k)\rangle\\
&+\int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}}\int d^2\mathbf{k}\,\frac{g^3N_c\rho^a(-k)k^i}{32\pi^{7/2}\,k^2}\left[\frac{\pi^2}{3}-\frac{67}{9}+\log^2\left(\frac{\Lambda}{k^+}\right)-\log^2\left(\frac{\vee}{k^+}\right)\right]|g_i^a(k^+,k)\rangle\\
&+\int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}}\int d^2\mathbf{k}\int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\,\frac{g^3N_c\rho^a(-\mathbf{k})\mathbf{k}^i}{32\pi^{7/2}\sqrt{\xi}(1-\xi)\mathbf{k}^2}\Bigg((\xi-2)\ln(\xi)+(\xi+1)\ln(1-\xi)-(\xi-2)\\
&\left[-\frac{2}{\epsilon}+\ln\left(\frac{\mathbf{k}^2}{\mu^2_{\overline{MS}}}\right)\right]\Bigg)\left|g_i^a(\xi k^+,\,\mathbf{k})\right\rangle.
\end{aligned} \tag{1.88}$$

Combining and simplifying we get

$$\begin{aligned}
|\Psi_{g\rho}\rangle = &\int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}}\int d^2\mathbf{k}\,\frac{g^3N_c\rho^a(-k)k^i}{32\pi^{7/2}\,k^2}\left[\left[-\frac{2}{\epsilon}+\log\left(\frac{k^2}{\mu^2_{\overline{MS}}}\right)\right]\times\left[\frac{11}{3}+\log\left(\frac{\Lambda}{k^+}\right)-2\,\log\left(\frac{\vee}{\Lambda}\right)\right]\right.\\
&\left.+\frac{\pi^2}{3}-\frac{67}{9}+\log^2\left(\frac{\Lambda}{k^+}\right)-\log^2\left(\frac{\vee}{k^+}\right)\right]|g_i^a(k^+,k)\rangle\\
&+\int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}}\int d^2\mathbf{k}\int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\,\frac{g^3N_c\rho^a(-\mathbf{k})\mathbf{k}^i}{32\pi^{7/2}\sqrt{\xi}(1-\xi)\mathbf{k}^2}\Bigg[(\xi-2)\ln(\xi)+(\xi+1)\ln(1-\xi)-(\xi-2)\\
&\left[-\frac{2}{\epsilon}+\ln\left(\frac{\mathbf{k}^2}{\mu^2_{\overline{MS}}}\right)\right]\Bigg]|g_i^a(\xi k^+,\,\mathbf{k})\rangle.
\end{aligned} \tag{1.89}$$

We transform Eq. (1.89) into coordinate space using Eqs. . (D.1) and (D.7).

$$\begin{aligned}
|\Psi_{g\rho}\rangle = &-\int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}}\int_{x,z}\frac{ig^3N_c\rho^a(x)(x-z)^i}{32\pi^{7/2}\,(x-z)^2}\left[-\frac{2}{\epsilon}-2\gamma-\log\left(\frac{(x-z)^2\,\bar\mu^2_{\rm MS}}{4}\right)\right]\times\\
&\left[\frac{11}{3}+\log\left(\frac{\Lambda}{k^+}\right)+\log\left[\frac{\vee}{k^+}-1\right]-\log\left(\frac{\vee}{\Lambda}\right)\right]|g_i^a(k^+,\,\mathbf{z})\rangle\\
&-\int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}}\int_{x,z}\frac{ig^3N_c\rho^a(x)(x-z)^i}{32\pi^{7/2}\,(x-z)^2}\left[\frac{\pi^2}{3}-\frac{67}{9}+\log^2\left[1-\frac{k^+}{\vee}\right]+\log\left[1-\frac{k^+}{\vee}\right]\log\left[\frac{\vee}{k^+}\right]\right]|g_i^a(k^+,\,\mathbf{z})\rangle
\end{aligned} \tag{1.90}$$

Correction. (X6) Written in the momentum k^+ of the outgoing gluon, the ξ integral of the $|\Psi^B_{g\rho}\rangle$ part runs over $\frac{k^+}{\vee}<\xi<\frac{k^+}{k^++\Lambda}$ (not up to 1 or $1-\frac{\Lambda}{k^+}$), and this part exists only for $k^+<\vee-\Lambda$. Exactly, for $\Lambda<k^+<\vee-\Lambda$ the two brackets become ($\delta=\Lambda/k^+$, $P\equiv-\frac{2}{\epsilon}-2\gamma-\ln\frac{(x-z)^2\bar\mu^2}{4}$)

$$P\Big[\frac{11}{3}+\ln\frac{\Lambda^2}{k^{+2}-\Lambda^2}+\ln\Big(1-\frac{k^+}{\vee}\Big)-8\delta+2\delta^2-\frac{4}{3}\delta^3\Big]+\frac{\pi^2}{3}-\frac{67}{9}+\ln^2\Big(1-\frac{k^+}{\vee}\Big)+\ln\Big(1-\frac{k^+}{\vee}\Big)\ln\frac{\vee}{k^+}+R_\delta,$$
$$R_\delta=\frac{46}{3}\delta-\frac{4}{3}\delta^2+\frac{8}{9}\delta^3+\Big(\frac{22}{3}-8\delta+2\delta^2-\frac{4}{3}\delta^3\Big)\ln(1-\delta)-\Big(8\delta-2\delta^2+\frac{4}{3}\delta^3\Big)\ln\delta-2\ln\delta\ln(1-\delta)-4\,\mathrm{Li}_2(\delta)-\ln^2(1+\delta)+\ln(1+\delta)\ln\delta,$$

(note $\frac{11}{3}+\ln\frac{\Lambda}{k^+}+\ln(\frac{\vee}{k^+}-1)-\ln\frac{\vee}{\Lambda}=\frac{11}{3}+2\ln\frac{\Lambda}{k^+}+\ln(1-\frac{k^+}{\vee})$). For $\vee-\Lambda<k^+<\vee$ only $|\Psi^A_{g\rho}\rangle$ contributes: $P\big[\frac{11}{3}-8\delta+2\delta^2-\frac{4}{3}\delta^3+\ln\frac{\delta}{1-\delta}-2\ln\frac{\vee}{\Lambda}\big]+\frac{\pi^2}{3}-\frac{67}{9}+\ln^2\delta-\ln^2\frac{\vee}{k^+}+R_\delta+\ln^2(1+\delta)-\ln(1+\delta)\ln\delta$.

$|\Psi_{g\rho\rho}\rangle$ is obtained from (1.34), (1.39), (1.45), (1.56),(1.68)

$$\begin{aligned}
|\Psi_{g\rho\rho}\rangle = &\int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}}\int d^2\mathbf{k}\,d^2\mathbf{p}\,\frac{ig^3f^{abc}\left\{\rho^b(-\mathbf{p}),\,\rho^c(-\mathbf{k}+\mathbf{p})\right\}}{32\pi^{9/2}\mathbf{p}^2\mathbf{k}^2}\Bigg\{\Bigg[\log\left(\frac{\Lambda}{k^+}\right)\left[\frac{k^2p^i}{(k-p)^2}-\frac{(k\cdot p)\,k^i}{(k-p)^2}\right]-2k^i\,\log\left(\frac{\vee}{\Lambda}\right)\\
&-2k^i\,\log\left(\frac{\Lambda\,k^2+k^+p^2}{\vee\,k^2+k^+p^2}\right)\Bigg]|g_i^a(k^+,\mathbf{k})\rangle\\
&-\int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi\,\frac{\xi^{3/2}}{(\xi(\mathbf{k}-\mathbf{p})^2+(1-\xi)\mathbf{k}^2)\,(1-\xi)}\left[\left(\frac{2}{\xi}\left(\mathbf{p}^2-\mathbf{k}\cdot\mathbf{p}\right)-\mathbf{p}^2+2\mathbf{k}\cdot\mathbf{p}\right)\mathbf{k}^i+\frac{1}{\xi}\mathbf{k}^2\mathbf{p}^i\right]|g_i^a(\xi k^+,\,\mathbf{k})\rangle\Bigg\}
\end{aligned} \tag{1.91}$$

Integrating wrt ξ and applying the limits, we get the following (X7: the exact limits are $\frac{k^+}{\vee}<\xi<\frac{k^+}{k^++\Lambda}$ for the last line, which exists only for $k^+<\vee-\Lambda$. In the result below $\ln\frac{\Lambda}{k^+}\to\ln\frac{\Lambda}{k^+-\Lambda}$ in the first term (X4), $-k^i\ln\frac{\vee}{k^+}\to-k^i\ln\frac{\vee}{k^++\Lambda}$, and add $-\frac{(2p^2-2\,k\cdot p+k^2)k^i+k^2p^i}{(k-p)^2}\ln\big[1+\frac{\Lambda}{k^+}\frac{k^2}{(k-p)^2}\big]$.)

$$\begin{aligned}
|\Psi_{g\rho\rho}\rangle = &\int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}}\int d^2\mathbf{k}\,d^2\mathbf{p}\,\frac{ig^3f^{abc}\left\{\rho^b(-\mathbf{p}),\,\rho^c(-\mathbf{k}+\mathbf{p})\right\}}{32\pi^{9/2}\mathbf{p}^2\mathbf{k}^2}\Bigg[\log\left(\frac{\Lambda}{k^+}\right)\left[\frac{k^2p^i}{(k-p)^2}-\frac{(k\cdot p)\,k^i}{(k-p)^2}\right]-2k^i\,\log\left(\frac{\vee}{\Lambda}\right)\\
&-2k^i\,\log\left(\frac{\Lambda\,k^2+k^+p^2}{\vee\,k^2+k^+p^2}\right)+\frac{(2p^2-2\,k\cdot p+k^2)\,k^i+k^2p^i}{(k-p)^2}\,\log\left[1+\left(\frac{\vee}{k^+}-1\right)\frac{k^2}{(k-p)^2}\right]\\
&-\frac{(p^2\,k^i+k^2\,p^i)}{(k-p)^2}\,\log\left(\frac{\vee-k^+}{\Lambda}\right)-k^i\,\log\left(\frac{\vee}{k^+}\right)\Bigg]|g_i^a(k)\rangle
\end{aligned} \tag{1.92}$$

By Fourier transform, we get the wave function in coordinate space

$$\begin{aligned}
|\Psi_{g\rho\rho}\rangle = &\frac{1}{2\pi}\int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}}\int d^2\mathbf{k}\,d^2\mathbf{p}\int_{\mathbf{z},\mathbf{x},\mathbf{y}} e^{i\mathbf{k}\cdot(\mathbf{z}-\mathbf{y})}e^{i\mathbf{p}\cdot(\mathbf{y}-\mathbf{x})}\frac{ig^3f^{abc}\left\{\rho^b(\mathbf{x}),\,\rho^c(\mathbf{y})\right\}}{32\pi^{9/2}\mathbf{p}^2\mathbf{k}^2}\Bigg\{\log\left(\frac{\Lambda}{k^+}\right)\left[\frac{k^2p^i}{(k-p)^2}-\frac{k\cdot p\,k^i}{(k-p)^2}\right]\\
&-2k^i\,\log\left(\frac{\Lambda\,k^2+k^+p^2}{\vee\,k^2+k^+p^2}\right)-2k^i\,\log\left(\frac{\vee}{\Lambda}\right)+\frac{(2p^2-2\,k\cdot p+k^2)\,k^i+k^2p^i}{(k-p)^2}\,\log\left[1+\left(\frac{\vee}{k^+}-1\right)\frac{k^2}{(k-p)^2}\right]\\
&-\frac{(p^2\,k^i+k^2\,p^i)}{(k-p)^2}\,\log\left(\frac{\vee-k^+}{\Lambda}\right)-k^i\,\log\left(\frac{\vee}{k^+}\right)\Bigg]|g_i^a(k^+,z)\rangle
\end{aligned} \tag{1.93}$$

Setting Λ to be 0 wherever possible, we get (X7) Not needed: keep $\log\frac{\Lambda k^2+k^+p^2}{\vee k^2+k^+p^2}$ as in the line above.

$$\begin{aligned}
|\Psi_{g\rho\rho}\rangle = &\frac{1}{2\pi}\int_\Lambda^\vee \frac{dk^+}{\sqrt{k^+}}\int d^2\mathbf{k}\,d^2\mathbf{p}\int_{\mathbf{z},\mathbf{x},\mathbf{y}} e^{i\mathbf{k}\cdot(\mathbf{z}-\mathbf{y})}e^{i\mathbf{p}\cdot(\mathbf{y}-\mathbf{x})}\frac{ig^3f^{abc}\left\{\rho^b(\mathbf{x}),\,\rho^c(\mathbf{y})\right\}}{32\pi^{9/2}\mathbf{p}^2\mathbf{k}^2}\Bigg[\log\left(\frac{\Lambda}{k^+}\right)\left[\frac{k^2p^i}{(k-p)^2}-\frac{k\cdot p\,k^i}{(k-p)^2}\right]\\
&-2k^i\,\log\left(\frac{k^+p^2}{\vee\,k^2+k^+p^2}\right)-2k^i\,\log\left(\frac{\vee}{\Lambda}\right)+\frac{(2p^2-2\,k\cdot p+k^2)\,k^i+k^2p^i}{(k-p)^2}\,\log\left[1+\left(\frac{\vee}{k^+}-1\right)\frac{k^2}{(k-p)^2}\right]\\
&-\frac{(p^2\,k^i+k^2\,p^i)}{(k-p)^2}\,\log\left(\frac{\vee-k^+}{\Lambda}\right)-k^i\,\log\left(\frac{\vee}{k^+}\right)\Bigg]|g_i^a(k^+,z)\rangle
\end{aligned} \tag{1.94}$$

$|\Psi_{g\rho\rho\rho}\rangle$ is obtained from (1.71) and (1.74)

$$|\Psi_{g\rho\rho\rho}\rangle = \log\left(\frac{\vee}{\Lambda}\right)\int_{\Lambda}^{\vee}\frac{dk^+}{\sqrt{k^+}}\int d^2\mathbf{k}\,d^2\mathbf{p}\,\frac{g^3\,\rho^b(\mathbf{p})\,\rho^b(-\mathbf{p})\,\rho^a(-\mathbf{k})\,\mathbf{k}^i}{16\pi^{9/2}\mathbf{k}^2\mathbf{p}^2}\,|g_i^a(k^+,\mathbf{k})\rangle \tag{1.95}$$

We transform Eq. (1.95) into co-ordinate space using Eq. (D.1)

$$|\Psi_{g\rho\rho\rho}\rangle = -\log\left(\frac{\vee}{\Lambda}\right)\int_{\Lambda}^{\vee}\frac{dk^+}{\sqrt{k^+}}\int_{\mathbf{w},\mathbf{x},\mathbf{y},\mathbf{u},\mathbf{z}}\frac{ig^3\,\rho^b(\mathbf{x})\,\rho^b(\mathbf{y})\,\rho^a(\mathbf{w})\,(x-u)\cdot(y-u)\,(w-z)^i}{16\pi^{9/2}(x-u)^2(y-u)^2(w-z)^2}\left|g_i^a(k^+,\mathbf{z})\right\rangle. \tag{1.96}$$

Normalization condition for vacuum incoming state is given as

$$\begin{aligned}\mathbb{N} = 1-\frac{1}{2}\Bigg[&\left\langle\Psi_{g\rho}^{\mathrm{LO}}\left|\Psi_{g\rho}^{\mathrm{LO}}\right.\right\rangle+\left\langle\Psi_{g\rho}^{\mathrm{LO}}\left|\Psi_{g\rho}\right.\right\rangle+\left\langle\Psi_{g\rho}\left|\Psi_{g\rho}^{\mathrm{LO}}\right.\right\rangle+\left\langle\Psi_{g\,\rho}^{LO}\left|\Psi_{g\,\rho\rho}\right.\right\rangle+\left\langle\Psi_{g\rho\rho}\left|\Psi_{g\rho}^{\mathrm{LO}}\right.\right\rangle+\langle\Psi_{gg\rho}|\Psi_{gg\rho}\rangle+\langle\Psi_{gg\rho}|\Psi_{gg\rho\rho}\rangle\\&+\langle\Psi_{gg\rho\rho}|\Psi_{gg\rho}\rangle+\langle\Psi_{gg\rho\rho}|\Psi_{gg\rho\rho}\rangle+\left\langle\Psi_{g\rho}^{\mathrm{LO}}\left|\Psi_{g\rho\rho\rho}\right.\right\rangle+\left\langle\Psi_{g\rho\rho\rho}\left|\Psi_{g\rho}^{\mathrm{LO}}\right.\right\rangle\Bigg]-\frac{1}{8}+\frac{3}{8}\left\langle\Psi_{g\rho}^{\mathrm{LO}}\left|\Psi_{g\rho}^{\mathrm{LO}}\right.\right\rangle\left\langle\Psi_{g\rho}^{\mathrm{LO}}\left|\Psi_{g\rho}^{\mathrm{LO}}\right.\right\rangle,\end{aligned} \tag{1.97}$$

Since we are looking for terms that contribute to single gluon production and to the vacuum expectation value of the Hamiltonian in the dressed state, we will require the normalization only up to order g^2.

$$\mathbb{N} = 1-\frac{g^2}{8\pi^3}\int_{\Lambda}^{\vee}dp^+\int d^2p\,\frac{\rho^a(-\mathbf{p})\rho^a(\mathbf{p})}{p^+\mathbf{p}^2} \tag{1.98}$$

Integrating wrt p^+

$$\mathbb{N} = 1-\frac{g^2}{8\pi^3}\ln\left(\frac{\vee}{\Lambda}\right)\int d^2p\frac{\rho^a(-\mathbf{p})\rho^a(\mathbf{p})}{\mathbf{p}^2} \tag{1.99}$$

By performing Fourier transform we get the normalization coefficient as

$$\mathbb{N} = 1-\frac{g^2}{8\pi^3}\ln\left(\frac{\vee}{\Lambda}\right)\int_{\mathbf{x},\mathbf{y},\mathbf{z}}\frac{X\cdot Y}{X^2Y^2}\rho^a(\mathbf{x})\rho^a(\mathbf{y}) \tag{1.100}$$

2 Calculation of the single gluon incoming state

2.1 No outgoing gluon

To order g^2, only one diagram contributes: the absorption of the incoming gluon by the valence ρ:

diagram `g_tovacum` (not included)

$$|(\Psi_i^a(k))_\rho\rangle = |0\rangle\,\frac{\langle 0|\,H_g\,|g_i^a(k)\rangle}{E_g(k)}. \tag{2.1}$$

Using the matrix elements from Eq. (C.1),

$$|(\Psi_i^a(k))_\rho\rangle = \frac{g}{2\pi^{3/2}}\frac{k^i\rho^a(\mathbf{k})}{\sqrt{k^+}\mathbf{k}^2}\,|0\rangle\,, \tag{2.2}$$

Here we used he LCPT transition element of Eq. (C.1). The vertical dashed line in the diagram denotes the intermediate state. In the transverse position space, we obtain

$$\left|\left(\Psi_i^a(k^+,\mathbf{z}')\right)_\rho\right\rangle = \frac{ig}{2\pi^{3/2}\sqrt{k^+}}\int_x\frac{X'^i}{X'^2}\rho^a(\mathbf{x})\,|0\rangle\,, \tag{2.3}$$

with the help of Eq. (D.1).

2.2 One outgoing gluon

2.2.1 Single ρ part; instantaneous contributions

We start with the instantaneous interactions and first consider the process when the plus momentum flows from the valence degree of freedom.

diagram `gtog_inst` (not included)

$$|(\Psi_i^a(k))^1_{g\rho\ \mathrm{inst.}}\rangle = \int_{k^++\Lambda}^{\vee}dp^+\int d^2p\,|g_j^b(p)\rangle\,\frac{\langle g_j^b(p)|\,H_{gg\,\mathrm{inst.}}\,|g_i^a(k)\rangle}{E_g(k)-E_g(p)} \tag{2.4}$$

Using the matrix elements from Eq. (C.5)

$$|(\Psi_i^a(k))^1_{g\rho\ \mathrm{inst.}}\rangle = i\frac{g^2f^{abc}}{(2\pi)^3}\int_{k^++\Lambda}^{\vee}dp^+\int d^2p\frac{\sqrt{p^+k^+}(k^++p^+)}{(p^+-k^+)^2}\frac{\rho^c(-\mathbf{p}+\mathbf{k})}{p^+\mathbf{k}^2-k^+\mathbf{p}^2}\delta_{ij}\,|g_i^b(p^+,\mathbf{p})\rangle \tag{2.5}$$

Using $p^+=\xi k^+$,

$$|(\Psi_i^a(k))^1_{g\rho\ \mathrm{inst.}}\rangle = i\frac{g^2f^{abc}}{(2\pi)^3}\int_{1+\frac{\Lambda}{k^+}}^{\frac{\vee}{k^+}}d\xi\int d^2p\frac{\sqrt{\xi}(1+\xi)}{(1-\xi)^2}\frac{\rho^c(-\mathbf{p}+\mathbf{k})}{\xi\mathbf{k}^2-\mathbf{p}^2}\,|g_i^b(\xi k^+,\mathbf{p})\rangle \tag{2.6}$$

Performing Fourier transform with the help of Eq. (D.3), we get

$$|\left(\Psi_i^a(k^+,\mathbf{z}')\right)^1_{g\rho\ \mathrm{inst.}}\rangle = i\frac{g^2f^{abc}}{(2\pi)^3}\int_{1+\frac{\Lambda}{k^+}}^{\frac{\vee}{k^+}}d\xi\int_{x,z}\frac{\sqrt{\xi}(1+\xi)}{(1-\xi)^2}\frac{\rho^c(\mathbf{x})}{\xi X^2-X'^2}\,|g_i^b(\xi k^+,\mathbf{z})\rangle \tag{2.7}$$

Note that the integration region over the ratio $\xi=p^+/k^+$ is not conventional, as $\xi>1$. Similar for the momentum flowing into the valence degree of freedom we get identical expression modulo the integration limits:

$$|(\Psi_i^a(k))^2_{g\rho\ \mathrm{inst.}}\rangle = i\frac{g^2f^{abc}}{(2\pi)^3}\int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}}d\xi\int d^2p\frac{\sqrt{\xi}(1+\xi)}{(1-\xi)^2}\frac{\rho^c(-\mathbf{p}+\mathbf{k})}{\xi\mathbf{k}^2-\mathbf{p}^2}\,|g_i^b(\xi k^+,\mathbf{p})\rangle \tag{2.8}$$

and

$$|\left(\Psi_i^a(k^+,\mathbf{z}')\right)^2_{g\rho\ \mathrm{inst.}}\rangle = i\frac{g^2f^{abc}}{(2\pi)^3}\int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}}d\xi\int_{x,z}\frac{\sqrt{\xi}(1+\xi)}{(1-\xi)^2}\frac{\rho^c(\mathbf{x})}{\xi X^2-X'^2}\,|g_i^b(\xi k^+,\mathbf{z})\rangle \tag{2.9}$$

diagram `kinem` (not included)

Figure 1: Kinematic constraint on soft gluon plus momentum. The intersection of shaded regions defines allowed range of integration with respect to p^+. The red region is due to constrain $\Lambda<p^+-k^+<\vee$, the green region $-\Lambda<p^+<\vee$ and $\Lambda<k^+<\vee$.

diagram `gandgtog` (not included)

2.2.2 Single ρ part

Non-instantaneous interaction also leads to two possible contribution to g^2 order. We consider first the diagram when the valence particle transfers plus momentum to the gluon (see Eq. (C.3) for three gluon interaction vertex)

$$|(\Psi_i^a(k))^1_{g\rho}\rangle = \int d^3r d^3p d^3q \, |g_j^b(p)\rangle \, \frac{\langle g_j^b(p)| \, H_{ggg} \, |g_l^c(q) g_m^d(r)\rangle \, \langle g_l^c(q) g_m^d(r)| \, H_g \, |g_i^a(k)\rangle}{2(E_g(k) - E_{gg}(q,r))(E_g(k) - E_g(p))} \tag{2.10}$$

Using the matrix element from Eqs. . (C.2) and (C.3)

$$|(\Psi_i^a(k))^1_{g\rho}\rangle = -+\frac{ig^2 f^{abc}}{8\pi^3} \int_{k^+ + \Lambda}^{\vee} dp^+ \int d^2p \, \frac{\rho^c(k-p)(p-k)^l}{\sqrt{p^+(p^+-k^+)k^+}(p^+-k^+)^{3/2}} \frac{1}{\left(\frac{k^2}{k^+} - \frac{p^2}{p^+}\right)\frac{(k-p)^2}{(k^+-p^+)}} \Bigg[\left(2k^j - p^j + \frac{p^+ - 2k^+}{p^+} p^j\right)$$
$$\delta_{il} + \left(2p^i - k^i - \frac{2p^+ - k^+}{k^+} k^i\right)\delta_{jl} + \left(\frac{p^+ + k^+}{p^+ - k^+}(p-k)^l - k^l - p^l\right)\delta_{ij} \Bigg] \, |g_j^b(p)\rangle \tag{2.11}$$

Using $\xi = \frac{p^+}{k^+}$, we get

$$|(\Psi_i^a(k))^1_{g\rho}\rangle = -\frac{ig^2 f^{abc}}{4\pi^3} \int_{1+\frac{\Lambda}{k^+}}^{\frac{\vee}{k^+}} d\xi \int d^2p \, \frac{\sqrt{\xi}}{1-\xi} \frac{(p^m - k^m)(p^l - \xi k^l)}{(\mathbf{p}-\mathbf{k})^2(\mathbf{p}^2 - \xi\mathbf{k}^2)} \rho^c(-(\mathbf{p}-\mathbf{k})) \left(\delta_{il}\delta_{jm} - \frac{1}{\xi}\delta_{jl}\delta_{im} - \frac{1}{1-\xi}\delta_{ij}\delta_{lm}\right)$$
$$|g_j^b(\xi k^+, \mathbf{p})\rangle \tag{2.12}$$

Performing the Fourier transform with help of Eq. (D.4), we obtain

$$|\left(\Psi_i^a(k^+, \mathbf{z}')\right)^1_{g\rho}\rangle = +-\frac{ig^2 f^{abc}}{4\pi^3} \int_{1+\frac{\Lambda}{k^+}}^{\frac{\vee}{k^+}} d\xi \int_{x,z} \frac{\sqrt{\xi}}{1-\xi} \frac{X'^m - \xi X^m}{X'^2 - \xi X^2} \frac{X'^l - X^l}{(X' - X)^2} \rho^c(\mathbf{x})$$
$$\times \left(\delta_{il}\delta_{jm} - \frac{1}{\xi}\delta_{jl}\delta_{im} - \frac{1}{1-\xi}\delta_{ij}\delta_{lm}\right) \, |g_j^b(\xi k^+, \mathbf{z})\rangle \tag{2.13}$$

The process when gluon transfers plus momentum two the valence degree of freedom gives

diagram `gto3g_torho` (not included)

$$|(\Psi_i^a(k))^2_{g\rho}\rangle = \int d^3r \int d^3q \int d^3p \, |g_j^b(p)\rangle \, \frac{\langle g_j^b(p)| \, H_g \, |g_l^c(q) g_m^d(r)\rangle \, \langle g_l^c(q) g_m^d(r)| \, H_{ggg} \, |g_i^a(k)\rangle}{2(E_g(k) - E_{gg}(q,r))(E_g(k) - E_g(p))} \tag{2.14}$$

Using the matrix element from Eqs. . (C.2) and (C.3)

$$|(\Psi_i^a(k))^2_{g\rho}\rangle = \int_{\Lambda}^{k^+ - \Lambda} dp^+ \int d^2p \, \frac{ig^2 f^{abc}(k-p)^l \rho^c(k-p)}{8\pi^3 \sqrt{k^+(k^+-p^+)p^+}(k^+-p^+)^{3/2}} \frac{1}{\frac{k^2}{k^+} - \frac{(k-p)^2}{(k^+-p^+)} - \frac{p^2}{p^+}} \times \frac{1}{\frac{k^2}{k^+} - \frac{p^2}{p^+}}$$
$$\left[\left(2p^i - k^i + \frac{k^+ - 2p^+}{k^+} k^i\right)\delta_{jl} + \left(2k^j - p^j - \frac{2k^+ - p^+}{p^+} p^j\right)\delta_{il} + \left(\frac{k^+ + p^+}{(k^+ - p^+)} + (k-p)^l - p^l - k^l\right)\delta_{ij}\right] \, |g_j^b(p)\rangle \tag{2.15}$$

Having $p^+ = \xi k^+$,

$$|(\Psi_i^a(k))^2_{g\rho}\rangle = \frac{ig^2 f^{abc}}{4\pi^3} \int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi \frac{\xi^{3/2}}{1-\xi} \int d^2p \, \frac{(k^m - p^m)(p^l - \xi k^l)}{(\mathbf{p}^2 - \xi\mathbf{k}^2)(\mathbf{p} - \xi\mathbf{k})^2} \rho^c(\mathbf{k} - \mathbf{p})$$
$$\times \left(\delta_{il}\delta_{jm} - \frac{1}{\xi}\delta_{jl}\delta_{im} - \frac{1}{1-\xi}\delta_{lm}\delta_{ij}\right) \, |g_j^b(\xi k^+, \mathbf{p})\rangle \tag{2.16}$$

Using Eq. (D.5), we do the Fourier transform

$$|\left(\Psi_i^a(k^+, \mathbf{z}')\right)^2_{g\rho}\rangle = -\frac{ig^2 f^{abc}}{4\pi^3} \int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}} d\xi \frac{\xi^{3/2}}{1-\xi} \int_{x,z} \frac{X'^l - X^l}{X'^2 - \xi X^2} \frac{X'^m - \xi X^m}{(X' - \xi X)^2} \rho^c(\mathbf{x})$$
$$\times \left(\delta_{il}\delta_{jm} - \frac{1}{\xi}\delta_{jl}\delta_{im} - \frac{1}{1-\xi}\delta_{lm}\delta_{ij}\right) \, |g_j^b(\xi k^+, \mathbf{z})\rangle \tag{2.17}$$

In all four contributions, one might get worried about the presence of the $\frac{1}{(1-\xi)^2}$ singularity. Although, it is outside of the integration limits of ξ, the limits do pinch it leading to a potentially dangerous contribution proportional to $\frac{k^+}{\Lambda}$. However, a simple analysis of Eqs. . (2.7) and (2.13) (as well as Eqs. . (2.9) and (2.17)) near $\xi = 1$ demonstrates that this integration boundary singularity $\frac{1}{(1-\xi)^2}$ cancels.

2.2.3 Two ρ part

In the expressions below we explicitly remove contributions when the outgoing gluon is identical to the incoming gluon.

diagram `gtovactog_tworho` (not included)

$$|(\Psi_i^a(k))^1_{g\rho\rho}\rangle = \int_{\Lambda}^{\vee} dp^+ \int d^2p \, |g_j^b(p^+, \mathbf{p})\rangle \, \frac{\langle g_j^b(p)| \, H_g \, |0\rangle \, \langle 0| \, H_g \, |g_i^a(k)\rangle}{E_g(k)(E_g(k) - E_g(p))} \tag{2.18}$$

$$|(\Psi_i^a(k))^1_{g\rho\rho}\rangle = \frac{g^2}{4\pi^3} \int_{\frac{\Lambda}{k^+}}^{\frac{\vee}{k^+}} d\xi \int d^2p \frac{k^i p^j \rho^b(-\mathbf{p})\rho^a(\mathbf{k})}{\sqrt{\xi}(\xi\mathbf{k}^2 - \mathbf{p}^2)\mathbf{k}^2} \, |g_j^b(\xi k^+, \mathbf{p})\rangle \ . \tag{2.19}$$

Here we used the matrix elements of Eq. (C.1). To find the Fourier transform in the transverse coordinate we use Eq. (D.2):

$$|\left(\Psi_i^a(k^+, \mathbf{z}')\right)^1_{g\rho\rho}\rangle = \frac{g^2}{4\pi^3} \int_{\frac{\Lambda}{k^+}}^{\frac{\vee}{k^+}} d\xi \int_{x,y,z} \frac{X'^i Y^j \rho^b(y)\rho^a(x)}{\sqrt{\xi}(\xi Y^2 - X'^2)Y^2} \, |g_j^b(\xi k^+, \mathbf{z})\rangle \ . \tag{2.20}$$

$$|(\Psi_i^a(k))^2_{g\rho\rho}\rangle = \frac{1}{2} \int_{\Lambda}^{\vee} dr^+ dq^+ dp^+ \int d^2r d^2q d^2p \, |g_j^b(p)\rangle \, \frac{\langle g_j^b(p)| \, H_g \, |g_l^c(q), g_m^d(r)\rangle \, \langle g_l^c(q), g_m^d(r)| \, H_g \, |g_i^a(k)\rangle}{(E_g(k) - E_{gg}(q,r))(E_g(k) - E_g(p))}$$

diagram `gg_tworho` (not included)

With the help of Eq. (C.2), we obtain

$$|(\Psi_i^a(k))^2_{g\rho\rho}\rangle = -\frac{g^2}{4\pi^3}\int_{\frac{\Lambda}{k^+}}^{\frac{\vee}{k^+}} d\xi \int d^2p\sqrt{\xi}\frac{k^i p^j \rho^a(\mathbf{k})\rho^b(-\mathbf{p})}{p^2(\xi\mathbf{k}^2-\mathbf{p}^2)}\,|g_j^b(\xi k^+,\mathbf{p})\rangle \tag{2.21}$$

Taking Fourier transform in the transverse space we get

$$|\left(\Psi_i^a(k^+,\mathbf{z}')\right)^2_{g\rho\rho}\rangle = -\frac{g^2}{4\pi^3}\int_{\frac{\Lambda}{k^+}}^{\frac{\vee}{k^+}} d\xi\sqrt{\xi}\int_{x,y,z}\frac{X'^i Y^j\rho^a(x)\rho^b(y)}{(\xi Y^2-X'^2)X'^2}\,|g_j^b(\xi k^+,\mathbf{z})\rangle\,, \tag{2.22}$$

where again we applied Eq. (D.2). In the case of the matrix elements with two ρ we have an ambiguity due to the commutator [1]

$$[\rho^a(\mathbf{x}),\rho^b(\mathbf{y})] = if^{abc}\rho^c(\mathbf{x})\delta^2(\mathbf{x}-\mathbf{y}) \tag{2.23}$$

which reduces the number of valence degrees of freedom by one. We thus choose to represent

$$\rho^a(\mathbf{x})\rho^b(\mathbf{y}) = \frac{1}{2}\{\rho^a(\mathbf{x}),\rho^b(\mathbf{y})\} + \frac{1}{2}[\rho^a(\mathbf{x}),\rho^b(\mathbf{y})] \tag{2.24}$$

where the first term depends on two valence degrees of freedom and is irreducible. Therefore the sum of $|(\Psi_i^a(k))^1_{g\rho\rho}\rangle$ and $|(\Psi_i^a(k))^2_{g\rho\rho}\rangle$ can be rewritten as

$$|(\Psi_i^a(k))^1_{g\rho\rho}\rangle + |(\Psi_i^a(k))^2_{g\rho\rho}\rangle = |(\Psi_i^a(k))^1_{g\{\rho,\rho\}}\rangle + |(\Psi_i^a(k))^2_{g[\rho,\rho]}\rangle \tag{2.25}$$

with

$$|(\Psi_i^a(k))_{g\{\rho,\rho\}}\rangle = -\frac{g^2}{8\pi^3}\int d^3p\frac{k^i p^j}{\mathbf{k}^2\mathbf{p}^2}\frac{1}{\sqrt{k^+p^+}}\{\rho^a(\mathbf{k}),\rho^b(-\mathbf{p})\}\,|g_j^b(p^+,\mathbf{p})\rangle \tag{2.26}$$

$$|(\Psi_i^a(k))_{g\{\rho,\rho\}}\rangle = -\frac{g^2}{8\pi^3}\int_{\frac{\Lambda}{k^+}}^{\frac{\vee}{k^+}}\frac{d\xi}{\sqrt{\xi}}\int d^2p\frac{k^i p^j}{\mathbf{k}^2\mathbf{p}^2}\{\rho^a(\mathbf{k}),\rho^b(-\mathbf{p})\}\,|g_j^b(\xi k^+,\mathbf{p})\rangle \tag{2.27}$$

and

$$|(\Psi_i^a(k))_{g[\rho,\rho]}\rangle = \frac{ig^2 f^{abc}}{8\pi^3}\int\frac{dp^+}{\sqrt{k^+p^+}}\int d^2p\frac{k^i p^j}{\mathbf{k}^2\mathbf{p}^2}\frac{\mathbf{p}^2k^+ + p^+\mathbf{k}^2}{\mathbf{p}^2k^+ - p^+\mathbf{k}^2}\rho^c(\mathbf{k}-\mathbf{p})\,|g_j^b(p^+,\mathbf{p})\rangle \tag{2.28}$$

$$|(\Psi_i^a(k))_{g[\rho,\rho]}\rangle = \frac{ig^2 f^{abc}}{8\pi^3}\int_{\frac{\Lambda}{k^+}}^{\frac{\vee}{k^+}}\frac{d\xi}{\sqrt{\xi}}\int d^2p\frac{k^i p^j}{\mathbf{k}^2\mathbf{p}^2}\frac{\mathbf{p}^2+\xi\mathbf{k}^2}{\mathbf{p}^2-\xi\mathbf{k}^2}\rho^c(\mathbf{k}-\mathbf{p})\,|g_j^b(\xi k^+,\mathbf{p})\rangle \tag{2.29}$$

Similarly in the coordinate space

$$|\left(\Psi_i^a(k^+,\mathbf{z}')\right)_{g\{\rho,\rho\}}\rangle = -\frac{g^2}{8\pi^3}\int_{\frac{\Lambda}{k^+}}^{\frac{\vee}{k^+}}\frac{d\xi}{\sqrt{\xi}}\int_{x,y,z}\frac{X'^i Y^j}{X'^2Y^2}\{\rho^a(\mathbf{x}),\rho^b(\mathbf{y})\}\,|g_j^b(\xi k^+,\mathbf{z})\rangle \tag{2.30}$$

and

$$|\left(\Psi_i^a(k^+,\mathbf{z}')\right)_{g[\rho,\rho]}\rangle = -\frac{ig^2 f^{abc}}{8\pi^3}\int_{\frac{\Lambda}{k^+}}^{\frac{\vee}{k^+}}\frac{d\xi}{\sqrt{\xi}}\int_{x,z}\frac{X'^i X^j}{X'^2X^2}\frac{(X'^2+\xi X^2)}{(\xi X^2-X'^2)}\rho^c(\mathbf{x})\,|g_j^b(\xi k^+,\mathbf{z})\rangle \tag{2.31}$$

2.3 Two outgoing gluons

2.3.1 ρ independent part

diagram `g_togg` (not included)

$$|(\Psi_i^a(k))_{gg}\rangle = \int d^3p\,d^3q\;|g_l^c(q),g_j^b(p)\rangle\,\frac{\langle g_l^c(q),g_j^b(p)|\,H_{ggg}\,|g_i^a(k)\rangle}{2(E_g(k)-E_{gg}(p,q))} \tag{2.32}$$

Using Eq. (C.3),

$$\begin{aligned}|(\Psi_i^a(k))_{gg}\rangle = \int d^3p\int d^3q\,\frac{igf^{abc}\delta^{(3)}(k-p-q)}{8\pi^{3/2}\sqrt{k^+p^+q^+}\left(\frac{k^2}{k^+}-\frac{p^2}{p^+}-\frac{q^2}{q^+}\right)}\Bigg[\left(p^i-q^i+\frac{q^+-p^+}{k^+}k^i\right)\delta_{jl}+\left(k^j+q^j-\frac{k^++q^+}{p^+}p^j\right)\delta_{il}\\ +\left(\frac{k^++p^+}{q^+}q^l-p^l-k^l\right)\delta_{ij}\Bigg]\,|g_l^c(q),g_j^b(p)\rangle\end{aligned} \tag{2.33}$$

Simplifying we get

$$\begin{aligned}|(\Psi_i^a(k))_{gg}\rangle = -\frac{igf^{abc}\sqrt{k^+}}{4\pi^{3/2}}\int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}}d\xi\sqrt{\xi(1-\xi)}\int d^2p\,\frac{p^m-\xi k^m}{(\mathbf{p}-\xi\mathbf{k})^2}\left[\delta_{im}\delta_{jl}-\frac{1}{\xi}\delta_{jm}\delta_{il}-\frac{1}{1-\xi}\delta_{lm}\delta_{ij}\right]\\ |g_l^c((1-\xi)k^+,\mathbf{k}-\mathbf{p}),g_j^b(\xi k^+,\mathbf{p})\rangle\;.\end{aligned} \tag{2.34}$$

The Fourier transform can be obtained by shifting integration variable p and using Eq. (D.1):

$$\begin{aligned}|\Psi_i^a(k^+,z')\rangle = -\frac{gf^{abc}}{4\pi^{3/2}}\sqrt{k^+}\int d^2z\int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}}d\xi\sqrt{\frac{1-\xi}{\xi}}\frac{(z-z')^m}{(z-z')^2}\left[\delta_{im}\delta_{jl}-\frac{1}{\xi}\delta_{jm}\delta_{il}-\frac{1}{1-\xi}\delta_{lm}\delta_{ij}\right]\\ |g_l^c((1-\xi)k^+,z),g_j^b(\xi k^+,\frac{z'-(1-\xi)z}{\xi})\rangle\end{aligned} \tag{2.35}$$

diagram `gtog_plusg` (not included)

2.3.2 One ρ part

$$|(\Psi_i^a(k))_{gg\rho}\rangle = \int d^3q\, d^3p\, |g_l^c(q), g_j^b(p)\rangle \frac{\langle g_l^c(q), g_j^b(p)|\, H_g\, |g_i^a(k)\rangle}{2(E_g(k) - E_{gg}(p,q))} \tag{2.36}$$

Using the matrix elements from Eq. (C.2)

$$|(\Psi_i^a(k))_{gg\rho}\rangle = -\frac{g}{2\pi^{3/2}} \int_\Lambda^\vee \frac{dp^+}{\sqrt{p^+}} \int d^2p \frac{p^j}{\mathbf{p}^2} \rho^b(-\mathbf{p})\, |g_i^a(k), g_j^b(p)\rangle \tag{2.37}$$

Using $\xi = \frac{p^+}{k^+}$, we get

$$|(\Psi_i^a(k))_{gg\rho}\rangle = -\frac{g}{2\pi^{3/2}} \sqrt{k^+} \int_{\frac{\Lambda}{k^+}}^{\frac{\vee}{k^+}} \frac{d\xi}{\sqrt{\xi}} \int d^2p \frac{p^j}{\mathbf{p}^2} \rho^b(-\mathbf{p})\, |g_i^a(k), g_j^b(\xi k^+, \mathbf{p})\rangle\ . \tag{2.38}$$

In the transverse coordinate space, this reads

$$|\big(\Psi_i^a(k^+, \mathbf{z}')\big)_{gg\rho}\rangle = \frac{ig}{2\pi^{3/2}} \sqrt{k^+} \int_{\frac{\Lambda}{k^+}}^{\frac{\vee}{k^+}} \frac{d\xi}{\sqrt{\xi}} \int_{x,z} \frac{X^j}{X^2} \rho^b(x)\, |g_i^a(k^+, \mathbf{z}'), g_j^b(\xi k^+, \mathbf{z})\rangle\ .$$

(or)

$$|\big(\Psi_i^a(k^+, \mathbf{z}')\big)_{gg\rho}\rangle = \frac{ig}{2\pi^{3/2}} \int_\Lambda^\vee \frac{dp^+}{\sqrt{p^+}} \int_{x,z} \frac{X^j}{X^2} \rho^b(x)\, |g_i^a(k^+, \mathbf{z}'), g_j^b(\xi k^+, \mathbf{z})\rangle\ . \tag{2.39}$$

2.4 Normalization

The final result for the single gluon wave function to order g^2 is given by the sum of contribution presented in the previous section, i.e.

$$|\Psi_i^a(k)\rangle = \mathbb{N}\, |g_i^a(k)\rangle \tag{2.40}$$

$$+ |(\Psi_i^a(k))_\rho\rangle + |(\Psi_i^a(k))_{gg}\rangle + |(\Psi_i^a(k))_{gg\rho}\rangle$$

$$+ |(\Psi_i^a(k))^1_{g\rho\ \text{inst.}}\rangle + |(\Psi_i^a(k))^2_{g\rho\ \text{inst.}}\rangle + |(\Psi_i^a(k))^1_{g\rho}\rangle + |(\Psi_i^a(k))^2_{g\rho}\rangle + |(\Psi_i^a(k))^1_{g\rho\rho}\rangle + |(\Psi_i^a(k))^2_{g\rho\rho}\rangle\,. \tag{2.41}$$

In the second (third) line of this equation, we list all contribution of order g (g^2). The first line is proportional to non-interacting gluon state. The proportionality factor $\mathbb{N}$ is to be determined from the normalization condition of the interacting gluon state:

$$\langle \Psi_{i'}^{a'}(k')|\, \Psi_i^a(k)\rangle = \delta^{aa'}\delta_{ii'}\delta^3(k-k'). \tag{2.42}$$

For our purposes we need to determine $\mathbb{N}$ only to g^2 order. Therefore, the number of terms contributing to normalization is rather limited, e.g. we cannot have a product of any terms withing the third line or the product of two states from the second and the third lines. The following terms cancel in the normalization

$$\langle g_{i'}^{a'}(k')|\, |(\Psi_i^a(k))^1_{g\rho\ \text{inst.}}\rangle + \langle g_{i'}^{a'}(k')|\, |(\Psi_i^a(k))^2_{g\rho\ \text{inst.}}\rangle + \langle (\Psi_{i'}^{a'}(k'))^1_{g\rho\ \text{inst.}}|\, |g_i^a(k)\rangle + \langle (\Psi_{i'}^{a'}(k'))^2_{g\rho\ \text{inst.}}|\, |g_i^a(k)\rangle = 0$$

$$\langle g_{i'}^{a'}(k')|\, |(\Psi_i^a(k))^1_{g\rho\rho}\rangle + \langle (\Psi_{i'}^{a'}(k'))^1_{g\rho\rho}|\, |g_i^a(k)\rangle + \langle (\Psi_{i'}^{a'}(k'))_\rho|\, |(\Psi_i^a(k))_\rho\rangle = 0$$

$$\langle g_{i'}^{a'}(k')|\, |(\Psi_i^a(k))^1_{g\rho}\rangle + \langle (\Psi_{i'}^{a'}(k'))^1_{g\rho}|\, |g_i^a(k)\rangle + \langle g_{i'}^{a'}(k')|\, |(\Psi_i^a(k))^2_{g\rho}\rangle + \langle (\Psi_{i'}^{a'}(k'))^2_{g\rho}|\, |g_i^a(k)\rangle = 0$$

$$\langle (\Psi_{i'}^{a'}(k'))_{gg\rho}|\, |(\Psi_i^a(k))_{gg}\rangle + \langle (\Psi_i^a(k))_{gg}|\, |(\Psi_{i'}^{a'}(k'))_{gg\rho}\rangle = 0$$

$$\langle (\Psi_{i'}^{a'}(k'))_{gg}|\, |(\Psi_i^a(k))_{gg}\rangle = -\frac{g^2 N_c}{8\pi^3}\left(2\ln\left(\frac{\Lambda}{k^+}\right) + \frac{11}{6}\right)\int \frac{d^2\tilde{p}}{\tilde{p}^2} = 0$$

The remaining terms are from

$$\langle g_i^a(k)|\, |(\Psi_i^a(k))^2_{g\rho\rho}\rangle + \langle (\Psi_{i'}^{a'}(k'))^2_{g\rho\rho}|\, |g_i^a(k)\rangle = -\frac{g^2}{4\pi^3\sqrt{k^+k'^+}} \frac{k^i k'^{i'}}{\mathbf{k}^2\mathbf{k}'^{2'}} \rho^{a'}(-\mathbf{k}')\rho^a(\mathbf{k})$$

$$\langle (\Psi_i^a(k))_{gg\rho}|\, |(\Psi_i^a(k))_{gg\rho}\rangle = \frac{g^2}{4\pi^3\sqrt{k^+k'^+}} \frac{k^i k'^{i'}}{\mathbf{k}^2\mathbf{k}'^{2'}} \rho^a(\mathbf{k})\rho^{a'}(-\mathbf{k}') + \delta^{aa'}\delta_{ii'}\delta^3(k-k')\frac{g^2}{4\pi^3}\int_\Lambda^\vee \frac{dp^+}{p^+}\int d^2p \frac{\rho^a(-\mathbf{p})\rho^a(\mathbf{p})}{\mathbf{p}^2}$$

Putting together and therefore the normalization coefficient is given by

$$\mathbb{N} = 1 - \frac{g^2}{8\pi^3}\int_\Lambda^\vee \frac{dp^+}{p^+}\int d^2p \frac{\rho^a(-\mathbf{p})\rho^a(\mathbf{p})}{\mathbf{p}^2} \tag{2.43}$$

(or)

$$\mathbb{N} = 1 - \frac{g^2}{8\pi^3}\ln\left(\frac{\vee}{\Lambda}\right)\int d^2p \frac{\rho^a(-\mathbf{p})\rho^a(\mathbf{p})}{\mathbf{p}^2} \tag{2.44}$$

By performing Fourier transform we get the normalization coefficient as

$$\mathbb{N} = 1 - \frac{g^2}{8\pi^3}\ln\left(\frac{\vee}{\Lambda}\right)\int_{\mathbf{x},\mathbf{y},\mathbf{z}} \frac{X\cdot Y}{X^2Y^2}\rho^a(\mathbf{x})\rho^a(\mathbf{y}) \tag{2.45}$$

[1] In momentum space, $[\rho^a(\mathbf{k}), \rho^b(-\mathbf{p})] = if^{abc}\rho^c(\mathbf{k}-\mathbf{p})$.

3 Parametrization of the Evolution operator (Ω)

To parameterize the unitary evolution operator Ω it is convenient to perform normal ordering of the creation-annihilation operators of the soft sector

$$\begin{aligned}\Omega = \mathbb{N} &\\ &+ \int \frac{d^3p}{(2\pi)^3}\left[\mathbb{A}^b_{1j}(p)a^b_j(p) + \mathbb{A}^b_{2j}(p)a^{\dagger b}_j(p)\right]\\ &+ \int \frac{d^3p}{(2\pi)^3}\frac{d^3q}{(2\pi)^3}\left[\mathbb{B}^{cb}_{1kj}(p,q)a^b_j(p)a^c_k(q) + \mathbb{B}^{cb}_{2kj}(p,q)a^{\dagger b}_j(p)a^{\dagger c}_k(q) + \mathbb{B}^{cb}_{3kj}(p,q)a^{\dagger c}_k(q)a^b_j(p)\right]\\ &+ \int \frac{d^3p}{(2\pi)^3}\frac{d^3q}{(2\pi)^3}\left[\mathbb{C}^{dcb}_{1lkj}(p,q)a^{\dagger d}_l(q)a^{\dagger c}_k(p-q)a^b_j(p) + \mathbb{C}^{dcb}_{2lkj}(p,q)a^{\dagger b}_j(p)a^c_k(p-q)a^d_l(q)\right] + \dots\end{aligned} \tag{3.1}$$

The symmetry of the combinations of the creation annihilation operators infers the following identities on the coefficients:

$$\mathbb{B}^{cb}_{1kj}(p,q) = \mathbb{B}^{bc}_{1jk}(q,p), \tag{3.2}$$
$$\mathbb{B}^{cb}_{2kj}(p,q) = \mathbb{B}^{bc}_{2jk}(q,p), \tag{3.3}$$
$$\mathbb{C}^{dcb}_{1lkj}(p,q) = \mathbb{C}^{cdb}_{1klj}(p,p-q), \tag{3.4}$$
$$\mathbb{C}^{dcb}_{2lkj}(p,q) = \mathbb{C}^{cdb}_{2klj}(p,p-q)\,. \tag{3.5}$$

$$\begin{aligned}\Omega^\dagger = \mathbb{N} &\\ &+ \int \frac{d^3p}{(2\pi)^3}\left[\mathbb{A}^{\dagger b}_{1j}(p)a^{\dagger b}_j(p) + \mathbb{A}^{\dagger b}_{2j}(p)a^b_j(p)\right]\\ &+ \int \frac{d^3p}{(2\pi)^3}\frac{d^3q}{(2\pi)^3}\left[\mathbb{B}^{\dagger cb}_{1kj}(p,q)a^{\dagger b}_j(p)a^{\dagger c}_k(q) + \mathbb{B}^{\dagger cb}_{2kj}(p,q)a^b_j(p)a^c_k(q) + \mathbb{B}^{\dagger cb}_{3kj}(p,q)a^{\dagger b}_j(p)a^c_k(q)\right]\\ &+ \int \frac{d^3p}{(2\pi)^3}\frac{d^3q}{(2\pi)^3}\left[\mathbb{C}^{\dagger dcb}_{1lkj}(p,q)a^{\dagger b}_j(p)a^c_k(p-q)a^d_l(q) + \mathbb{C}^{\dagger dcb}_{2lkj}(p,q)a^{\dagger d}_l(q)a^{\dagger c}_k(p-q)a^b_j(p)\right] + \dots\end{aligned} \tag{3.6}$$

Using the unitarity condition of $\Omega^\dagger\Omega = 1$, establish the relation between daggered and undaggered coefficients. We will keep only the terms required to obtain the coefficients at particular orders for the NLO cross section. Note that coefficients have different orders in g, and some of them mix the orders, e.g., $\mathbb{A}_1$ contains both g and g^3 contributions. For our calculation, we have to consider Ω to g^3 order. However, we stress that not all possible g^3 order contributions are included. In this expansion, we kept only the coefficients that have two gluons after applying the evolution to the vacuum of a single gluon state. Higher gluon numbers do not contribute to the single gluon production cross-section. The coefficients $\mathbb{A}_1$, $\mathbb{A}_2, \mathbb{A}^\dagger_1$ and $\mathbb{A}^\dagger_2$ contain terms of order g and g^3. Thus, we express them according to their respective orders as follows:

$$\mathbb{A}^b_{1j}(p) = \mathbb{A}^{(1)\,b}_{1j}(p) + \mathbb{A}^{(3)\,b}_{1j}(p) \tag{3.7}$$
$$\mathbb{A}^b_{2j}(p) = \mathbb{A}^{(1)\,b}_{2j}(p) + \mathbb{A}^{(3)\,b}_{2j}(p) \tag{3.8}$$

Similarly, we can write it for $\mathbb{A}^\dagger_1$ and $\mathbb{A}^\dagger_2$ as well. For $\mathbb{N}$, we have

$$\mathbb{N} = \mathbb{N}^{(0)} + \mathbb{N}^{(2)} = 1 + \mathbb{N}^{(2)}\,. \tag{3.9}$$

We will only have the terms that are required for computing single gluon production.

$$\begin{aligned}\Omega^\dagger\Omega = 1 = \mathbb{N}^2 &+ \int \frac{d^3p}{(2\pi)^3}\mathbb{A}^{\dagger\,(1)\,b}_{2j}(p)\mathbb{A}^{(1)\,b}_{2j}(p)\\ &+ \int \frac{d^3p}{(2\pi)^3}\,a^{\dagger\,b}_j(p)\Bigg[\mathbb{N}^{\dagger\,(2)}\mathbb{A}^{(1)\,b}_{2j}(p) + \mathbb{A}^{\dagger\,(1)\,b}_{1j}(p)\mathbb{N}^{(2)} + 2\int \frac{d^3q}{(2\pi)^3}\,\mathbb{A}^{\dagger\,(1)\,c}_{2k}(q)\mathbb{B}^{cb}_{2\,kj}(p,q) + \int \frac{d^3q}{(2\pi)^3}\,\mathbb{B}^{\dagger\,cb}_{3kj}(p,q)\mathbb{A}^{(1)\,c}_{2k}(q)\\ &+ 2\int \frac{d^3q}{(2\pi)^3}\,\mathbb{C}^{\dagger\,dcb}_{1lkj}(p,q)\mathbb{B}^{cd}_{2kl}(q,p-q) + \mathbb{A}^{(3)\,b}_{2j}(p) + \mathbb{A}^{\dagger\,(3)\,b}_{1j}(p) + \mathbb{A}^{(1)\,b}_{2j}(p) + \mathbb{A}^{\dagger\,(1)\,b}_{1j}(p)\Bigg]\\ &+ \int \frac{d^3p}{(2\pi)^3}\frac{d^3q}{(2\pi)^3}\,a^{\dagger\,b}_j(p)a^{\dagger\,c}_k(q)\Bigg[\mathbb{B}^{cb}_{2kj}(p,q) + \frac{1}{2}\mathbb{A}^{(1)\,\dagger\,b}_{1j}(p)\mathbb{A}^{(1)\,c}_{2k}(q) + \frac{1}{2}\mathbb{A}^{(1)\,\dagger\,c}_{1k}(q)\mathbb{A}^{(1)\,b}_{2j}(p) + \mathbb{B}^{\dagger\,cb}_{1kj}(p,q)\\ &+ \mathbb{C}^{\dagger\,cbd}_{2kjl}(q+p,q)\mathbb{A}^{(1)\,d}_{2l}(q+p)\Bigg]\\ &+ \int \frac{d^3p}{(2\pi)^3}\frac{d^3q}{(2\pi)^3}\,a^{\dagger\,b}_j(p)a^c_k(q)\Bigg[\mathbb{B}^{\dagger cb}_{3kj}(p,q) + \mathbb{B}^{bc}_{3jk}(q,p) + \mathbb{A}^{\dagger b}_{1j}(p)\mathbb{A}^c_{1k}(q) + \mathbb{A}^{\dagger c}_{2k}(p)\mathbb{A}^b_{2j}(q)\\ &+ 2\mathbb{A}^{\dagger d}_{2l}(q-p)\mathbb{C}^{bdc}_{1jlk}(q,p) + 2\mathbb{C}^{\dagger cdb}_{1klj}(p,q)\mathbb{A}^d_{2l}(p-q)\Bigg]\\ &+ \int \frac{d^3p}{(2\pi)^3}\frac{d^3q}{(2\pi)^3}\,a^{\dagger\,d}_l(q)a^{\dagger\,c}_k(p-q)a^b_j(p)\Bigg[\mathbb{C}^{\dagger\,dcb}_{2lkj}(p,q) + \mathbb{C}^{dcb}_{1lkj}(p,q)\Bigg]\\ &+ \dots\end{aligned} \tag{3.10}$$

where we used the following identity

$$\left[a^a_i(k), a^{\dagger b}_j(p)\right] = a^a_i(k)a^{\dagger b}_j(p) - a^{\dagger b}_j(p)a^a_i(k) = (2\pi)^3\delta^{ab}\delta_{ij}\delta^{(3)}(k-p) \tag{3.11}$$

to normal order relevant operators and we already took into account that $\mathbb{N} = \mathbb{N}^\dagger$. These gives following five relation between the coefficients:

$$\mathbb{N}^2 = 1 - \int \frac{d^3p}{(2\pi)^3}\mathbb{A}^{\dagger b}_{2j}(p)\mathbb{A}^b_{2j}(p) \quad \text{or} \quad \mathbb{N} = 1 - \frac{1}{2}\int \frac{d^3p}{(2\pi)^3}\mathbb{A}^b_{1j}(p)\mathbb{A}^{\dagger\,b}_{1j}(p) \tag{3.12}$$
$$\mathbb{A}^{(1)\,\dagger\,b}_{1j}(p) = -\mathbb{A}^{(1)\,b}_{2j}(p) \tag{3.13}$$
$$\begin{aligned}\mathbb{A}^{(3)\,\dagger\,b}_{1j}(p) = -\mathbb{A}^{(3)\,b}_{2j}(p) - \mathbb{N}^{\dagger\,(2)}\mathbb{A}^{(1)\,b}_{2j}(p) + \mathbb{A}^{(1)\,b}_{2j}(p)\mathbb{N}^{(2)} + \int \frac{d^3q}{(2\pi)^3}\Bigg[&2\mathbb{A}^{(1)\,c}_{1k}(q)\mathbb{B}^{cb}_{2kj}(p,q) - \mathbb{B}^{\dagger\,cb}_{3kj}(p,q)\mathbb{A}^{(1)\,c}_{2k}(q)\\ &- 2\,\mathbb{C}^{\dagger\,dcb}_{1lkj}(p,q)\mathbb{B}^{cd}_{2kl}(q,p-q)\Bigg]\end{aligned} \tag{3.14}$$
$$\mathbb{B}^{\dagger\,cb}_{1kj}(p,q) = \frac{1}{2}\left[-2\,\mathbb{B}^{cb}_{2kj}(p,q) + \mathbb{A}^{(1)\,b}_{2j}(p)\mathbb{A}^{(1)\,c}_{2k}(q) + \mathbb{A}^{(1)\,c}_{2k}(q)\mathbb{A}^{(1)\,b}_{2j}(p) + 2\,\mathbb{C}^{cbd}_{1kjl}(q+p,q)\mathbb{A}^{(1)\,d}_{2l}(q+p)\right] \tag{3.15}$$
$$\mathbb{C}^{\dagger dcb}_{2lkj}(p,q) = -\mathbb{C}^{dcb}_{1lkj}(p,q) \tag{3.16}$$

The fact that $A^{(1)}$ can at most be linear in ρ at g order we find (E5) This is not correct: $\mathbb{N}^{(2)} \propto \int K(x,y)\rho^a(x)\rho^a(y)$ does not commute with $\rho^b(u)$, $[\int_{x,y} K\rho^a\rho^a, \rho^b(u)] = if^{abc}\int_x K(x,u)\{\rho^a(x),\rho^c(u)\}$. Keep $-\mathbb{N}^{\dagger(2)}\mathbb{A}^{(1)}_2 + \mathbb{A}^{(1)}_2\mathbb{N}^{(2)} = -[\mathbb{N}^{(2)}, \mathbb{A}^{(1)}_2]$ in the relation above; the two equations below and the simplified relation are to be removed.

$$\mathbb{N}^{(2)}\mathbb{A}^{(1)c}_{1k}(q) = \mathbb{A}^{(1)c}_{1k}(q)\mathbb{N}^{(2)} \tag{3.17}$$
$$\mathbb{N}^{(2)}\mathbb{A}^{(1)c}_{2k}(q) = \mathbb{A}^{(1)c}_{2k}(q)\mathbb{N}^{(2)} \tag{3.18}$$

and thus the relation for $\mathbb{A}^{(3)\,\dagger\,b}_{1j}(p)$ simplifies to:

$$\mathbb{A}^{(3)\,\dagger\,b}_{1j}(p) = -\mathbb{A}^{(3)\,b}_{2j}(p) + \int \frac{d^3q}{(2\pi)^3}\left[2\,\mathbb{A}^{(1)\,c}_{1k}(q)\mathbb{B}^{cb}_{2kj}(p,q) - \mathbb{B}^{\dagger\,cb}_{3kj}(p,q)\mathbb{A}^{(1)\,c}_{2k}(q) - 2\,\mathbb{C}^{\dagger\,dcb}_{1lkj}(p,q)\mathbb{B}^{cd}_{2kl}(q,p-q)\right] \tag{3.19}$$

Partially implementing the constraints on the coefficients we get:

$$\begin{aligned}\Omega = \mathbb{N} &\\ &+ \int \frac{d^3p}{(2\pi)^3}\left[\mathbb{A}_{1j}^{(1)\,b}(p)a_j^b(p) - \mathbb{A}_{1j}^{\dagger\,(1)\,b}(p)a_j^{\dagger b}(p) + \mathbb{A}_{1j}^{(3)\,b}(p)a_j^b(p) + \mathbb{A}_{2j}^{(3)\,b}(p)a_j^{\dagger b}(p)\right]\\ &+ \int \frac{d^3p}{(2\pi)^3}\frac{d^3q}{(2\pi)^3}\left[\mathbb{B}_{1kj}^{cb}(p,q)a_j^b(p)a_k^c(q) + \mathbb{B}_{2kj}^{cb}(p,q)a_j^{\dagger b}(p)a_k^{\dagger c}(q) + \mathbb{B}_{3kj}^{cb}(p,q)a_k^{\dagger c}(q)a_j^b(p)\right]\\ &+ \int \frac{d^3p}{(2\pi)^3}\frac{d^3q}{(2\pi)^3}\left[\mathbb{C}_{1lkj}^{dcb}(p,q)a_l^{\dagger d}(q)a_k^{\dagger c}(p-q)a_j^b(p) - \mathbb{C}_{1lkj}^{\dagger\,dcb}(p,q)a_j^{\dagger b}(p)a_k^c(p-q)a_l^d(q)\right] + \dots\end{aligned} \tag{3.20}$$

The coefficients $\mathbb{A}_1^{(1)}, \mathbb{A}_2^{(3)}, \mathbb{B}_2, \mathbb{B}_3, \mathbb{C}_1$ are obtained directly from our LCPT calculations. The other coefficients, such as $\mathbb{B}_1$ and $\mathbb{A}_1^{(3)}$ follow from the constraints. In the transverse coordinate space, we have

$$\begin{aligned}\Omega = \mathbb{N} &\\ &+ \int \frac{dp^+}{2\pi}\int d^2x\left[\mathbb{A}_{1j}^{(1)\,b}(p^+,x)a_j^b(p^+,x) - \mathbb{A}_{1j}^{\dagger\,(1)\,b}(p^+,x)a_j^{\dagger b}(p^+,x) + \mathbb{A}_{1j}^{(3)\,b}(p^+,x)a_j^b(p^+,x) + \mathbb{A}_{2j}^{(3)\,b}(p^+,-x)a_j^{\dagger b}(p^+,x)\right]\\ &+ \int \frac{dp^+}{2\pi}\frac{dq^+}{2\pi}\int d^2x\,d^2y\Big[\mathbb{B}_{1kj}^{cb}(p^+,x;q^+,y)a_j^b(p^+,x)a_k^c(q^+,y) + \mathbb{B}_{2kj}^{cb}(p^+,-x;q^+,-y)a_j^{\dagger b}(p^+,x)a_k^{\dagger c}(q^+,y)\\ &+ \mathbb{B}_{3kj}^{cb}(p^+,x;q^+,-y)a_k^{\dagger c}(q^+,y)a_j^b(p^+,x)\Big]\\ &+ \int \frac{dp^+}{2\pi}\frac{dq^+}{2\pi}\int d^2x\,d^2y\,d^2z\left[\mathbb{C}_{1lkj}^{dcb}(p^+,z-y;q^+,y-x)a_l^{\dagger d}(q^+,x)a_k^{\dagger c}(p^+-q^+,y)a_j^b(p^+,z)\right.\\ &\left.- \mathbb{C}_{1lkj}^{\dagger\,dcb}(p^+,z-y;q^+,y-x)a_j^{\dagger b}(p^+,z)a_k^c(p^+-q^+,y)a_l^d(q^+,x)\right] + \dots\end{aligned} \tag{3.21}$$

Moreover, the coefficients that were constrained by the requirement of unitarity of the evolution operator, i.e., $\Omega^\dagger\Omega = \mathcal{I}$, are as follows in their respective transverse coordinate space.

$$\mathbb{A}_{1j}^{(1)\,\dagger\,b}(p^+,z) = -\mathbb{A}_{2j}^{(1)\,b}(p^+,-z) \tag{3.22}$$

$$\begin{aligned}\mathbb{A}_{1j}^{(3)\,\dagger\,b}(p^+,x) = -\mathbb{A}_{2j}^{(3)\,b}(p^+,-x) + \int\frac{dq^+}{2\pi}\int d^2y\left[2\,\mathbb{A}_{1k}^{(1)\,c}(q^+,y)\mathbb{B}_{2kj}^{cb}(p^+,-x;\,q^+,-y) - \mathbb{B}_{3kj}^{\dagger\,cb}(p^+,x;q^+,-y)\mathbb{A}_{2k}^{(1)\,c}(q^+,-y)\right.\\ \left.- 2\int d^2z\,\mathbb{C}_{1lkj}^{\dagger\,dcb}(p^+,x-z;q^+,z-y)\mathbb{B}_{2lk}^{dc}(p^+-q^+,-z;q^+,-y)\right]\end{aligned} \tag{3.23}$$

$$\begin{aligned}\mathbb{B}_{1kj}^{\dagger\,cb}(p^+,x;q^+,y) = \frac{1}{2}\Bigg[&-2\,\mathbb{B}_{2kj}^{cb}(p^+,-x;q^+,-y) + \mathbb{A}_{2j}^{(1)\,b}(p^+,-x)\mathbb{A}_{2k}^{(1)\,c}(q^+,-y) + \mathbb{A}_{2k}^{(1)\,c}(q^+,-y)\mathbb{A}_{2j}^{(1)\,b}(p^+,-x)\\ &+ 2\int d^2z\left[\mathbb{C}_{1kjl}^{cbd}(q^+ + p^+, z-x; q^+, x-y)\mathbb{A}_{2l}^{(1)\,d}(q^+ + p^+, -z)\right]\Bigg]\end{aligned} \tag{3.24}$$

$$\mathbb{C}_{2lkj}^{\dagger\,dcb}(p^+,z;q^+,x) = -\mathbb{C}_{1lkj}^{dcb}(p^+,-z;q^+,-x) \tag{3.25}$$

Where we used the common convention for the Fourier transforms

$$a(k^+,k) = \int d^2z\,e^{-ikz}a(k^+,z), \tag{3.26}$$

$$a^\dagger(k^+,k) = \int d^2z\,e^{ikz}a^\dagger(k^+,z), \tag{3.27}$$

$$f(k) = \int d^2x\,e^{ikx}f(x)\,. \tag{3.28}$$

In order to find the coefficients, we compute the "evolved" wave function and match it with the LCPT wave functions. There are three relevant contributions

- `Action on vacuum`

$$\begin{aligned}\Omega\,|0\rangle = \mathbb{N}\,|0\rangle &\\ &+ (2\pi)^{3/2}\int\frac{d^3p}{(2\pi)^3}\left[\mathbb{A}_{2j}^{(3)\,b}(p) - \mathbb{A}_{1j}^{\dagger\,(1)\,b}(p)\right]|g_j^b(p)\rangle\\ &+ (2\pi)^3\int\frac{d^3p}{(2\pi)^3}\frac{d^3q}{(2\pi)^3}\mathbb{B}_{2kj}^{cb}(p,q)\,|g_j^b(p)g_k^c(q)\rangle\end{aligned} \tag{3.29}$$

In the last line it is convenient to introduce alternative coordinates, $k^+ = p^+ + q^+$ and so $\xi = \frac{p^+}{k^+}$. This will give us

$$\begin{aligned}\Omega\,|0\rangle = \mathbb{N}\,|0\rangle &\\ &+ (2\pi)^{3/2}\int\frac{d^3p}{(2\pi)^3}\left[\mathbb{A}_{2j}^{(3)\,b}(p) - \mathbb{A}_{1j}^{\dagger\,(1)\,b}(p)\right]|g_j^b(p)\rangle\\ &+ 2\pi\int dk^+k^+\int d\xi\int\frac{d^2p}{(2\pi)^2}\frac{d^2q}{(2\pi)^2}\,\mathbb{B}_{2kj}^{cb}(\xi k^+,\mathbf{p};(1-\xi)k^+,\mathbf{q})\,|g_j^b(\xi k^+,\mathbf{p})g_k^c((1-\xi)k^+,\mathbf{q})\rangle\end{aligned} \tag{3.30}$$

In transverse coordinate space, we get

$$\begin{aligned}\Omega\,|0\rangle = \mathbb{N}\,|0\rangle &\\ &+ \frac{1}{(2\pi)^{1/2}}\int dp^+\int d^2x\left[\mathbb{A}_{2j}^{(3)\,b}(p^+,-x) - \mathbb{A}_{1j}^{\dagger\,(1)\,b}(p^+,x)\right]|g_j^b(p^+,x)\rangle\\ &+ \frac{1}{2\pi}\int dp^+\,dq^+\int d^2z\,d^2x\,\mathbb{B}_{2kj}^{cb}(p^+,-z;q^+,-x)\,|g_j^b(p^+,z)g_k^c(q^+,x)\rangle\end{aligned} \tag{3.31}$$

Similarly, changing to the momentum fraction,

$$\begin{aligned}\Omega\,|0\rangle = \mathbb{N}\,|0\rangle &\\ &+ \frac{1}{(2\pi)^{1/2}}\int dp^+\int d^2x\left[\mathbb{A}_{2j}^{(3)\,b}(p^+,-x) - \mathbb{A}_{1j}^{\dagger\,(1)\,b}(p^+,x)\right]|g_j^b(p^+,x)\rangle\\ &+ \frac{1}{2\pi}\int dk^+k^+\int d\xi\int d^2z d^2x\,\mathbb{B}_{2kj}^{cb}(\xi k^+,-z;(1-\xi)k^+,-x)\,|g_j^b(\xi k^+,z),\,g_k^c((1-\xi)k^+,x)\rangle\end{aligned} \tag{3.32}$$

These relations will help us fix coefficients $\mathbb{N}$, $\mathbb{A}^{(1)}$, $\mathbb{A}^{(3)}$ and $\mathbb{B}_2$. The remaining two coefficients can be fixed by considering Ω acting on a single gluon state.

- Action on single gluon state

$$\begin{aligned}\Omega\,|g_i^a(k)\rangle = &\frac{1}{(2\pi)^{3/2}}\left[\mathbb{A}_{1i}^{(1)\,a}(k)+\mathbb{A}_{1i}^{(3)\,a}(k)\right]|0\rangle\\ &+\mathbb{N}\,|g_i^a(k)\rangle\\ &+\int\frac{d^3p}{(2\pi)^3}\mathbb{B}_{3ji}^{ba}(k,p)\,|g_j^b(p)\rangle\\ &+(2\pi)^{3/2}\int\frac{d^3p}{(2\pi)^3}\left[\mathbb{A}_{2j}^{(3)\,b}(p)-\mathbb{A}_{1j}^{\dagger\,(1)\,b}(p)\right]|g_i^a(k)g_j^b(p)\rangle\\ &+(2\pi)^{3/2}\int\frac{d^3p}{(2\pi)^3}\mathbb{C}_{1jki}^{bca}(k,p)\,|g_j^b(p)g_k^c(k-p)\rangle\end{aligned}\tag{3.33}$$

We ignore final states with three or more gluons since we only need terms with up to two gluons in the final state for our problem. As before it is convenient to introduce the momentum fraction $p^+=\xi k^+$,

$$\begin{aligned}\Omega\,|g_i^a(k)\rangle = &\frac{1}{(2\pi)^{3/2}}\left[\mathbb{A}_{1i}^{(1)\,a}(k)+\mathbb{A}_{1i}^{(3)\,a}(k)\right]|0\rangle\\ &+\mathbb{N}\,|g_i^a(k)\rangle\\ &+\int\frac{d\xi}{2\pi}k^+\int\frac{d^2p}{(2\pi)^2}\mathbb{B}_{3ji}^{ba}(k^+,k;\xi k^+,p)\,|g_j^b(\xi k^+,p)\rangle\\ &+(2\pi)^{3/2}\int\frac{d\xi}{2\pi}k^+\int\frac{d^2p}{(2\pi)^2}\left[\mathbb{A}_{2j}^{(3)\,b}(\xi k^+,p)-\mathbb{A}_{1j}^{\dagger\,(1)\,b}(\xi k^+,p)\right]|g_i^a(k^+,k)g_j^b(\xi k^+,p)\rangle\\ &+(2\pi)^{3/2}\int\frac{d\xi}{2\pi}\,k^+\int\frac{d^2p}{(2\pi)^2}\mathbb{C}_{1jki}^{bca}(k^+,k;\xi k^+,p)\,|g_j^b(\xi k^+,p)g_k^c((1-\xi)k^+,k-p)\rangle+\ldots\end{aligned}\tag{3.34}$$

In transverse coordinate space we get,

$$\begin{aligned}\Omega\,|g_i^a(k^+,z)\rangle = &\frac{1}{(2\pi)^{1/2}}\left[\mathbb{A}_{1i}^{(1)\,a}(k^+,z)+\mathbb{A}_{1i}^{(3)\,a}(k^+,z)\right]|0\rangle\\ &+\mathbb{N}\,|g_i^a(k^+,z)\rangle\\ &+\frac{1}{2\pi}\int dp^+\int d^2x\,\mathbb{B}_{3ji}^{ba}(k^+,z;p^+,-x)\,|g_j^b(p^+,x)\rangle\\ &+\frac{1}{(2\pi)^{1/2}}\int dp^+\int d^2x\left[\mathbb{A}_{2j}^{(3)\,b}(p^+,-x)-\mathbb{A}_{1j}^{\dagger\,(1)\,b}(p^+,x)\right]|g_i^a(k^+,z)g_j^b(p^+,x)\rangle\\ &+\frac{1}{(2\pi)^{1/2}}\int dp^+\int d^2x d^2y\,\mathbb{C}_{1jki}^{bca}(k^+,z-y;p^+,y-x)\,|g_j^b(p^+,x)g_k^c(k^+-p^+,y)\rangle+\ldots\end{aligned}\tag{3.35}$$

(or)

$$\begin{aligned}\Omega\,|g_i^a(k^+,z)\rangle = &\frac{1}{(2\pi)^{1/2}}\left[\mathbb{A}_{1i}^{(1)\,a}(k^+,z)+\mathbb{A}_{1i}^{(3)\,a}(k^+,z)\right]|0\rangle\\ &+\mathbb{N}\,|g_i^a(k^+,z)\rangle\\ &+\frac{1}{2\pi}\int d\xi\,k^+\int d^2x\,\mathbb{B}_{3ji}^{ba}(k^+,z;\xi k^+,-x)\,|g_j^b(\xi k^+,x)\rangle\\ &+\frac{1}{(2\pi)^{1/2}}\int d\xi\,k^+\int d^2x\left[\mathbb{A}_{2j}^{(3)\,b}(\xi k^+,-x)-\mathbb{A}_{1j}^{\dagger\,(1)\,b}(\xi k^+,x)\right]|g_i^a(k^+,z)g_j^b(\xi k^+,x)\rangle\\ &+\frac{1}{(2\pi)^{1/2}}\int d\xi\,k^+\int d^2x\,d^2y\,\mathbb{C}_{1jki}^{bca}(k^+,z-y;\xi k^+,y-x)\,|g_j^b(\xi k^+,x)g_k^c((1-\xi)k^+,y)\rangle+\ldots\end{aligned}\tag{3.36}$$

Note that the factors of π are from the normalization of the creation annihilation operators, see Appendix A. We are now in the position to match the coefficients of the evolution Ω operator by comparing it with the light cone wave functions derived from the vacuum, and single gluon incoming states in the sections 1 and 2. The constraints originating from the incoming vacuum state lead to the following relations. $\mathbb{N}$ is obtained by comparing the first line of Eq. (3.29) with Eq. (1.98) for momentum space. Thus, we get

$$\mathbb{N}=\mathbb{N}_0+\mathbb{N}_2=1-\frac{g^2}{(2\pi)^3}\,\log\left(\frac{\vee}{\Lambda}\right)\int d^2p\,\frac{\rho^b(-p)\rho^b(p)}{p^2}\tag{3.37}$$

In transverse co-ordinates,

$$\mathbb{N}=1-\frac{g^2}{(2\pi)^3}\,\log\left(\frac{\vee}{\Lambda}\right)\int_{x,y,z}\frac{(x-z)\cdot(y-z)}{(x-z)^2(y-z)^2}\rho^a(x)\rho^a(y)\tag{3.38}$$

where $\mathbb{N}_0=1$ and $\mathbb{N}_2$ is the term proportional to g^2. $\mathbb{A}_2^{(3)}$ and $\mathbb{A}_1^{\dagger\,(1)}$ are obtained by comparing second line of Eq. (3.29) with Eq. (1.77) for momentum space.

$$-\mathbb{A}_{2i}^{a}(k^+,\mathbf{k})=\mathbb{A}_{1i}^{\dagger\,a}(k^+,\mathbf{k})=\sqrt{2}\frac{g\,k^i\rho^a(-\mathbf{k})}{\sqrt{k^+}\,\mathbf{k}^2}\tag{3.39}$$

In transverse co-ordinates,

$$-\mathbb{A}_{2i}^{a}(k^+,-\mathbf{z})=\mathbb{A}_{1i}^{\dagger\,a}(k^+,\mathbf{z})=-\frac{1}{\sqrt{2}\pi}\frac{ig}{\sqrt{k^+}}\int_x\frac{(\mathbf{x}-\mathbf{z})^i}{(\mathbf{x}-\mathbf{z})^2}\rho^a(\mathbf{x})\tag{3.40}$$

$\mathbb{A}_2^{(3)}$ is obtained by comparing the second line of Eq. (3.29) with Eq. (1.89) and Eq. (1.91) for momentum space.

$$\begin{aligned}\mathbb{A}_{2i}^{(3)\,a}(k^+,k) = &\text{(remove)}\frac{g^3N_c\rho^a(-k)k^i}{8\sqrt{2k^+}\pi^2\,k^2}\left[\left[-\frac{2}{\epsilon}+\log\left(\frac{k^2}{\mu^2_{\overline{MS}}}\right)\right]\times\left[\frac{11}{3}+\log\left(\frac{\Lambda}{k^+}\right)-2\log\left(\frac{\vee}{\Lambda}\right)\right]+\frac{\pi^2}{3}-\frac{67}{9}+\log^2\left(\frac{\Lambda}{k^+}\right)-\log^2\left(\frac{\vee}{k^+}\right)\right]\\ &\text{(remove)}+\frac{g^3N_c\rho^a(-k)k^i}{8\sqrt{2k^+}\pi^2\,k^2}\left[\frac{\pi^2}{3}-\frac{67}{9}+\log^2\left[1-\frac{k^+}{\vee}\right]+\log\left[1-\frac{k^+}{\vee}\right]\log\left[\frac{\vee}{k^+}\right]\right]\\ &\text{(E4, new)}+\frac{g^3N_c\rho^a(-k)k^i}{8\sqrt{2k^+}\pi^2\,k^2}\left[\left[-\frac{2}{\epsilon}+\log\left(\frac{k^2}{\mu^2_{\overline{MS}}}\right)\right]\left[\frac{11}{3}+\log\frac{\Lambda}{k^+}+\log\left(\frac{\vee}{k^+}-1\right)-\log\frac{\vee}{\Lambda}\right]\right.\\ &\left.+\frac{\pi^2}{3}-\frac{67}{9}+\log^2\left[1-\frac{k^+}{\vee}\right]+\log\left[1-\frac{k^+}{\vee}\right]\log\left[\frac{\vee}{k^+}\right]\right]\quad\text{(exact form: X6)}\\ &+\int d^2\mathbf{p}\,\frac{ig^3f^{abc}\left\{\rho^b(-\mathbf{p}),\,\rho^c(-\mathbf{k}+\mathbf{p})\right\}}{8\sqrt{2}\pi^3\,\mathbf{p}^2\mathbf{k}^2\,\sqrt{k^+}}\left[\log\left(\frac{\Lambda}{k^+}\right)\left[\frac{k^2p^i}{(k-p)^2}-\frac{k\cdot p\,k^i}{(k-p)^2}\right]-2\,k^i\,\log\left(\frac{\vee}{\Lambda}\right)\right.\\ &\left.-2\,k^i\,\log\left(\frac{k^+p^2}{\vee\,k^2+k^+p^2}\right)\right]\\ &-\int d^2\mathbf{p}\,\frac{ig^3f^{abc}\left\{\rho^b(-\mathbf{p}),\,\rho^c(-\mathbf{k}+\mathbf{p})\right\}}{8\sqrt{2}\pi^3\,\mathbf{p}^2\mathbf{k}^2\,\sqrt{k^+}}\int_{\frac{\Lambda}{k^+}}^{1-\frac{\Lambda}{k^+}}d\xi\,\frac{\xi}{(\xi(\mathbf{k}-\mathbf{p})^2+(1-\xi)\mathbf{k}^2)\,(1-\xi)}\end{aligned}$$

$$\left[\left(\frac{2}{\xi}\left(\mathbf{p}^2-\mathbf{k}\cdot\mathbf{p}\right)-\mathbf{p}^2+2\mathbf{k}\cdot\mathbf{p}\right)\mathbf{k}^i+\frac{1}{\xi}\mathbf{k}^2\mathbf{p}^i\right]$$
$$+\int d^2\mathbf{p}\,\frac{g^3\,\rho^b(\mathbf{p})\,\rho^b(-\mathbf{p})\,\rho^a(-\mathbf{k})\,\mathbf{k}^i}{4\sqrt{2}\pi^3\,\mathbf{k}^2\mathbf{p}^2\sqrt{k^+}}\,\log\left(\frac{\vee}{\Lambda}\right) \tag{3.41}$$

$\mathbb{A}_2^{(3)}$ in coordinate space is obtained by comparing second line of Eq. (3.32) with Eq. (1.90), Eq. (1.94) and Eq. (1.96)

$$\begin{aligned}\mathbb{A}_{2i}^{(3)\,a}(k^+,-z)=&-\int_x\frac{ig^3N_c\rho^a(x)(x-z)^i}{16\sqrt{2}\pi^3\,(x-z)^2\,\sqrt{k^+}}\left[\left[-\frac{2}{\epsilon}-2\gamma-\log\left(\frac{(x-z)^2\,\mu^2}{4}\right)\right]\times\left[\frac{11}{3}+\log\left(\frac{\Lambda}{k^+}\right)+\log\left(\frac{\vee}{k^+}-1\right)-\log\left(\frac{\vee}{\Lambda}\right)\right]\right.\\&\left.-\frac{67}{9}+\frac{\pi^2}{3}+\log\left(1-\frac{k^+}{\vee}\right)^2+\log\left(1-\frac{k^+}{\vee}\right)\log\left(\frac{\vee}{k^+}\right)\right]\\&+\int d^2\mathbf{p}\,d^2\mathbf{k}\int_{\mathbf{x},\mathbf{y}}e^{i\mathbf{k}\cdot(\mathbf{z}-\mathbf{y})}e^{i\mathbf{p}\cdot(\mathbf{y}-\mathbf{x})}\frac{ig^3f^{abc}\left\{\rho^b(\mathbf{x}),\,\rho^c(\mathbf{y})\right\}}{32\sqrt{2k^+}\,\pi^5\mathbf{p}^2\mathbf{k}^2}\left[\log\left(\frac{\Lambda}{k^+}\right)\left[\frac{k^2p^i}{(k-p)^2}-\frac{k\cdot p\,k^i}{(k-p)^2}\right]-2\,k^i\,\log\left(\frac{\vee}{\Lambda}\right)\right.\\&-2k^i\,\log\left(\frac{k^+p^2}{\vee\,k^2+k^+p^2}\right)+\frac{(2p^2-2\,k\cdot p+k^2)\,k^i+k^2p^i}{(k-p)^2}\,\log\left[1+\left(\frac{\vee}{k^+}-1\right)\frac{k^2}{(k-p)^2}\right]\\&\left.-\frac{(p^2\,k^i+k^2\,p^i)}{(k-p)^2}\,\log\left(\frac{\vee-k^+}{\Lambda}\right)-k^i\,\log\left(\frac{\vee}{k^+}\right)\right]\\&-\int_{\mathbf{w},\,\mathbf{x},\,\mathbf{y},\,\mathbf{u}}\frac{ig^3\,\rho^b(\mathbf{x})\,\rho^b(\mathbf{y})\,\rho^a(\mathbf{w})\,(x-u)\cdot(y-u)\,(w-z)^i}{8\sqrt{2}\pi^4\,(x-u)^2(y-u)^2(w-z)^2\,\sqrt{k^+}}\,\log\left(\frac{V\vee}{\Lambda}\right)\end{aligned} \tag{3.42}$$

$\mathbb{B}_2$ is obtained by comparing third line of Eq. (3.29) with Eq. (1.80) and Eq. (1.84) for momentum space.

$$\begin{aligned}\mathbb{B}_{2ji}^{ba}(k^+,k;p^+,p)=&\frac{ig^2f^{abc}p^jk^i\rho^c(-p-k)\,\sqrt{p^+k^+}}{2\,(p^+k^2+k^+p^2)}\left[\frac{1}{k^+\,p^2}-\frac{1}{k^2\,p^+}\right]\,\Theta\left(k^+-\Lambda-p^+\right)\Theta(p^+-\Lambda)\Theta(k^+-\Lambda)\Theta(\vee-k^+-p^+)\\&+\frac{g^2p^jk^i\left\{\rho^b(-p),\rho^a(-k)\right\}}{2\,p^2k^2\,\sqrt{p^+k^+}}\,\Theta\left(k^+-\Lambda-p^+\right)\Theta(p^+-\Lambda)\Theta(k^+-\Lambda)\Theta(\vee-k^+-p^+)\\&-\frac{ig^2f^{abc}\rho^c(-p-k)\,(p+k)^l\,(p^mk^+-p^+k^m)}{(p+k)^2\,\sqrt{p^+k^+}\,(p^++k^+)\left(\frac{p^2}{p^+}+\frac{k^2}{k^+}\right)}\left[\frac{\delta_{lm}\delta_{ji}}{(p^++k^+)}-\frac{\delta_{im}\delta_{jl}}{k^+}-\frac{\delta_{jm}\delta_{il}}{p^+}\right]\,\Theta\left(k^+-\Lambda-p^+\right)\Theta(p^+-\Lambda)\Theta(k^+-\Lambda)\Theta(\vee-k^+-p^+)\\&-\frac{ig^2f^{abc}\rho^c(-p-k)\,(p^+-k^+)\delta_{ji}}{2\,\sqrt{p^+k^+}\,(p^++k^+)^2\left(\frac{k^2}{k^+}+\frac{p^2}{p^+}\right)}\,\Theta\left(k^+-\Lambda-p^+\right)\Theta(p^+-\Lambda)\Theta(k^+-\Lambda)\Theta(\vee-k^+-p^+)\end{aligned} \tag{3.43}$$

$\mathbb{B}_2$ is obtained in coordinate space by comparing third line of eq. (3.32) with Eq. (1.82) and Eq. (1.85).

$$\begin{aligned}\mathbb{B}_{2ji}^{bc}(k^+,-\mathbf{w};p^+,-\mathbf{z})=&\int_{\mathbf{x}}\frac{ig^2f^{cba}\rho^a(\mathbf{x})\sqrt{p^+k^+}}{4\pi^2\,[p^+(x-z)^2+k^+(x-w)^2]\,(p^++k^+)}\left[\frac{\delta^{ij}}{2(z-w)^2}\left((x-z)^2-(x-w)^2\right)\right.\\&+\frac{(p^++k^+)}{k^+}\left(\frac{(x-z)^j(z-w)^i}{(z-w)^2}-\frac{(x-z)^j(x-w)^i}{2(x-w)^2}\right)\\&\left.+\frac{(p^++k^+)}{p^+}\left(\frac{(x-w)^i(z-w)^j}{(z-w)^2}+\frac{(x-z)^j(x-w)^i}{2(x-z)^2}\right)\right]\times\Theta\left(k^+-\Lambda-p^+\right)\Theta(p^+-\Lambda)\Theta(k^+-\Lambda)\Theta(\vee-k^+-p^+)\\&-\int_{\mathbf{x},\,\mathbf{y}}\frac{g^2\left\{\rho^c(\mathbf{y}),\,\rho^b(\mathbf{x})\right\}(y-w)^i(x-z)^j}{8\pi^2\sqrt{p^+k^+}(y-w)^2(x-z)^2}\times\Theta\left(k^+-\Lambda-p^+\right)\Theta(p^+-\Lambda)\Theta(k^+-\Lambda)\Theta(\vee-k^+-p^+)\end{aligned} \tag{3.44}$$

In terms of longitudinal momentum fraction,

$$\begin{aligned}\mathbb{B}_{2ji}^{ba}(\xi k^+,-\mathbf{z};(1-\xi)k^+,-\mathbf{u})=&\int_{\mathbf{x}}\frac{ig^2f^{abc}\rho^c(\mathbf{x})\sqrt{\xi(1-\xi)}}{4\pi^2k^+\left((1-\xi)(x-u)^2+\xi(x-z)^2\right)}\left(\frac{\delta^{ij}}{2(z-u)^2}\left((x-z)^2-(x-u)^2\right)\right.\\&\left.+\frac{(x-u)^i(z-u)^j}{\xi(z-u)^2}+\frac{(x-z)^j(z-u)^i}{(1-\xi)(z-u)^2}+\frac{(x-z)^j(x-u)^i}{2\xi(x-z)^2}-\frac{(x-z)^j(x-u)^i}{2(1-\xi)(x-u)^2}\right)\times\Theta\left(1-\frac{\Lambda}{k^+}-\xi\right)\\&-\int_{\mathbf{x},\,\mathbf{y}}\frac{g^2\left\{\rho^a(\mathbf{y}),\,\rho^b(\mathbf{x})\right\}(y-z)^i(x-u)^j}{8\pi^2k^+\sqrt{\xi(1-\xi)}(y-z)^2(x-u)^2}\times\Theta\left(1-\frac{\Lambda}{k^+}-\xi\right)\end{aligned} \tag{3.45}$$

The constraints originating from the incoming single gluon state provide the following *additional constraints*. $\mathbb{B}_3$ in momentum space is obtained by comparing third line of Eq. (3.33) with Eqs. (2.5), (2.11), (2.15), (2.26) and (2.28)

$$\begin{aligned}\mathbb{B}_{3ji}^{ba}(k^+,k;p^+,p)=&\frac{ig^2f^{abc}\sqrt{p^+k^+}(k^++p^+)}{(p^+-k^+)^2}\frac{\rho^c(-\mathbf{p}+\mathbf{k})}{p^+\mathbf{k}^2-k^+\mathbf{p}^2}\delta_{ij}\left[\Theta\left(p^+-k^+-\Lambda\right)+\Theta\left(k^+-\Lambda-p^+\right)\right]\\&-\frac{g^2\,k^ip^j}{k^2p^2\,\sqrt{k^+p^+}}\left[\{\rho^a(k),\rho^b(-p)\}-if^{abc}\rho^c(k-p)\frac{p^2k^++k^2p^+}{p^2k^+-k^2p^+}\right]\\&+\frac{2i\,g^2f^{abc}\,\rho^c(k-p)(k-p)^l\,\sqrt{k^+p^+}}{(k^+-p^+)^2}\times\frac{(k^mp^+-k^+p^m)}{(k^2p^+-p^2k^+)}\left[\frac{\delta_{jm}\delta_{il}}{p^+}-\frac{\delta_{im}\delta_{jl}}{k^+}-\frac{\delta_{lm}\delta_{ij}}{(p^+-k^+)}\right]\times\\&\left[+-\frac{(k^+-p^+)}{(k-p)^2}\,\Theta(p^+-k^+-\Lambda)+\frac{1}{\left(\frac{k^2}{k^+}-\frac{(k-p)^2}{(k^+-p^+)}-\frac{p^2}{p^+}\right)}\,\Theta(k^+-\Lambda-p^+)\right]\end{aligned} \tag{3.46}$$

$\mathbb{B}_3$ in coordinate space is obtained by comparing third line of Eq. (3.36) with eqs. (2.7), (2.9), (2.13), (2.17), (2.30) and (2.31)

$$\begin{aligned}\mathbb{B}_{3ji}^{ba}(k^+,z;p^+,-u)=&i\frac{g^2f^{abc}}{(2\pi)^2}\int_x\frac{\sqrt{p^+k^+}(p^++k^+)}{(k^+-p^+)^2}\frac{\rho^c(\mathbf{x})\delta_{ij}}{p^+(x-u)^2-k^+(x-z)^2}\left[\Theta\left(p^+-k^+-\Lambda\right)+\Theta\left(k^+-\Lambda-p^+\right)\right]\\&+\frac{ig^2f^{abc}}{2\pi^2}\frac{1}{k^+}\int_x\frac{\sqrt{k^+p^+}}{k^+-p^+}\frac{(k^+(x-z)^m-p^+(x-u)^m)((x-z)^k-(x-u)^k)}{k^+(x-z)^2-p^+(x-u)^2}\rho^c(\mathbf{x})\left(\delta_{ik}\delta_{jm}-\frac{k^+}{p^+}\delta_{jk}\delta_{im}\right.\\&\left.-\frac{k^+}{k^+-p^+}\delta_{ij}\delta_{km}\right)\times\left[+-\frac{1}{((x-z)-(x-u))^2}\Theta\left(p^+-k^+-\Lambda\right)-\frac{p^+k^+}{(k^+(x-z)-p^+(x-u))^2}\Theta\left(k^+-\Lambda-p^+\right)\right]\\&-\frac{g^2}{4\pi^2\sqrt{p^+k^+}}\int_{x,y}\frac{(x-z)^i(y-u)^j}{(x-z)^2(y-u)^2}\{\rho^a(\mathbf{x}),\rho^b(\mathbf{y})\}\\&-\frac{ig^2f^{abc}}{4\pi^2\sqrt{p^+k^+}}\int_x\frac{(x-z)^i(x-u)^j}{(x-z)^2(x-u)^2}\frac{(k^+(x-z)^2+p^+(x-u)^2)}{(p^+(x-u)^2-k^+(x-z)^2)}\rho^c(\mathbf{x})\end{aligned} \tag{3.47}$$

In terms of longitudinal momentum fraction,

$$\mathbb{B}_{3ji}^{ba}(k^+,z;\xi k^+,-u)=i\,\frac{g^2f^{abc}}{(2\pi)^2k^+}\int_x\frac{\sqrt{\xi}(1+\xi)}{(1-\xi)^2}\frac{\rho^c(\mathbf{x})\delta_{ij}}{\xi(x-u)^2-(x-z)^2}\left[\Theta\left(\xi-1-\frac{\Lambda}{k^+}\right)+\Theta\left(1-\frac{\Lambda}{k^+}-\xi\right)\right]$$

$$
\begin{aligned}
&+\frac{ig^2f^{abc}}{2\pi^2k^+}\int_x\frac{\sqrt{\xi}}{1-\xi}\frac{((x-z)^m-\xi(x-u)^m)((x-z)^k-(x-u)^k)}{(x-z)^2-\xi(x-u)^2}\rho^c(\mathbf{x})\Bigg(\delta_{ik}\delta_{jm}-\frac{1}{\xi}\delta_{jk}\delta_{im}\\
&-\frac{1}{1-\xi}\delta_{ij}\delta_{km}\Bigg)\times\left[+-\frac{1}{((x-z)-(x-u))^2}\,\Theta\left(\xi-1-\frac{\Lambda}{k^+}\right)-\frac{\xi}{((x-z)-\xi(x-u))^2}\Theta\left(1-\frac{\Lambda}{k^+}-\xi\right)\right]\\
&-\frac{g^2}{4\pi^2k^+\sqrt{\xi}}\int_{x,y}\frac{(x-z)^i(y-u)^j}{(x-z)^2(y-u)^2}\{\rho^a(\mathbf{x}),\rho^b(\mathbf{y})\}\\
&-\frac{ig^2f^{abc}}{4\pi^2k^+\sqrt{\xi}}\int_x\frac{(x-z)^i(x-u)^j}{(x-z)^2(x-u)^2}\frac{((x-z)^2+\xi(x-u)^2)}{(\xi(x-u)^2-(x-z)^2)}\rho^c(\mathbf{x})
\end{aligned}
\tag{3.48}
$$

$\mathbb{C}_1$ in momentum space is obtained by equating fifth line of Eq. (3.33) with Eq. (2.32).

$$
\mathbb{C}^{bca}_{1jki}(k^+,k;p^+,p)=-\frac{igf^{abc}\,(p^mk^+-p^+k^m)\,\sqrt{k^+p^+(k^+-p^+)}}{\sqrt{2}\,(pk^+-p^+k)^2}\left[\frac{\delta_{im}\delta_{jk}}{k^+}-\frac{\delta_{ik}\delta_{mj}}{p^+}-\frac{\delta_{km}\delta_{ij}}{(k^+-p^+)}\right]\Theta\left(k^+-\Lambda-p^+\right)
\tag{3.49}
$$

In terms of longitudinal momentum fraction, we get

$$
\mathbb{C}^{bca}_{1jki}(k^+,k;\xi k^+,p)=-\frac{igf^{abc}}{\sqrt{2k^+}}\sqrt{\xi(1-\xi)}\,\frac{p^m-\xi k^m}{(\mathbf{p}-\xi\mathbf{k})^2}\times\left[\delta_{im}\delta_{jk}-\frac{1}{\xi}\delta_{jm}\delta_{ik}-\frac{1}{1-\xi}\delta_{km}\delta_{ij}\right]\Theta\left(1-\frac{\Lambda}{k^+}-\xi\right)
\tag{3.50}
$$

Performing the Fourier transform to the transverse coordinate space we get

$$
\mathbb{C}^{bca}_{1jki}(k^+,z;\xi k^+,x)=-\frac{gf^{abc}}{2\pi\sqrt{2k^+}}\sqrt{\xi(1-\xi)}\delta^{(2)}(z+\xi x)\frac{x^m}{x^2}\left[\delta_{im}\delta_{jk}-\frac{1}{\xi}\delta_{jm}\delta_{ik}-\frac{1}{1-\xi}\delta_{km}\delta_{ij}\right]\Theta\left(1-\frac{\Lambda}{k^+}-\xi\right)
\tag{3.51}
$$

(or)

$$
\mathbb{C}^{bca}_{1jki}(k^+,z;p^+,x)=-\frac{gf^{abc}}{2\pi\sqrt{2k^+}}\,\frac{\sqrt{p^+\,(k^+-p^+)}}{k^+}\,\delta^{(2)}\left(z+\frac{p^+}{k^+}x\right)\frac{x^m}{x^2}\left[\delta_{im}\delta_{jk}-\frac{k^+}{p^+}\delta_{jm}\delta_{ik}-\frac{k^+}{k^+-p^+}\delta_{km}\delta_{ij}\right]\Theta\left(k^+-p^+-\Lambda\right)
\tag{3.52}
$$

4 Production cross section

The production cross-section is given by

$$
\frac{d^3N}{d^3k}=\frac{1}{(2\pi)^3}\int d^2wd^2w'e^{-ik(w'-w)}\,\langle 0|\,\Omega^\dagger S^\dagger\Omega a_i^{a\dagger}(k^+,w)a_i^a(k^+,w')\Omega^\dagger S\Omega\,|0\rangle
\tag{4.1}
$$

To simplify the derivation we insert trivial combinations $SS^\dagger$ where it is appropriate:

$$
\frac{d^3N}{d^3k}=\frac{1}{(2\pi)^3}\int d^2wd^2w'e^{-ik(w'-w)}\,\langle 0|\,\Omega^\dagger S^\dagger\Omega S^\dagger Sa_i^{a\dagger}(k^+,w)SS^\dagger a_i^a(k^+,w')SS^\dagger\Omega^\dagger S\Omega\,|0\rangle
\tag{4.2}
$$

To simplify the notation we define

$$
\bar\Omega^\dagger=S^\dagger\Omega^\dagger S
\tag{4.3}
$$

and similarly for creation annihilation operators. Introducing complete set of states

$$
1=\int dq^+\int d^2v\,|g_n^f(q^+,v)\rangle\,\langle g_n^f(q^+,v)|+\int dq^+dr^+\int d^2vd^2u\,|g_n^f(q^+,v),g_m^e(r^+,u)\rangle\,\langle g_n^f(q^+,v),g_m^e(r^+,u)|
\tag{4.4}
$$

$$
\begin{aligned}
\frac{d^3N}{d^3k}=\frac{1}{(2\pi)^3}\int_{w,w'}e^{-ik(w'-w)}\,\langle 0|\,\Omega^\dagger\bar\Omega\Bigg[&\int dq^{+\prime}\int d^2v'\,|g_{n'}^{f'}(q^{+\prime},v')\rangle\,\langle g_{n'}^{f'}(q^{+\prime},v')|+\frac{1}{2}\int dq^{+\prime}dr^{+\prime}\int d^2v'd^2u'\\
&|g_{n'}^{f'}(q^{+\prime},v'),g_{m'}^{e'}(r^{+\prime},u')\rangle\,\langle g_{n'}^{f'}(q^{+\prime},v'),g_{m'}^{e'}(r^{+\prime},u')|\Bigg]\bar a_i^{a\dagger}(k^+,w)\bar a_i^a(k^+,w')\Bigg[\int dq^+\int d^2v\,|g_n^f(q^+,v)\rangle\,\langle g_n^f(q^+,v)|\\
&+\frac{1}{2}\int dq^+dr^+\int d^2vd^2u\,|g_n^f(q^+,v),g_m^e(r^+,u)\rangle\,\langle g_n^f(q^+,v),g_m^e(r^+,u)|\Bigg]\bar\Omega^\dagger\Omega\,|0\rangle
\end{aligned}
\tag{4.5}
$$

We can write this as

$$
\frac{d^3N}{d^3k}=\frac{1}{(2\pi)^3}\Bigg[\underbrace{\mathcal{A}_i^{\dagger a}(k)\mathcal{A}_i^a(k)}_{\text{Group I}}+\int dq^+\int_v\underbrace{\mathcal{A}_{ij}^{\dagger ab}(k;q^+,v)\mathcal{A}_{ij}^{ab}(k;q^+,v)}_{\text{Group II}}\Bigg]
\tag{4.6}
$$

where

$$
\mathcal{A}_i^a(k)=\sqrt{2\pi}\int d^2w'\,e^{-ikw'}U^{ab}(w')\,\langle g_i^b(k^+,w')|\,\bar\Omega^\dagger\Omega\,|0\rangle
\tag{4.7}
$$

is the amplitude corresponding to one gluon at the final state and

$$
\mathcal{A}_{ij}^{ab}(k;q^+,v)=\sqrt{2\pi}\int d^2w'\,e^{-ikw'}U^{ac}(w')\,\langle g_j^b(q^+,v)g_i^c(k^+,w')|\,\bar\Omega^\dagger\Omega\,|0\rangle
\tag{4.8}
$$

is the amplitude corresponding to two gluons at the final state. The scattering matrix operator present in $\bar\Omega$ rotates the soft and valence degrees of freedom according to

$$
\begin{aligned}
\bar\rho^a(x)&=S^\dagger\rho^a(x)S=U^{ab}(x)\rho^b(x)\,,\\
\bar a_i^a(k^+,x)&=S^\dagger a_i^a(k^+,x)S=U^{ab}(x)a_i^b(k^+,x)\,,\\
\bar a_i^{a\dagger}(k^+,x)&=S^\dagger a_i^{a\dagger}(k^+,x)S=U^{ab}(x)a_i^{b\dagger}(k^+,x)\,.
\end{aligned}
\tag{4.9}
$$

Ω (and $\bar\Omega$) has eight terms in total. However not all of the combinations of these terms would contribute either to $\mathcal{A}_i^a(k^+,k)$ or $\mathcal{A}_{ij}^{ab}(k^+,k)$. For example, Ω, since it acts on vacuum, can only lead to a non-trivial contribution the terms involving the coefficients $\mathbb{N}$, $\mathbb{A}^\dagger$, and $\mathbb{B}^\dagger$ are considered. Take for example contribution defined by $\mathbb{N}$, $\bar\Omega$ produces non-trivial result only if terms involving $\bar{\mathbb{A}}^\dagger$, and $\bar{\mathbb{B}}^\dagger$ are considered.

$$
\mathcal{A}_i^a=\sum_{n=1}^4\mathcal{A}_{ni}^a
\tag{4.10}
$$

$\mathcal{A}_{ni}^a$ are given below

- $\bar{\mathbb{A}}\mathbb{N}$

$$
\mathcal{A}^a_{1_ai}(k)=\int_{w'}e^{-ikw'}\left[\bar{\mathbb{A}}_{1i}^{\dagger\,(1)\,a}(k^+,w')+\bar{\mathbb{A}}_{1i}^{\dagger\,(1)\,a}(k^+,w')\mathbb{N}_2+\bar{\mathbb{A}}_{1i}^{\dagger\,(3)\,a}(k^+,w')\right]
\tag{4.11}
$$

- $\bar{\mathbb{N}}\mathbb{A}$

$$\mathcal{A}^{a}_{1_b i}(k) = \int_{w'} e^{-ikw'} U^{ab}(w') \Big[- \mathbb{A}^{\dagger\,(1)\,b}_{1i}(k^+, w') - \bar{\mathbb{N}}_2 \mathbb{A}^{\dagger\,(1)\,b}_{1i}(k^+, w') + \mathbb{A}^{(3)\,b}_{2i}(k^+, -w') \Big] \tag{4.12}$$

Combining eqs. (4.11) and (4.12) we get,

$$\mathcal{A}^{a}_{1i}(k) = \mathcal{A}^{a}_{1_a i}(k) + \mathcal{A}^{a}_{1_b i}(k)$$

$$\begin{aligned}\mathcal{A}^{a}_{1i}(k) = \int_{w'} e^{-ikw'} \Bigg\{ \Big[- U^{ab}(w') \mathbb{A}^{\dagger\,(1)\,b}_{1i}(k^+, w') + \bar{\mathbb{A}}^{\dagger\,(1)\,a}_{1i}(k^+, w') \Big] + \Big[- U^{ab}(w') \bar{N}_2 \mathbb{A}^{\dagger\,(1)\,b}_{1i}(k^+, w') \\ + \bar{\mathbb{A}}^{\dagger\,(1)\,a}_{1i}(k^+, w') \mathbb{N}_2 \Big] + \Big[U^{ab}(w') \mathbb{A}^{(3)\,b}_{2i}(k^+, -w') + \bar{\mathbb{A}}^{\dagger\,(3)\,a}_{1i}(k^+, w') \Big] \Bigg\}\end{aligned} \tag{4.13}$$

- $\bar{\mathbb{B}}_3\mathbb{A}$

$$\mathcal{A}^{a}_{2i}(k) = - \int \frac{dp^+}{2\pi} \int_{x,w'} e^{-ikw'} \bar{\mathbb{B}}^{\dagger ba}_{3ji}(k^+, w'; p^+, -x) U^{bc}(x) \mathbb{A}^{\dagger\,(1)\,c}_{1j}(p^+, x) \tag{4.14}$$

- $\bar{\mathbb{A}}\mathbb{B}_2$

$$\mathcal{A}^{a}_{3i}(k) = -2 \int \frac{dp^+}{2\pi} \int_{x,w'} e^{-ikw'}\, U^{ab}(w') U^{dc}(x) \bar{\mathbb{A}}^{(1)\,d}_{1j}(p^+, x) \mathbb{B}^{cb}_{2ji}(k^+, -w'; p^+, -x) \tag{4.15}$$

- $\bar{\mathbb{C}}\mathbb{B}_2$

$$\mathcal{A}^{a}_{4i}(k) = 2 \int \frac{dp^+}{2\pi} \int_{z,x,w'} e^{-ikw'}\, \bar{\mathbb{C}}^{\dagger\, dca}_{1jki}(k^+, w' - z; p^+, z - x) U^{db}(x) U^{ce}(z) \mathbb{B}^{be}_{2jk}(k^+ - p^+, -z; p^+, -x) \tag{4.16}$$

$$\mathcal{A}^{ab}_{i}(k) = \sum_{n=1}^{3} \mathcal{A}^{ab}_{ni} \tag{4.17}$$

Combining all the contributions we get,

$$\begin{aligned}\mathcal{A}^{a}_{i}(k) = \int_{w'} e^{-ikw'} \Bigg\{ & \Big[\bar{\mathbb{A}}^{\dagger\,(1)\,a}_{1i}(k^+, w') - U^{ab}(w') \mathbb{A}^{\dagger\,(1)\,b}_{1i}(k^+, w') \Big] + \Big[\bar{\mathbb{A}}^{\dagger\,(1)\,a}_{1i}(k^+, w') \mathbb{N}_2 - U^{ab}(w') \bar{\mathbb{N}}_2 \mathbb{A}^{\dagger\,(1)\,b}_{1i}(k^+, w') \Big] \\ & + \Big[U^{ab}(w') \mathbb{A}^{(3)\,b}_{2i}(k^+, -w') - \bar{\mathbb{A}}^{(3)\,a}_{2i}(k^+, -w') \Big] \\ & + \int \frac{dp^+}{2\pi} \int_{x} \Big[\bar{\mathbb{B}}^{\dagger\, ba}_{3ji}(k^+, w'; p^+, -x) \bar{\mathbb{A}}^{\dagger\,(1)\,b}_{1j}(p^+, x) - \bar{\mathbb{B}}^{\dagger ba}_{3ji}(k^+, w'; p^+, -x) U^{bc}(x) \mathbb{A}^{\dagger\,(1)\,c}_{1j}(p^+, x) \Big] \\ & + 2 \int \frac{dp^+}{2\pi} \int_{x} \Big[\bar{\mathbb{A}}^{(1)\,b}_{1j}(p^+, x) \bar{\mathbb{B}}^{ba}_{2ji}(k^+, -w';\, p^+, -x) - U^{ad}(w') U^{bc}(x) \bar{\mathbb{A}}^{(1)\,b}_{1j}(p^+, x) \mathbb{B}^{cd}_{2ji}(k^+, -w'; p^+, -x) \Big] \\ & + 2 \int \frac{dp^+}{2\pi} \int_{x,z} \Big[- \bar{\mathbb{C}}^{\dagger\, bca}_{1jki}(k^+, w' - z; p^+, z - x) \bar{\mathbb{B}}^{bc}_{2jk}(k^+ - p^+, -z; p^+, -x) \\ & \qquad + \bar{\mathbb{C}}^{\dagger\, dca}_{1jki}(k^+, w' - z; p^+, z - x) U^{db}(x) U^{ce}(z) \mathbb{B}^{be}_{2jk}(k^+ - p^+, -z; p^+, -x) \Big] \Bigg\}\end{aligned} \tag{4.18}$$

$$\begin{aligned}\mathcal{A}^{a}_{i}(k) = \int_{w'} e^{-ikw'} \Bigg\{ & \Big[- \bar{\mathbb{A}}^{(1)\,a}_{2i}(k^+, -w') + U^{ab}(w') \mathbb{A}^{(1)\,b}_{2i}(k^+, -w') \Big] \\ & + \Big[- \bar{\mathbb{A}}^{(1)\,a}_{2i}(k^+, -w') \mathbb{N}_2 + U^{ab}(w') \bar{\mathbb{N}}_2 \mathbb{A}^{(1)\,b}_{2i}(k^+, -w') \Big] \\ & + \Big[U^{ab}(w') \mathbb{A}^{(3)\,b}_{2i}(k^+, -w') - \bar{\mathbb{A}}^{(3)\,a}_{2i}(k^+, -w') \Big] \\ & + \int \frac{dp^+}{2\pi} \int_{x} \Big[- \bar{\mathbb{B}}^{\dagger\, ba}_{3ji}(k^+, w'; p^+, -x) \bar{\mathbb{A}}^{(1)\,b}_{2j}(p^+, -x) + \bar{\mathbb{B}}^{\dagger ba}_{3ji}(k^+, w'; p^+, -x) U^{bc}(x) \mathbb{A}^{(1)\,c}_{2j}(p^+, -x) \Big] \\ & + 2 \int \frac{dp^+}{2\pi} \int_{x} \Big[\bar{\mathbb{A}}^{(1)\,b}_{1j}(p^+, x) \bar{\mathbb{B}}^{ba}_{2ji}(k^+, -w';\, p^+, -x) - U^{ad}(w') U^{bc}(x) \bar{\mathbb{A}}^{(1)\,b}_{1j}(p^+, x) \mathbb{B}^{cd}_{2ji}(k^+, -w'; p^+, -x) \Big] \\ & + 2 \int \frac{dp^+}{2\pi} \int_{x,z} \Big[- \bar{\mathbb{C}}^{\dagger\, bca}_{1jki}(k^+, w' - z; p^+, z - x) \bar{\mathbb{B}}^{bc}_{2jk}(k^+ - p^+, -z; p^+, -x) \\ & \qquad + \bar{\mathbb{C}}^{\dagger\, dca}_{1jki}(k^+, w' - z; p^+, z - x) U^{db}(x) U^{ce}(z) \mathbb{B}^{be}_{2jk}(k^+ - p^+, -z; p^+, -x) \Big] \Bigg\}\end{aligned} \tag{4.19}$$

Group (II):

- $\bar{\mathbb{B}}_1\mathbb{N}$

$$\mathcal{A}^{ab}_{1_a ij}(k; q^+, v) = \frac{2}{\sqrt{2\pi}} \int_{w'} e^{-ikw'} U^{bc}(v) U^{\dagger bc}(v) \bar{\mathbb{B}}^{\dagger ca}_{1ji}(k^+, w'; q^+, v) \tag{4.20}$$

- $\bar{\mathbb{N}}\mathbb{B}_2$

$$\mathcal{A}^{ab}_{1_b ij}(k; q^+, v) = \frac{2}{\sqrt{2\pi}} \int_{w'} e^{-ikw'} U^{ac}(w') \mathbb{B}^{bc}_{2ji}(k^+, -w'; q^+, -v) \tag{4.21}$$

Combining eqs. (4.20) and (4.21) we get,

$$\mathcal{A}^{ab}_{1ij}(k) = \mathcal{A}^{ab}_{1_a ij}(k) + \mathcal{A}^{ab}_{1_b ij}(k)$$

$$\mathcal{A}^{ab}_{1ij}(k; q^+, v) = \frac{2}{\sqrt{2\pi}} \int_{w'} e^{-ikw'} \Big[U^{ac}(w') \mathbb{B}^{bc}_{2ji}(k^+, -w'; q^+, -v) + U^{bc}(v) U^{\dagger bc}(v) \bar{\mathbb{B}}^{\dagger ca}_{1ji}(k^+, w'; q^+, v) \Big] \tag{4.22}$$

- $\bar{\mathbb{A}}\mathbb{A}$

$$\mathcal{A}^{ab}_{2ij}(k; q^+, v) = - \frac{1}{\sqrt{2\pi}} \int_{w'} e^{-ikw'} \Big[U^{ac}(w') \bar{\mathbb{A}}^{\dagger\,(1)\,d}_{1j}(q^+, v) U^{db}(v) \mathbb{A}^{\dagger\,(1)\,c}_{1i}(k^+, w') + \bar{\mathbb{A}}^{\dagger\,(1)\,a}_{1i}(k^+, w') \mathbb{A}^{\dagger\,(1)\,b}_{1j}(q^+, v) \Big] \tag{4.23}$$

- $\bar{\mathbb{C}}\mathbb{A}$

$$\mathcal{A}^{ab}_{3ij}(k;q^+,v)=\frac{2}{\sqrt{2\pi}}\int_{w',z}e^{-ikw'}\bar{\mathbb{C}}^{dac}_{1jik}(q^++k^+,z-w';q^+,w'-v)U^{bd}(v)U^{\dagger bd}(v)U^{ce}(z)\mathbb{A}^{\dagger\,(1)\,e}_{1k}(q^++k^+,z) \tag{4.24}$$

$$\begin{aligned}\mathcal{A}^{ab}_{ij}(k;q^+,v)=\frac{2}{\sqrt{2\pi}}\int_{w'}e^{-ikw'}\Bigg\{&U^{ac}(w')\mathbb{B}^{bc}_{2ji}(k^+,-w';q^+,-v)-U^{bc}(v)U^{\dagger bc}(v)\bar{\mathbb{B}}^{ca}_{2ji}(k^+,-w';q^+,-v)\\&+\frac{1}{2}\Big[U^{\dagger\,bc}(v)\bar{\mathbb{A}}^{\dagger\,(1)\,c}_{1j}(q^+,v)\bar{\mathbb{A}}^{\dagger\,(1)\,a}_{1i}(k^+,w')-U^{ac}(w')\bar{\mathbb{A}}^{\dagger\,(1)\,d}_{1j}(q^+,v)U^{\dagger db}(v)U^{db}(v)\mathbb{A}^{\dagger\,(1)\,c}_{1i}(k^+,w')\Big]\\&+\frac{1}{2}\Big[U^{\dagger bc}(v)\bar{\mathbb{A}}^{\dagger\,(1)\,a}_{1i}(k^+,w')\bar{\mathbb{A}}^{\dagger\,(1)\,c}_{1j}(q^+,v)-\bar{\mathbb{A}}^{\dagger\,(1)\,a}_{1i}(k^+,w')\mathbb{A}^{\dagger\,(1)\,b}_{1j}(q^+,v)\Big]\\&+\int_z\Big[\bar{\mathbb{C}}^{dac}_{1jik}(q^++k^+,z-w';q^+,w'-v)U^{\dagger\,bd}(v)U^{ce}(z)\mathbb{A}^{\dagger\,(1)\,e}_{1k}(q^++k^+,z)\\&-U^{\dagger\,bd}(v)\bar{\mathbb{C}}^{dac}_{1jik}(q^++k^+,z-w';q^+,w'-v)\bar{\mathbb{A}}^{\dagger\,(1)\,c}_{1k}(q^++k^+,z)\Big]\Bigg\}\end{aligned} \tag{4.25}$$

$$\begin{aligned}\mathcal{A}^{ab}_{ij}(k;q^+,v)=\frac{2}{\sqrt{2\pi}}\int_{w'}e^{-ikw'}\Bigg\{&U^{ac}(w')\mathbb{B}^{bc}_{2ji}(k^+,-w';q^+,-v)-U^{\dagger\,bc}(v)\bar{\mathbb{B}}^{ca}_{2ji}(k^+,-w';q^+,-v)\\&+\frac{1}{2}\Big[U^{\dagger\,bc}(v)\bar{\mathbb{A}}^{(1)\,c}_{2j}(q^+,-v)\bar{\mathbb{A}}^{(1)\,a}_{2i}(k^+,-w')-U^{ac}(w')\bar{\mathbb{A}}^{(1)\,d}_{2j}(q^+,-v)U^{\dagger db}(v)U^{db}(v)\mathbb{A}^{(1)\,c}_{2i}(k^+,-w')\Big]\\&+\frac{1}{2}\Big[U^{\dagger bc}(v)\bar{\mathbb{A}}^{(1)\,a}_{2i}(k^+,-w')\bar{\mathbb{A}}^{(1)\,c}_{2j}(q^+,-v)-\bar{\mathbb{A}}^{(1)\,a}_{2i}(k^+,-w')\mathbb{A}^{(1)\,b}_{2j}(q^+,-v)\Big]\\&-\int_z\Big[\bar{\mathbb{C}}^{dac}_{1jik}(q^++k^+,z-w';q^+,w'-v)U^{\dagger\,bd}(v)U^{ce}(z)\mathbb{A}^{(1)\,e}_{2k}(q^++k^+,-z)\\&-U^{\dagger\,bd}(v)\bar{\mathbb{C}}^{dac}_{1jik}(q^++k^+,z-w';q^+,w'-v)\bar{\mathbb{A}}^{(1)\,c}_{2k}(q^++k^+,-z)\Big]\Bigg\}\end{aligned} \tag{4.26}$$

From here onward, we will denote $\mathbb{A}_2$ as $\mathbb{A}$ and $\mathbb{A}^\dagger_2$ as $\mathbb{A}^\dagger$. We compute the production cross-section using the above amplitudes. It is calculated to order g^4. It is as follows:

$$\frac{d^3N}{d^3k}=\underbrace{\frac{d^3N_{\rm LO}}{d^3k}}_{\rm Order(g^2)}+\underbrace{\frac{d^3N^{\rm (I)}_{\rm NLO}}{d^3k}+\frac{d^3N^{\rm (II)}_{\rm NLO}}{d^3k}}_{\rm Order(g^4)} \tag{4.27}$$

4.1 Production cross section at the leading order

The amplitude corresponding to leading-order production is

$$\mathcal{A}^a_i(k)=\int_{w'}e^{-ikw'}\Big[-U^{ab}(w')\mathbb{A}^{\dagger\,(1)\,b}_{1i}(k^+,w')+\bar{\mathbb{A}}^{\dagger\,(1)\,a}_{1i}(k^+,w')\Big] \tag{4.28}$$

and leading order (g^2) production cross-section is given by

$$\frac{d^3N_{\rm LO}}{d^3k}=\frac{1}{(2\pi)^3}\int_{w,w'}e^{-ik(w'-w)}\Big[-\mathbb{A}^{(1)\,c}_{1i}(k^+,w)U^{\dagger ca}(w)+\bar{\mathbb{A}}^{(1)\,a}_{1i}(k^+,w)\Big]\times\Big[-U^{ab}(w')\mathbb{A}^{\dagger\,(1)\,b}_{1i}(k^+,w')+\bar{\mathbb{A}}^{\dagger\,(1)\,a}_{1i}(k^+,w')\Big] \tag{4.29}$$

where,

$$\bar{\mathbb{A}}^{(1)\,a}_{1i}(k^+,w')=\frac{ig}{\sqrt{2k^+\pi}}\int_{x'}\frac{(x'-w')^i}{(x'-w')^2}U^{ab}(x')\rho^b(x') \tag{4.30}$$

$$\frac{d^3N_{\rm LO}}{d^3k}=\frac{\alpha_s}{4\pi^4(N_c^2-1)\,k^+}\int_{x,w,x',w'}e^{-ik(w'-w)}\frac{(x-w)}{(x-w)^2}\cdot\frac{(x'-w')}{(x'-w')^2}\langle\rho^b(x)\rho^b(x')\rangle{\rm Tr}\Big\{\left[U^\dagger(x)-U^\dagger(w)\right]\left[U(x')-U(w')\right]\Big\} \tag{4.31}$$

4.2 Production cross section at the Next to leading order

The NLO part (g^4) is split into two groups. The first (second) group consists of the contributions arising from the terms A proportional to the vacuum (single gluons) state:

$$\begin{aligned}\frac{d^3N^{\rm (I)}_{\rm NLO}}{d^3k}=&\frac{1}{(2\pi)^3}\int_{w,w'}e^{-ik(w'-w)}\\&\times\Bigg\{\Big[-\bar{\mathbb{A}}^{\dagger\,(1)\,a}_i(k^+,-w')+\mathbb{A}^{\dagger\,(1)\,b'}_i(k^+,-w')U^{b'a}(w')\Big]\times\Big[-\bar{\mathbb{A}}^{(1)\,a}_i(k^+,-w)\mathbb{N}_2+U^{ab}(w)\bar{\mathbb{N}}_2\mathbb{A}^{(1)\,b}_i(k^+,-w)\Big]\\&+\Big[-\bar{\mathbb{A}}^{\dagger\,(1)\,a}_i(k^+,-w')+\mathbb{A}^{\dagger\,(1)\,b'}_i(k^+,-w')U^{b'a}(w')\Big]\times\Big[U^{ab}(w)\mathbb{A}^{(3)\,b}_i(k^+,-w)-\bar{\mathbb{A}}^{(3)\,a}_i(k^+,-w)\Big]\\&+\int\frac{dp^+}{2\pi}\int_x\Big[-\bar{\mathbb{A}}^{\dagger\,(1)\,a}_i(k^+,-w')+\mathbb{A}^{\dagger\,(1)\,b'}_i(k^+,-w')U^{b'a}(w')\Big]\times\\&\quad\Big[-\bar{\mathbb{B}}^{\dagger\,ba}_{3ji}(k^+,w;p^+,-x)\bar{\mathbb{A}}^{(1)\,b}_j(p^+,-x)+\bar{\mathbb{B}}^{\dagger\,ba}_{3ji}(k^+,w;p^+,-x)U^{bc}(x)\mathbb{A}^{(1)\,c}_j(p^+,-x)\Big]\\&+2\int\frac{dp^+}{2\pi}\int_x\Big[-\bar{\mathbb{A}}^{\dagger\,(1)\,a}_i(k^+,-w')+\mathbb{A}^{\dagger\,(1)\,b'}_i(k^+,-w')U^{b'a}(w')\Big]\times\\&\quad\Big[U^{ac}(w)U^{bd}(x)\bar{\mathbb{A}}^{\dagger\,(1)\,b}_j(p^+,-x)\mathbb{B}^{cd}_{2ji}(k^+,-w;p^+,-x)-\bar{\mathbb{A}}^{\dagger\,(1)\,b}_j(p^+,-x)\bar{\mathbb{B}}^{ba}_{2ji}(k^+,-w;\,p^+,-x)\Big]\\&+2\int\frac{dp^+}{2\pi}\int_{x,z}\Big[-\bar{\mathbb{A}}^{\dagger\,(1)\,a}_i(k^+,-w')+\mathbb{A}^{\dagger\,(1)\,b'}_i(k^+,-w')U^{b'a}(w')\Big]\times\\&\quad\Big[-\bar{\mathbb{C}}^{\dagger\,bca}_{1jki}(k^+,w-z;p^+,z-x)\bar{\mathbb{B}}^{bc}_{2jk}(k^+-p^+,-z;p^+,-x)+\bar{\mathbb{C}}^{\dagger\,dca}_{1jki}(k^+,w-z;p^+,z-x)\\&\quad U^{db}(x)U^{ce}(z)\mathbb{B}^{be}_{2jk}(k^+-p^+,-z;p^+,-x)\Big]\Bigg\}+\text{c.c.}\end{aligned} \tag{4.32}$$

Correction. (G4) The red/blue Wilson-line placements below matter only for a general adjoint U ($U^\dagger=U^T\neq U$); with the convention $U^{bc}=U^{cb}$ both forms are equal.

$$\frac{d^3N^{\rm (II)}_{\rm NLO}}{d^3k}=\frac{4}{(2\pi)^3}\int\frac{dp^+}{2\pi}\int_{z,w,w'}e^{-ik(w'-w)}$$

$$
\Bigg\{\Bigg[\mathbb{B}^{\dagger\,bc'}_{2ji}(k^+,-w';p^+,-z)U^{c'a}(w')U^{ac'}(w')-\bar{\mathbb{B}}^{\dagger\,c'a}_{2ji}(k^+,-w';p^+,-z)U^{bc'}(z)U^{c'b}(z)\Bigg]\times
$$

$$
\Bigg[U^{ac}(w)\mathbb{B}^{bc}_{2ji}(k^+,-w;p^+,-z)-U^{\dagger\,bc}(z)\bar{\mathbb{B}}^{ca}_{2ji}(k^+,-w;p^+,-z)\Bigg]
$$

$$
+\frac{1}{4}\Bigg[\bar{\mathbb{A}}^{\dagger\,(1)\,a}_{i}(k^+,-w')\bar{\mathbb{A}}^{\dagger\,(1)\,c'}_{j}(p^+,-z)U^{c'b}(z)-\mathbb{A}^{\dagger\,(1)\,c'}_{i}(k^+,-w')U^{bd'}(z)U^{d'b}(z)\bar{\mathbb{A}}^{\dagger\,(1)\,d'}_{j}(p^+,-z)U^{c'a}(w')U^{ac'}(w')\Bigg]\times
$$

$$
\Bigg[U^{bc}(z)U^{cb}(z)\bar{\mathbb{A}}^{(1)\,c}_{j}(p^+,-z)\bar{\mathbb{A}}^{(1)\,a}_{i}(k^+,-w)-U^{ac}(w)\bar{\mathbb{A}}^{(1)\,d}_{j}(p^+,-z)U^{db}(z)\mathbb{A}^{(1)\,c}_{i}(k^+,-w)\Bigg]
$$

$$
+\frac{1}{4}\Bigg[\bar{\mathbb{A}}^{\dagger\,(1)\,c'}_{j}(p^+,-z)\bar{\mathbb{A}}^{\dagger\,(1)\,a}_{i}(k^+,-w')U^{bc'}(z)U^{c'b}(z)-\mathbb{A}^{\dagger\,(1)\,b}_{j}(p^+,-z)\bar{\mathbb{A}}^{\dagger\,(1)\,a}_{i}(k^+,-w')\Bigg]\times
$$

$$
\Bigg[U^{bc}(z)U^{cb}(z)\bar{\mathbb{A}}^{(1)\,a}_{i}(k^+,-w)\bar{\mathbb{A}}^{(1)\,c}_{j}(p^+,-z)-\bar{\mathbb{A}}^{(1)\,a}_{i}(k^+,-w)\mathbb{A}^{(1)\,b}_{j}(p^+,-z)\Bigg]
$$

$$
+\int_{y,y'}\Bigg[\mathbb{A}^{\dagger\,(1)\,e'}_{k'}(p^++k^+,-y')U^{\dagger\,e'c'}(y')U^{d'b}(z)\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^++k^+,y'-w';p^+,w'-z)-\bar{\mathbb{A}}^{\dagger\,(1)\,c'}_{k'}(p^++k^+,-y')
$$

$$
\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^++k^+,y'-w';p^+,w'-z)U^{d'b}(z)\Bigg]\times\Bigg[\bar{\mathbb{C}}^{dac}_{1jik}(p^++k^+,y-w;p^+,w-z)U^{bd}(z)U^{db}(z)U^{ce}(y)\mathbb{A}^{(1)\,e}_{k}(p^++k^+,-y)
$$

$$
-U^{bd}(z)U^{db}(z)\bar{\mathbb{C}}^{dac}_{1jik}(p^++k^+,y-w;p^+,w-z)\bar{\mathbb{A}}^{(1)\,c}_{k}(p^++k^+,-y)\Bigg]\Bigg\}
$$

$$
+\Bigg\{\frac{4}{(2\pi)^3}\int\frac{dp^+}{2\pi}\int_{z,w,w'}e^{-ik(w'-w)}
$$

$$
\Bigg\{\frac{1}{2}\Bigg[\bar{\mathbb{A}}^{\dagger\,(1)\,a}_{i}(k^+,-w')\bar{\mathbb{A}}^{\dagger\,(1)\,c'}_{j}(p^+,-z)U^{c'b}(z)-\mathbb{A}^{\dagger\,(1)\,c'}_{i}(k^+,-w')U^{bd'}(z)U^{d'b}(z)\bar{\mathbb{A}}^{\dagger\,(1)\,d'}_{j}(p^+,-z)U^{c'a}(w')U^{ac'}(w')\Bigg]\times
$$

$$
\Bigg[U^{ac}(w)\mathbb{B}^{bc}_{2ji}(k^+,-w;p^+,-z)-U^{\dagger\,bc}(z)\bar{\mathbb{B}}^{ca}_{2ji}(k^+,-w;p^+,-z)\Bigg]
$$

$$
+\frac{1}{2}\Bigg[\bar{\mathbb{A}}^{\dagger\,(1)\,c'}_{j}(p^+,-z)\bar{\mathbb{A}}^{\dagger\,(1)\,a}_{i}(k^+,-w')U^{bc'}(z)U^{c'b}(z)-\mathbb{A}^{\dagger\,(1)\,b}_{j}(p^+,-z)\bar{\mathbb{A}}^{\dagger\,(1)\,a}_{i}(k^+,-w')\Bigg]\times
$$

$$
\Bigg[U^{ac}(w)\mathbb{B}^{bc}_{2ji}(k^+,-w;p^+,-z)-U^{\dagger\,bc}(z)\bar{\mathbb{B}}^{ca}_{2ji}(k^+,-w;p^+,-z)\Bigg]
$$

$$
-\int_{y'}\Bigg[\mathbb{A}^{\dagger\,(1)\,e'}_{k'}(p^++k^+,-y')U^{\dagger\,e'c'}(y')U^{d'b}(z)\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^++k^+,y'-w';p^+,w'-z)-\bar{\mathbb{A}}^{\dagger\,(1)\,c'}_{k'}(p^++k^+,-y')
$$

$$
\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^++k^+,y'-w';p^+,w'-z)U^{d'b}(z)\Bigg]\times\Bigg[U^{ac}(w)\mathbb{B}^{bc}_{2ji}(k^+,-w;p^+,-z)-U^{\dagger\,bc}(z)\bar{\mathbb{B}}^{ca}_{2ji}(k^+,-w;p^+,-z)\Bigg]
$$

$$
+\frac{1}{4}\Bigg[\bar{\mathbb{A}}^{\dagger\,(1)\,c'}_{j}(p^+,-z)\bar{\mathbb{A}}^{\dagger\,(1)\,a}_{i}(k^+,-w')U^{bc'}(z)U^{c'b}(z)-\mathbb{A}^{\dagger\,(1)\,b}_{j}(p^+,-z)\bar{\mathbb{A}}^{\dagger\,(1)\,a}_{i}(k^+,-w')\Bigg]\times
$$

$$
\Bigg[U^{bc}(z)U^{cb}(z)\bar{\mathbb{A}}^{(1)\,c}_{j}(p^+,-z)\bar{\mathbb{A}}^{(1)\,a}_{i}(k^+,-w)-U^{ac}(w)\bar{\mathbb{A}}^{(1)\,d}_{j}(p^+,-z)U^{db}(z)\mathbb{A}^{(1)\,c}_{i}(k^+,-w)\Bigg]
$$

$$
-\frac{1}{2}\int_{y'}\Bigg[\mathbb{A}^{\dagger\,(1)\,e'}_{k'}(p^++k^+,-y')U^{\dagger\,e'c'}(y')U^{d'b}(z)\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^++k^+,y'-w';p^+,w'-z)-\bar{\mathbb{A}}^{\dagger\,(1)\,c'}_{k'}(p^++k^+,-y')
$$

$$
\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^++k^+,y'-w';p^+,w'-z)U^{d'b}(z)\Bigg]\times\Bigg[U^{bc}(z)U^{cb}(z)\bar{\mathbb{A}}^{(1)\,c}_{j}(p^+,-z)\bar{\mathbb{A}}^{(1)\,a}_{i}(k^+,-w)
$$

$$
-U^{ac}(w)\bar{\mathbb{A}}^{(1)\,d}_{j}(p^+,-z)U^{db}(z)\mathbb{A}^{(1)\,c}_{i}(k^+,-w)\Bigg]
$$

$$
-\frac{1}{2}\int_{y'}\Bigg[\mathbb{A}^{\dagger\,(1)\,e'}_{k'}(p^++k^+,-y')U^{\dagger\,e'c'}(y')U^{d'b}(z)\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^++k^+,y'-w';p^+,w'-z)-\bar{\mathbb{A}}^{\dagger\,(1)\,c'}_{k'}(p^++k^+,-y')
$$

$$
\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^++k^+,y'-w';p^+,w'-z)U^{d'b}(z)\Bigg]\times\Bigg[U^{bc}(z)U^{cb}(z)\bar{\mathbb{A}}^{(1)\,a}_{i}(k^+,-w)\bar{\mathbb{A}}^{(1)\,c}_{j}(p^+,-z)
$$

$$
-\bar{\mathbb{A}}^{(1)\,a}_{i}(k^+,-w)\mathbb{A}^{(1)\,b}_{j}(p^+,-z)\Bigg]\Bigg\}+\text{c.c.}\Bigg\}
\tag{4.33}
$$

A State definitions and normalization

Following conventional notation, $|0\rangle$ denotes the vacuum state of the free Hamiltonian on the soft Hilbert space. The energy of this state will be set to zero, $E_0 = 0$. We define the vacuum state as

$$a_i^a(k^+, \mathbf{k})\,|0\rangle = 0, \qquad b_i^\alpha(k^+, \mathbf{k})\,|0\rangle = 0, \qquad d_i^\alpha(k^+, \mathbf{k})\,|0\rangle = 0, \tag{A.1}$$

for any $\Lambda < k^+ < \vee$. Additionally, we will use the following three eigenstates of free Hamiltonian:

- One gluon state:

$$|g_i^a(k)\rangle \equiv \frac{a_i^{a\dagger}(k)}{(2\pi)^{3/2}}\,|0\rangle \tag{A.2}$$

or

$$\left|g_i^a(k^+, \mathbf{z})\right\rangle \equiv \frac{a_i^{a\dagger}(k^+, \mathbf{z})}{(2\pi)^{1/2}}\,|0\rangle \tag{A.3}$$

Note that $|g_i^a(k^+, \mathbf{z})\rangle$ is different from a simple Fourier transform by a factor of 2π:

$$|g(k^+, \mathbf{z})\rangle = 2\pi \int \frac{d^2k}{(2\pi)^2} e^{-i\mathbf{kz}} |g(k)\rangle\,. \tag{A.4}$$

The normalization of the state is

$$\left\langle g_j^b(p)\,|g_i^a(k)\,\right\rangle = \delta^{ab}\delta_{ij}\delta^{(2)}(\mathbf{k}-\mathbf{p})\delta(k^+ - p^+)$$

and

$$\left\langle g_j^b(p^+, \mathbf{z}')\,\left|g_i^a(k^+, \mathbf{z})\right.\right\rangle = \delta^{ab}\,\delta_{ij}\,\delta^{(2)}(\mathbf{z}-\mathbf{z}')\,\delta(k^+ - p^+).$$

Thus

$$a_i^a(k^+, \mathbf{k})\left|g_j^b(q)\right\rangle = (2\pi)^{3/2}\delta^{ab}\delta_{ij}\delta^{(3)}(k-q)\,|0\rangle \tag{A.5}$$

For completeness we note $E_g(k) = \frac{\mathbf{k}^2}{2k^+}$.

- Two gluon state:

$$\begin{aligned}\left|g_i^a(k)\,g_j^b(p)\right\rangle &\equiv \frac{a_i^{a\dagger}(k)\,a_j^{b\dagger}(p)}{(2\pi)^3}\,|0\rangle \\ \left|g_i^a(k^+, \mathbf{z})\,g_j^b(p^+, \mathbf{z}')\right\rangle &\equiv \frac{a_i^{a\dagger}(k^+, \mathbf{z})\,a_j^{b\dagger}(p^+, \mathbf{z}')}{2\pi}\,|0\rangle\,.\end{aligned} \tag{A.6}$$

with the energy $E_{gg}(k, p) \equiv \frac{\mathbf{k}^2}{2k^+} + \frac{\mathbf{p}^2}{2p^+}$.

- Quark anti-quark state:

$$\begin{aligned}\left|\bar{q}_{\lambda_2}^\beta(p)\,q_{\lambda_1}^\alpha(k)\right\rangle &\equiv \frac{b_{\lambda_1}^{\alpha\dagger}(k)\,d_{\lambda_2}^{\beta\dagger}(p)}{(2\pi)^3}\,|0\rangle \\ \left|\bar{q}_{\lambda_2}^\beta(p^+, \mathbf{z}')\,q_{\lambda_1}^\alpha(k^+, \mathbf{z})\right\rangle &\equiv \frac{b_{\lambda_1}^{\alpha\dagger}(k^+, \mathbf{z})\,d_{\lambda_2}^{\beta\dagger}(p^+, \mathbf{z}')}{2\pi}\,|0\rangle\,,\end{aligned} \tag{A.7}$$

with the energy $E_{q\bar{q}}(k, p) = \frac{\mathbf{k}^2}{2k^+} + \frac{\mathbf{p}^2}{2p^+}$.

B The CGC Hamiltonian: Eikonal Approximation for QCD Hamiltonian

In order to introduce the eikonal approximation for the LC QCD Hamiltonian, we split the gluon fields into the soft and valence modes:

$$\underline{A}_i^a(x) = \int_\Lambda^{e^{\delta Y}\Lambda} \frac{dk^+}{2\pi} \int \frac{d^2\mathbf{k}}{(2\pi)^2} \frac{1}{\sqrt{2k^+}} \left(a_i^a(k^+,\mathbf{k}) e^{-ik\cdot x} + a_i^{a\dagger}(k^+,\mathbf{k}) e^{ik\cdot x} \right), \tag{B.1}$$

and

$$\overline{A}_i^a(x) = \int_{e^{\delta Y}\Lambda}^{\infty} \frac{dk^+}{2\pi} \int \frac{d^2\mathbf{k}}{(2\pi)^2} \frac{1}{\sqrt{2k^+}} \left(a_i^a(k^+,\mathbf{k}) e^{-ik\cdot x} + a_i^{a\dagger}(k^+,\mathbf{k}) e^{ik\cdot x} \right). \tag{B.2}$$

$$H_{int} = H_g + H_{gqq} + H_{ggg} + H_{gggg} + H_{ggqq} + H_{qq-inst} + H_{gg-inst} + H_{gq} + H_V. \tag{B.3}$$

After inserting $A_i^a(x) = \underline{A}_i^a(x) + \overline{A}_i^a(x)$ to H_{int}, we define the following components of the interaction Hamiltonian:

$$H_g \equiv -g \int dx^-\, d^2\mathbf{x}\, (\partial_i \underline{A}_i^a) \left(f^{abc} \frac{1}{\partial^+} (\overline{A}_j^b \partial^+ \overline{A}_j^c) + 2 \frac{1}{\partial^+} (\overline{\Psi}_+^\dagger t^a \overline{\Psi}_+) \right), \tag{B.4}$$

$$H_{ggg} \equiv -g f^{abc} \int dx^-\, d^2\mathbf{x} \left((\partial_i \underline{A}_j^a) \underline{A}_i^b \underline{A}_j^c + (\partial_i \underline{A}_i^a) \frac{1}{\partial^+} (\underline{A}_j^b \partial^+ \underline{A}_j^c) \right), \tag{B.5}$$

$$H_{gg-inst} \equiv g^2 f^{abc} \int dx^-\, d^2\mathbf{x}\, \frac{1}{\partial^+} (\underline{A}_i^b \partial^+ \underline{A}_i^c) \left(f^{ade} \frac{1}{\partial^+} (\overline{A}_j^d \partial^+ \overline{A}_j^e) + 2 \frac{1}{\partial^+} (\overline{\Psi}_+^\dagger t^a \overline{\Psi}_+) \right), \tag{B.6}$$

$$H_{gggg} \equiv \frac{g^2}{4} f^{abc} f^{ade} \int dx^-\, d^2\mathbf{x} \left(\underline{A}_i^b \underline{A}_j^c \underline{A}_i^d \underline{A}_j^e - 2 \frac{1}{\partial^+} (\underline{A}_i^b \partial^+ \underline{A}_i^c) \frac{1}{\partial^+} (\underline{A}_j^d \partial^+ \underline{A}_j^e) \right), \tag{B.7}$$

For operational purposes, we require the interacting Hamiltonian to be written in terms of its creation and annihilation operators. Thus, the interacting Hamiltonian in oscillator modes is as follows:

$$\mathsf{H}_g = \int_\Lambda^{e^{\delta Y}\Lambda} \frac{dk^+}{2\pi} \int \frac{d^2\mathbf{k}}{(2\pi)^2} \frac{g\mathbf{k}^i}{\sqrt{2}|k^+|^{3/2}} \left[a_i^{a\dagger}(k^+,\mathbf{k}) \rho^a(-\mathbf{k}) + a_i^a(k^+,\mathbf{k}) \rho^a(\mathbf{k}) \right], \tag{B.8}$$

$$\begin{aligned} \mathsf{H}_{g\,gg} = -\int_\Lambda^{e^{\delta Y}\Lambda} \frac{dk^+}{2\pi} \frac{dp^+}{2\pi} dq^+ \int \frac{d^2\mathbf{k}}{(2\pi)^2} \frac{d^2\mathbf{p}}{(2\pi)^2} d^2\mathbf{q}\, \frac{igf^{abc}}{2\sqrt{2k^+p^+q^+}} \\ \times \Bigg[\left(\mathbf{q}^i - \frac{q^+}{p^+ + q^+} \mathbf{k}^i \right) a_i^a(k) a_j^{b\dagger}(p) a_j^{c\dagger}(q) \delta^{(3)}(-k+p+q) \\ + \left(\mathbf{p}^i + \mathbf{q}^i + \frac{p^+ + q^+}{q^+ - p^+} \mathbf{k}^i \right) a_i^{a\dagger}(k) a_j^{b\dagger}(p) a_j^c(q) \delta^{(3)}(k+p-q), + h.c. \Bigg] \end{aligned} \tag{B.9}$$

$$\begin{aligned} \mathsf{H}_{gg-inst} \quad = \int_\Lambda^{e^{\delta Y}\Lambda} \frac{dp^+}{2\pi} \frac{dq^+}{2\pi} \int \frac{d^2\mathbf{p}}{(2\pi)^2} \frac{d^2\mathbf{q}}{(2\pi)^2} \frac{ig^2 f^{abc}}{\sqrt{p^+q^+}} \left(\frac{q^+ \rho^a(-\mathbf{p}-\mathbf{q})}{2(p^++q^+)^2}\, a_i^{b\dagger}(p^+,\mathbf{p})\, a_i^{c\dagger}(q^+,\mathbf{q}) \right. \\ \left. - h.c. - \frac{(p^++q^+)\rho^a(-\mathbf{p}+\mathbf{q})}{2(p^+-q^+)^2}\, a_i^{b\dagger}(p^+,\mathbf{p})\, a_i^c(q^+,\mathbf{q}) \right), \end{aligned} \tag{B.10}$$

C Matrix elements

- Gluon emission from valence ρ

$$\langle g_i^a(k)\,|\,H_g\,|\,0\rangle = \frac{gk^i\rho^a(-\mathbf{k})}{4\pi^{3/2}|k^+|^{3/2}} \tag{C.1}$$

- Valence adds an extra gluon to an existing one

$$\left\langle g_l^c(q)\,g_j^b(p)\,|H_g|\,g_i^a(k)\right\rangle \,=\, \delta^{ac}\delta_{il}\delta^{(3)}(k-q)\,\frac{gp^j\rho^b(-\mathbf{p})}{4\pi^{3/2}|p^+|^{3/2}} \,+\, \delta^{ab}\delta_{ij}\delta^{(3)}(k-p)\,\frac{gq^l\rho^c(-\mathbf{q})}{4\pi^{3/2}|q^+|^{3/2}} \tag{C.2}$$

- Three gluon interaction

$$\begin{aligned}\left\langle g_l^c(q)\,g_j^b(p)\,|H_{ggg}|\,g_i^a(k)\right\rangle = \frac{igf^{abc}\delta^{(3)}(k-p-q)}{8\pi^{3/2}\sqrt{k^+p^+q^+}}\Bigg[\left(p^i-q^i+\frac{q^+-p^+}{k^+}k^i\right)\delta_{jl}+\left(k^j+q^j-\frac{k^++q^+}{p^+}p^j\right)\delta_{il}\\ +\left(\frac{k^++p^+}{q^+}q^l-p^l-k^l\right)\delta_{ij}\Bigg]\end{aligned} \tag{C.3}$$

- Valence creates two gluons instantaneously

$$\left\langle g_j^c(q)\,g_i^b(p)\,|H_{gg-inst}|\,0\right\rangle \,=\, \frac{ig^2f^{abc}(q^+-p^+)\delta_{ij}\rho^a(-\mathbf{p}-\mathbf{q})}{2(2\pi)^3\sqrt{p^+q^+}(p^++q^+)^2} \tag{C.4}$$

- Instantaneous interaction between valence and gluon

$$\left\langle g_j^c(q)\,|H_{gg-inst}|\,g_i^b(p)\right\rangle \,=\, \frac{ig^2f^{abc}(p^++q^+)\delta_{ij}\rho^a(\mathbf{p}-\mathbf{q})}{2(2\pi)^3\sqrt{p^+q^+}(p^+-q^+)^2} \tag{C.5}$$

D Fourier Transforms

The list of necessary integrals is presented below. Some of the integrals are defined as Cauchy principal value.

$$\int\frac{d^2k}{2\pi}\,\frac{k^i}{\mathbf{k}^2}e^{i\mathbf{k}\cdot X}=\frac{iX^i}{X^2}, \tag{D.1}$$

$$\int\frac{d^2k}{2\pi}\,\frac{d^2p}{2\pi}\,\frac{k^i\mathbf{p}^j}{\mathbf{k}^2\left(a\mathbf{k}^2+b\mathbf{p}^2\right)}e^{i\mathbf{k}\cdot X+i\mathbf{p}\cdot Z}=-\frac{X^iZ^j}{Z^2\left(bX^2+aZ^2\right)}, \tag{D.2}$$

$$\int\frac{d^2k}{2\pi}\,\frac{d^2p}{2\pi}\,\frac{1}{\xi\mathbf{k}^2-\mathbf{p}^2}e^{i\mathbf{k}\cdot X+i\mathbf{p}\cdot Z}=\frac{1}{\xi Z^2-X^2}, \tag{D.3}$$

$$\int\frac{d^2k}{2\pi}\,\frac{d^2p}{2\pi}\,\frac{k^m-p^m}{(\mathbf{k}-\mathbf{p})^2}\,\frac{p^l-\xi k^l}{\mathbf{p}^2-\xi\mathbf{k}^2}e^{i\mathbf{k}\cdot X+i\mathbf{p}\cdot Y}=-\frac{X^m+\xi Y^m}{X^2-\xi Y^2}\,\frac{X^l+Y^l}{(X+Y)^2}, \tag{D.4}$$

$$\int\frac{d^2k}{2\pi}\,\frac{d^2p}{2\pi}\,\frac{k^m-p^m}{\mathbf{p}^2-\xi\mathbf{k}^2}\,\frac{p^l-\xi k^l}{(\mathbf{p}-\xi\mathbf{k})^2}e^{i\mathbf{k}\cdot X+i\mathbf{p}\cdot Y}=-\frac{X^l+Y^l}{X^2-\xi Y^2}\,\frac{X^m+\xi Y^m}{(X+\xi Y)^2}, \tag{D.5}$$

$$\int\frac{d^2\mathbf{k}}{2\pi}\,\frac{d^2\mathbf{p}}{2\pi}\,\frac{1}{\xi(1-\xi)\mathbf{k}^2+\mathbf{p}^2}e^{i\mathbf{k}\cdot X+i\mathbf{p}\cdot Z}=\frac{1}{X^2+\xi(1-\xi)Z^2}, \tag{D.6}$$

$$\int\frac{d^2\mathbf{k}}{2\pi}\,\frac{k^i}{k^2}\ln\left(\frac{k^2}{\mu^2}\right)e^{i\mathbf{k}\cdot X}=\frac{iX^i}{X^2}\left(-2\gamma-\ln\left(\frac{X^2\mu^2}{4}\right)\right) \tag{D.7}$$

$$\begin{aligned}\int\frac{d^2k}{2\pi}\,d^2p\int_{x,y,z}e^{ik\cdot(z-x)}\,e^{ip\cdot(x-y)}\,\frac{k^i}{k^2\,p^2}\,\log\left[\frac{p^2}{k^2}\right]=-i\pi\,\frac{\log(x-y)^2}{(x-y)^2}\,\left[\frac{(z-x)^i}{(z-x)^2}-\frac{(z-y)^i}{(z-y)^2}\right]\\ +i\log(x-y)^2\,\left[\frac{(z-x)^i\,\log(z-x)^2}{(z-x)^2}-\frac{(z-y)^i\,\log(z-y)^2}{(z-y)^2}\right]\end{aligned} \tag{D.8}$$

$$\int d^2k\,d^2p\int_{x,z}e^{ik\cdot(x-z')}\,e^{ip\cdot(z-x)}\,\frac{(p^m-k^m)\,(p^l-\xi\,k^l)}{(p-k)^2\,(p^2-\xi\,k^2)} \tag{D.9}$$

E BFKL evolution in the coordinate representation

$$\begin{aligned}\frac{d}{dY}\langle\rho^a(\mathbf{x})\rho^a(\mathbf{y})\rangle=-\frac{\alpha_sN_c}{2\pi^2}\Big\langle K(\mathbf{x},\mathbf{x},\mathbf{z})\rho^a(\mathbf{x})\rho^a(\mathbf{y})-2K(\mathbf{x},\mathbf{y},\mathbf{z})\rho^a(\mathbf{x})\rho^a(\mathbf{y})+K(\mathbf{y},\mathbf{y},\mathbf{z})\rho^a(\mathbf{y})\rho^a(\mathbf{x})\\ -2\delta^2(\mathbf{x}-\mathbf{y})K(\mathbf{u},\mathbf{v},\mathbf{x})\rho^a(\mathbf{u})\rho^a(\mathbf{v})\\ +\left(2K(\mathbf{x},\mathbf{z},\mathbf{y})-K(\mathbf{z},\mathbf{z},\mathbf{y})\right)\rho^a(\mathbf{x})\rho^a(\mathbf{z})+\left(2K(\mathbf{y},\mathbf{z},\mathbf{x})-K(\mathbf{z},\mathbf{z},\mathbf{x})\right)\rho^a(\mathbf{y})\rho^a(\mathbf{z})\Big\rangle\end{aligned} \tag{E.1}$$

Here (be careful with the order of the arguments!)

$$K(x,y,z)=\frac{X\cdot Y}{X^2Y^2} \tag{E.2}$$

A more explicit form of the evolution

$$\begin{aligned}\frac{d\langle\rho^f(x)\rho^f(y)\rangle}{dY}=-\frac{\alpha_sN_c}{2\pi^2}\Bigg[\int d^2z(K(x,x,z)-2K(x,y,z)+K(y,y,z))\langle\rho^f(x)\rho^f(y)\rangle\\ -2\delta^2(x-y)\int d^2zd^2z'K(z,z',x)\langle\rho^f(z)\rho^f(z')\rangle\\ +\int d^2z\,(2K(y,z,x)-K(z,z,x))\,\langle\rho^f(y)\rho^f(z)\rangle\\ +\int d^2z\,(2K(x,z,y)-K(z,z,y))\,\langle\rho^f(x)\rho^f(z)\rangle\Bigg]\end{aligned} \tag{E.3}$$

It is convenient to introduce the dipole kernel

$$K_D(x,y,z)=K(x,x,z)-2K(x,y,z)+K(y,y,z)\,. \tag{E.4}$$

Applying the BFKL evolution to the leading order inclusive gluon production cross-section (the integration variables were renamed to get a common gluon distribution in the projectile)

$$\frac{dN}{d^2wd^2w'dy}=\frac{\alpha_s}{4\pi^4(N_c^2-1)}\int_{x,y}\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\langle\rho^f(x)\rho^f(y)\rangle\mathrm{Tr}\left\{(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\right\} \tag{E.5}$$

we get

$$\frac{dN}{d^2wd^2w'dy}=-\frac{\alpha_s^2N_c\delta Y}{8\pi^6(N_c^2-1)}\int_{x,y,z}\langle\rho^f(x)\rho^f(y)\rangle$$
$$\Big[\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')K_D(x,y,z)\mathrm{Tr}\left\{(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\right\}$$
$$-2\,\mathcal{K}(z-w)\cdot\mathcal{K}(z-w')K(x,y,z)\mathrm{Tr}\left\{(U^\dagger(z)-U^\dagger(w))(U(z)-U(w'))\right\}$$
$$+\mathcal{K}(z-w)\cdot\mathcal{K}(y-w')(2K(x,y,z)-K(x,x,z))\mathrm{Tr}\left\{(U^\dagger(z)-U^\dagger(w))(U(y)-U(w'))\right\}$$
$$+\mathcal{K}(x-w)\cdot\mathcal{K}(z-w')(2K(x,y,z)-K(y,y,z))\mathrm{Tr}\left\{(U^\dagger(x)-U^\dagger(w))(U(z)-U(w'))\right\}\Big] \tag{E.6}$$

F JIMWLK evolution of the leading order single gluons production cross section.

$$H_{\text{JIMWLK}}=\frac{\alpha_s}{2\pi^2}\int_{t,u,v}\mathcal{K}(u,v,t)[U^\dagger(u)-U^\dagger(t)]_{ce}J_R^c(u)[U(v)-U(t)]_{ed}J_R^d(v) \tag{F.1}$$

where

$$[J_R^a(z),U(x)]=\delta^{(2)}(x-z)U(x)T^a,\qquad [J_R^a(z),U^\dagger(x)]=-\delta^{(2)}(x-z)T^aU^\dagger(x)\,, \tag{F.2}$$

We thus can rewrite this in the following form

$$H_{\text{JIMWLK}}=\frac{\alpha_s}{2\pi^2}\int_{t,u,v}\mathcal{K}(u,v,t)[(U^\dagger(u)-U^\dagger(t))(U(v)-U(t))]_{cd}J_R^c(u)J_R^d(v)$$
$$+\frac{\alpha_s}{2\pi^2}\int_{t,u,v}\mathcal{K}(u,v,t)(\delta^2(u-v)-\delta^2(u-t))[(U^\dagger(u)-U^\dagger(t))U(u)T^c]_{cd}J_R^d(v) \tag{F.3}$$

In the last term, the second delta-function sets the positions of the Wilson lines to the same point leading to vanishing contribution. Thus we have

$$H_{\text{JIMWLK}}=\frac{\alpha_s}{2\pi^2}\int_{t,u,v}\mathcal{K}(u,v,t)[(U^\dagger(u)-U^\dagger(t))(U(v)-U(t))]_{cd}J_R^c(u)J_R^d(v)$$
$$-\frac{\alpha_s}{2\pi^2}\int_{t,u}\mathcal{K}(u,u,t)\mathrm{tr}[(U^\dagger(u)-U^\dagger(t))U(u)T^d]J_R^d(u) \tag{F.4}$$

where we took into account the properties of the adjoint generator. The last term will be considered at the end. For now we will concentrate on the first term.

The leading order gluon production cross section is given by Eq. (E.5).

We need to obtain $\mathcal{M}=H_{\text{JIMWLK}}\times\mathcal{O}_g$. Thus, the principal part

$$J_R^c(u)J_R^d(v)\mathrm{Tr}\Big[(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\Big] \tag{F.5}$$

can be computed by transferring both right rotation operators to the right of all Wilson lines with the help of the commutator identities.

$$[J^d(v),\mathrm{Tr}\left\{(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\right\}]=(\delta^2(y-v)-\delta^2(w'-v))\mathrm{Tr}\left\{(U^\dagger(x)-U^\dagger(w))U(v)T^d\right\}$$
$$-(\delta^2(x-v)-\delta^2(w-v))\mathrm{Tr}\left\{U^\dagger(v)(U(y)-U(w'))T^d\right\}. \tag{F.6}$$

and

$$[J^c(u),[J^d(v),\mathrm{Tr}\left\{(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\right\}]]=$$
$$-(\delta^2(x-v)-\delta^2(w-v))(\delta^2(y-u)-\delta^2(w'-u))\mathrm{Tr}\left\{U^\dagger(v)U(u)T^cT^d\right\}$$
$$-(\delta^2(y-v)-\delta^2(w'-v))(\delta^2(x-u)-\delta^2(w-u))\mathrm{Tr}\left\{U^\dagger(u)U(v)T^dT^c\right\}$$
$$+\delta^2(u-v)(\delta^2(y-v)-\delta^2(w'-v))\mathrm{Tr}\left\{(U^\dagger(x)-U^\dagger(w))U(v)T^cT^d\right\}$$
$$+\delta^2(u-v)(\delta^2(x-v)-\delta^2(w-v))\mathrm{Tr}\left\{U^\dagger(v)(U(y)-U(w'))T^dT^c\right\}. \tag{F.7}$$

The second terms can be obtained from the first by $u\Leftrightarrow v$ and $c\Leftrightarrow d$. We thus have to explicitly consider the first term. The symmetry of the last two terms is not transparent and we would have to compute both of them explicitly. Lets start with the first one.

We will utilize the following transformations quite often:

$$\mathrm{Tr}\left[WT^cT^d\right]M_{cd}=W_{ab}T^c_{be}T^d_{ea}M_{cd}=T^e_{da}W_{ab}T^e_{bc}M_{cd}=\mathrm{Tr}\left[T^eWT^eM\right] \tag{F.8}$$
$$\mathrm{Tr}\left[WT^dT^c\right]M_{cd}=\mathrm{Tr}\left[T^eWT^eM^T\right] \tag{F.9}$$

For the first term we get (note that $d/dY=-H$)

$$\frac{dN^{(1)}}{d^2wd^2w'dy}=\frac{\alpha_s^2\delta Y}{8\pi^6(N_c^2-1)}\int_{x,y,t}\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\langle\rho^f(x)\rho^f(y)\rangle$$
$$\times\Big[\mathcal{K}(y,x,t)\mathrm{Tr}\left\{T^eU^\dagger(x)U(y)T^e(U^\dagger(x)-U^\dagger(t))(U(y)-U(t))(U^\dagger(y)-U^\dagger(t))(U(x)-U(t))\right\}$$
$$-\mathcal{K}(w',x,t)\mathrm{Tr}\left\{T^eU^\dagger(x)U(w')T^e(U^\dagger(x)-U^\dagger(t))(U(y)-U(t))(U^\dagger(w')-U^\dagger(t))(U(x)-U(t))\right\}$$
$$-\mathcal{K}(y,w,t)\mathrm{Tr}\left\{T^eU^\dagger(w)U(y)T^e(U^\dagger(x)-U^\dagger(t))(U(y)-U(t))(U^\dagger(y)-U^\dagger(t))(U(w)-U(t))\right\}$$
$$+\mathcal{K}(w',w,t)\mathrm{Tr}\left\{T^eU^\dagger(w)U(w')T^e(U^\dagger(x)-U^\dagger(t))(U(y)-U(t))(U^\dagger(w')-U^\dagger(t))(U(w)-U(t))\right\}\Big] \tag{F.10}$$

The prefactor in front of the square bracket is invariant under $x\Leftrightarrow y$ and $w\Leftrightarrow w'$. This symmetry can be used to show that the second term equals the third. We do not perform this simplification here and below.

Performing calculations for the second contribution:

$$\frac{dN^{(2)}}{d^2wd^2w'dy}=\frac{\alpha_s^2\delta Y}{8\pi^6(N_c^2-1)}\int_{x,y,t}\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\langle\rho^f(x)\rho^f(y)\rangle$$
$$\times\Big[\mathcal{K}(x,y,t)\mathrm{Tr}\left\{T^eU^\dagger(x)U(y)T^e(U^\dagger(y)-U^\dagger(t))(U(x)-U(t))\right\}$$
$$-\mathcal{K}(x,w',t)\mathrm{Tr}\left\{T^eU^\dagger(x)U(w')T^e(U^\dagger(y)-U^\dagger(t))(U(x)-U(t))(U^\dagger(w')-U^\dagger(t))(U(x)-U(t))\right\}$$
$$-\mathcal{K}(w,y,t)\mathrm{Tr}\left\{T^eU^\dagger(w)U(y)T^e(U^\dagger(y)-U^\dagger(t))(U(x)-U(t))(U^\dagger(y)-U^\dagger(t))(U(w)-U(t))\right\}$$
$$+\mathcal{K}(w,w',t)\mathrm{Tr}\left\{T^eU^\dagger(w)U(w')T^e(U^\dagger(y)-U^\dagger(t))(U(x)-U(t))(U^\dagger(w')-U^\dagger(t))(U(w)-U(t))\right\}\Big] \tag{F.11}$$

Note that $\mathcal{K}(x,y,z)=\mathcal{K}(y,x,z)$. (E7) In $dN^{(1)}$–$dN^{(4)}$ the matrix must sit at the points u,v of $\mathcal{K}(u,v,t)$ of each line: $(U^\dagger(u)-U^\dagger(t))(U(v)-U(t))$ in $dN^{(1)}$, its transpose $(U^\dagger(v)-U^\dagger(t))(U(u)-U(t))$ in $dN^{(2)}$, and $(U^\dagger(v)-U^\dagger(t))(U(v)-U(t))$ for $\mathcal{K}(v,v,t)$ in $dN^{(3)}$, $dN^{(4)}$.

Third term

$$\frac{dN^{(3)}}{d^2wd^2w'dy}=-\frac{\alpha_s^2\delta Y}{8\pi^6(N_c^2-1)}\int_{x,y,t}\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\langle\rho^f(x)\rho^f(y)\rangle$$

$$\times \Big[\mathcal{K}(y,y,t)\mathrm{Tr}\left\{T^e(U^\dagger(x)-U^\dagger(w))U(y)T^e(U^\dagger(x)-U^\dagger(t))(U(y)-U(t))(U^\dagger(y)-U^\dagger(t))(U(y)-U(t))\right\}$$
$$-\mathcal{K}(w',w',t)\mathrm{Tr}\left\{T^e(U^\dagger(x)-U^\dagger(w))U(w')T^e(U^\dagger(x)-U^\dagger(t))(U(y)-U(t))(U^\dagger(w')-U^\dagger(t))(U(w')-U(t))\right\}\Big] \tag{F.12}$$

The fourth term:

$$\frac{dN^{(4)}}{d^2wd^2w'dy}=-\frac{\alpha_s^2\delta Y}{8\pi^6(N_c^2-1)}\int_{x,y,t}\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\langle\rho^f(x)\rho^f(y)\rangle$$
$$\times\Big[\mathcal{K}(x,x,t)\mathrm{Tr}\left\{T^eU^\dagger(x)(U(y)-U(w'))T^e(U^\dagger(y)-U^\dagger(t))(U(x)-U(t))(U^\dagger(x)-U^\dagger(t))(U(x)-U(t))\right\}$$
$$-\mathcal{K}(w,w,t)\mathrm{Tr}\left\{T^eU^\dagger(w)(U(y)-U(w'))T^e(U^\dagger(y)-U^\dagger(t))(U(x)-U(t))(U^\dagger(w)-U^\dagger(t))(U(w)-U(t))\right\}\Big] \tag{F.13}$$

And finally we need to account for the second term in Eq. (F.4)

$$\frac{dN^{(5)}}{d^2wd^2w'dy}=-+\frac{\alpha_s^2\delta Y}{8\pi^6(N_c^2-1)}\int_{x,y,t}\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\langle\rho^f(x)\rho^f(y)\rangle$$
$$\times\Big[\mathcal{K}(y,y,t)\mathrm{Tr}\left\{(U^\dagger(y)-U^\dagger(t))U(y)T^d\right\}\mathrm{Tr}\left\{(U^\dagger(x)-U^\dagger(w))U(y)T^d\right\}$$
$$-\mathcal{K}(w',w',t)\mathrm{Tr}\left\{(U^\dagger(w')-U^\dagger(t))U(w')T^d\right\}\mathrm{Tr}\left\{(U^\dagger(x)-U^\dagger(w))U(w')T^d\right\}$$
$$-\mathcal{K}(x,x,t)\mathrm{Tr}\left\{(U^\dagger(x)-U^\dagger(t))U(x)T^d\right\}\mathrm{Tr}\left\{U^\dagger(x)(U(y)-U(w'))T^d\right\}$$
$$+\mathcal{K}(w,w,t)\mathrm{Tr}\left\{(U^\dagger(w)-U^\dagger(t))U(w)T^d\right\}\mathrm{Tr}\left\{U^\dagger(w)(U(y)-U(w'))T^d\right\}\Big] \tag{F.14}$$

The sum of the above gives the needed target evolution of the leading order single inclusive production. Note that the terms can be combined, but it does not appear to be very useful while matching the results from the diagrams.